August 26, 2026 08:30 AM GMT

Asia-Pacific Technology

China DRAM Going Mainstream; Initiate CXMT at OW

MORGAN STANLEY TAIWAN LIMITED+

Charlie Chan
Equity Analyst
Charlie.Chan@morganstanley.com +886 2 2730-1725

GREATER CHINA TECHNOLOGY SEMICONDUCTORS

Asia Pacific
Industry View **Attractive**

CXMT benefits from China's strong local AI/server demand for HBM/DDR5. Ongoing global DRAM supply tightness supports pricing but also spurs global brands to adopt CXMT's DRAM.

China's memory industry leader: CXMT is China's largest DRAM fab, with proprietary process nodes and an 11% global bit-shipment share in 2026, per our estimate. It is backed by the China Integrated Circuit Industry Investment Fund ("Big Fund") and counts Alibaba, Tencent, and Xiaomi among its strategic investors. CXMT's DRAM process trails Samsung and Micron by about two generations, but its LPDDR5 products already meet Chinese smartphone requirements. More importantly, CXMT is a key domestic supplier for China's AI infrastructure and national security, including in networking equipment and AI GPU servers.

Ongoing capacity expansion despite restricted access to US semi equipment: Our supply-chain checks suggest ASML DUV lithography tools may be the main equipment bottleneck under export controls. Meanwhile, domestic Chinese equipment vendors continue improving and are replacing US suppliers across several process steps. Assuming DRAM industry margins stay near 80%, 2026-28, we expect CXMT to add ~100kwpm annually, reaching 800kwpm capacity by 2031.

Key Debate #1: Can CXMT produce HBM3E soon? Our recent checks suggest HBM3E supply will be the major bottleneck for China'a AI GPU shipment in 2027. We believe CXMT can produce 3-4mn units of HBM3E stacks in 2027 despite low yields, sufficient to support ~0.8-1.0mn units of China AI GPU shipment.

Key Debate #2: Will CXMT affect global DRAM pricing before 2028? We remain constructive on 2027 global DRAM pricing and demand, given AI-driven demand, and we see no pricing impact before 2028 when CXMT should have enough capacity for large-scale shipments. However, Apple reportedly tested CXMT memory products for devices sold in China, and we have seen US PC brands (incl. HP) adopt CXMT DRAM, suggesting commercially acceptable quality, though any Apple supply relationship remains subject to geopolitical and US approval considerations.

Key Debate #3: How to evaluate CXMT? Worth a premium to global peers? Our PT Rmb88, based on our residual income (RI) model, implies an 18.5x 2027e P/E (vs global peers' 4.4x), reflecting CXMT's stronger growth profile. We expect CXMT to benefit from: 1) strong China domestic AI server market demand; and 2) continued capacity expansion even if prices moderate in a memory downcycle, given DRAM's strategic importance in China. **Downside risks include:** 1) earlier-than-expected global DRAM price declines; 2) CXMT delays in new products, HBM or 3D DRAM development; 3) tighter restrictions on China's fabs for semi equipment, materials, or servicing; and 4) weaker demand from major China customers.

Exhibit 1 : We initiate on CXMT (688825.SS) at OW

Stock Rating	**Overweight**
Industry View	**Attractive**
Price target	Rmb88.00
Up/downside to price target (%)	56
Shr price, close (Aug 25, 2026)	Rmb56.50
52-Week Range	Rmb61.80-38.11
Sh out, dil, curr (mn)	60,193
Mkt cap, curr (mn)	Rmb3,400,904.5
EV, curr (mn)	Rmb3,510,785.0
Avg daily trading value (mn)	Rmb35,437

Fiscal Year Ending	12/25	12/26e	12/27e	12/28e
EPS (Rmb)**	0.0	3.0	4.8	6.3
Prior EPS (Rmb)**	-	-	-	-
EPS (Rmb)§	-	2.5	4.2	5.2
Revenue, net (Rmb mn)	61,799.3	388,869.4	643,077.2	850,095.9
EBITDA (Rmb mn)	35,859.1	312,302.1	508,674.1	668,090.4
ModelWare net inc (Rmb mn)	1,874.9	189,693.0	319,242.9	423,535.0
P/E	-	18.8	11.8	8.9
P/BV	-	12.4	6.1	3.6
RNOA (%)	4.2	85.3	110.2	126.5
ROE (%)	4.5	334.2	105.0	67.9
EV/EBITDA	-	11.1	6.3	4.2
Div yld (%)	-	0.0	0.0	0.0
FCF yld ratio (%)**	-	3.3	7.2	10.3
Leverage (EOP) (%)	152.1	(49.8)	(84.8)	(101.3)

Unless otherwise noted, all metrics are based on Morgan Stanley ModelWare framework
** = Based on consensus methodology
§ = Consensus data is provided by Refinitiv Estimates
e = Morgan Stanley Research estimates

Contributors

MORGAN STANLEY TAIWAN LIMITED+
Charlie Chan
Equity Analyst
+886 2 2730-1725
Charlie.Chan@morganstanley.com

MORGAN STANLEY & CO. INTERNATIONAL PLC+
Shawn Kim
Equity Analyst
+44 20 7677-1018
Shawn.Kim@morganstanley.com

MORGAN STANLEY & CO. LLC
Joseph Moore
Equity Analyst
+1 212 761-7516
Joseph.Moore@morganstanley.com

MORGAN STANLEY & CO. INTERNATIONAL PLC+
Lee Simpson
Equity Analyst
+44 20 7425-3378
Lee.Simpson@morganstanley.com

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Kazuo Yoshikawa, CFA
Equity Analyst
+81 3 6836-8408
Kazuo.Yoshikawa@morganstanleymufg.com

MORGAN STANLEY & CO. LLC
Shane Brett
Equity Analyst
+1 212 761-1022
Shane.Brett@morganstanley.com

MORGAN STANLEY TAIWAN LIMITED+
Daniel Yen, CFA
Equity Analyst
+886 2 2730-2863
Daniel.Yen@morganstanley.com

MORGAN STANLEY ASIA LIMITED+
Duan Liu
Equity Analyst
+852 2239-7357
Duan.Liu@morganstanley.com

MORGAN STANLEY ASIA LIMITED+
Daisy Dai, CFA
Equity Analyst
+852 2848-7310
Daisy.Dai@morganstanley.com

MORGAN STANLEY TAIWAN LIMITED+
Tiffany Yeh
Equity Analyst
+886 2 7712-3032
Tiffany.Yeh@morganstanley.com

MORGAN STANLEY ASIA LIMITED+
Ethan Jia
Research Associate
+852 3963-2287
Ethan.Jia@morganstanley.com

MORGAN STANLEY MUFG SECURITIES CO., LTD.+
Suzune Tamura, CFA
Equity Analyst
+81 3 6836-8891
Suzune.Tamura@morganstanleymufg.com

MORGAN STANLEY & CO. INTERNATIONAL PLC+
Cindy Huang
Equity Analyst
+44 20 7425-2915
Cindy.Huang@morganstanley.com

2

Executive Summary: China DRAM Supply – Can It Expand Into The Global Market?

Tech Diffusion

A Morgan Stanley Research Key Theme of 2026

Multipolar World

A Morgan Stanley Research Key Theme of 2026

How big and how advanced is China's CXMT in DRAM?

Some key facts and trends in capacity expansion

For 2026, we estimate CXMT will achieve:

- 300k wpm of capacity, representing 13% of global DRAM wafer supply
- 41bn Gb of bit shipments, representing 11% of global bit supply
- US$57bn in revenue

We note that YMTC and JHICC (two other Chinese memory integrated device manufacturers (IDM)) are also building DRAM capacity. However, we expect CXMT to remain the domestic market leader in terms of both technologies and capacity.

We estimate CXMT's share of global DRAM bit shipments will increase to 15% by 2030, from 11% in 2026. With DRAM industry margins currently at 80%, as disclosed by the leading DRAM vendors, we expect CXMT to accelerate capacity expansion from 2H26, adding ~100k wafers per month (wpm) annually.

The technology advancement continues

Recent discussions with DRAM customers and semiconductor equipment vendors point to improving product competitiveness. Our checks indicate CXMT's DDR5 supports 5600MHz, with a roadmap to 6400MHz, broadening its relevance to mainstream server DRAM.

CXMT has also developed 3D stacking capability for a 12H HBM3E package. Thermal management and manufacturing challenges remain, and we forecast mass production in 1H27.

Major Chinese CSPs, including Alibaba, ByteDance, and Tencent, may already be purchasing CXMT DRAM for server applications, according to our checks. Adoption of CXMT's LPDDR5 is expanding among Chinese smartphone brands and select lower-end AI accelerator platforms, while Dell and HP have reportedly used CXMT DRAM in select China and ASEAN PC models. These checks support the view that CXMT is increasingly viewed as a source in mainstream DRAM, although advanced qualification remains a separate hurdle.

Exhibit 2: DDR5 spec comparison: CXMT vs. global peers

DDR5 Spec Comparison				
Manufacturer 16 Gb DDR5	Samsung 16 Gb DDR5	SK hynix 16 Gb DDR5	Micron 16 Gb DDR5	CXMT 16 Gb DDR5
Year First Revealed (Clock Speed, ex.)	2021DDR5-5600, DDR5-6000	2021DDR5-4800, DDR5-6000	2021DDR5-4800	2024DDR5-6000
Parent Product Examples	G.SKILL Trident Z5 DDR5 F5-5600U3636C16GX2-TZ5K (K4RAH086VB-BCQK)	SK Hynix 32 GB HMCG88MEBUA81N DDR5 UDIMM PC5-4800B module (H5CG48MEBD-X014)	TeamGroup ELITE DDR516 GB UDIMM (MT60B2G8HB-48B:A)	Gloway 16GB×2 (32GB) DDR5-6000 Module(H2G08SC522-6AF)
DRAM Generation	D1y	D1y	D1z	G4
Die Size (Scaled)	73.58 mm^2(8.94 mm × 8.23 mm)	75.21 mm^2(9.24 mm × 8.14 mm)	66.26 mm^2(8.43 mm × 7.86 mm)	66.99 mm^2(8.19 mm × 8.18 mm)
Bit Density	0.217 Gb/mm^2	0.213 Gb/mm^2	0.241 Gb/mm^2	0.239 Gb/mm^2
Peripheral Gate Process	HKMG (Gate-first)	Poly/W Gate	Poly/W Gate	Poly/W Gate

Source: Company data, Techinsights, Morgan Stanley Research.

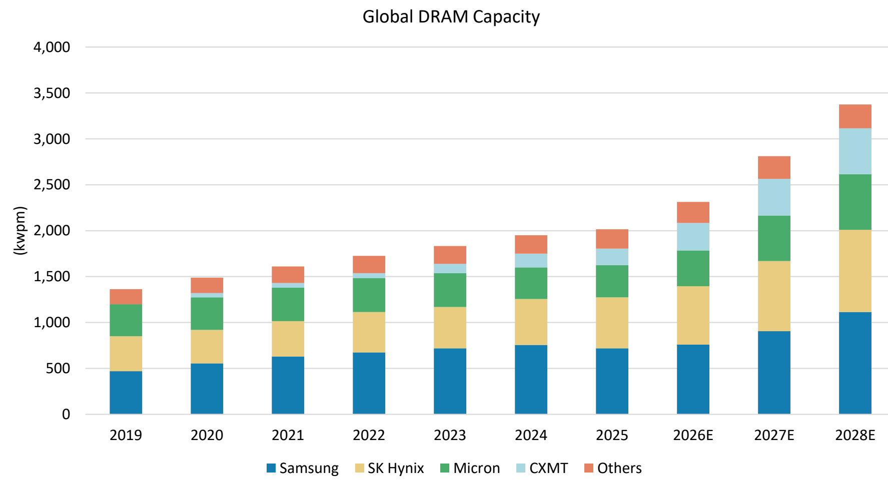
DUV is less efficient than EUV but works; 3D DRAM provides an opportunity for China to overcome the EUV limitation

We view ASML's deep ultraviolet (DUV) lithography tools as the primary bottleneck to this expansion. Our recent checks with DRAM customers and equipment vendors suggest that global investors may be underestimating CXMT's DRAM quality, particularly in server DRAM performance, high-bandwidth memory (HBM) generation and density, and adoption across China's AI compute ecosystem. Debate also continues on whether Apple could qualify CXMT's low-power double data rate (LPDDR) products, especially against a backdrop of tight global memory supply.

CXMT recently completed the migration to its Gen4b process, a 16nm-class node, using primarily domestic equipment, with the exception of DUV lithography tools, which we believe have achieved production yields above 80%. As a result, we expect CXMT to accelerate capacity expansion from 2H26. On our estimates, CXMT alone could expand capacity to 800k wafers per month by 2031, provided the company continues to secure DUV lithography tools from overseas suppliers. This would represent a slightly faster growth rate than the broader memory industry's expected capacity expansion, which we estimate will roughly double over the same period.

Our channel checks with equipment vendors suggest that CXMT may adopt vertical channel transistor (VCT) technology, a transitional architecture toward 3D DRAM, to advance to its Gen5 (~15nm-class) process node without relying on EUV lithography tools. CXMT will begin limited production of this next-generation DRAM technology in 2H28 (see Appendix 2: What is 3D DRAM Technology? Can China CXMT be the First Chip Maker to Adopt it?).

Exhibit 3: Global DRAM capacity to reach 3,374kwpm in 2028e , while CXMT may account for 15% of that (around 500kwpm by 2028)



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

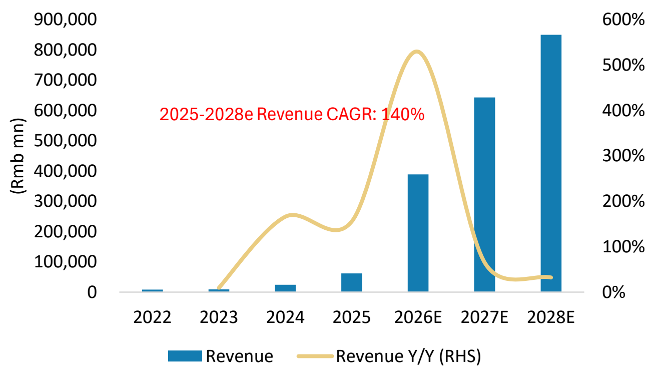
Initiate China CXMT at OW – Both scale and technology are advancing

We initiate coverage of CXMT with an Overweight rating and a price target of Rmb88, implying an 18.5x 2027e P/E, compared with global peers at 4.4x 2027e P/E. We believe this valuation premium is justified by CXMT's stronger growth outlook, supported by sustained capacity expansion and increasing demand from China's AI infrastructure ecosystem. CXMT's higher growth relative to global peers reflects two key factors: (1) strong demand from China's domestic AI server market, including both GPU- and CPU-based systems; and (2) continued capacity expansion driven by localization requirements. We expect CXMT to continue expanding capacity even if DRAM pricing moderates.

Bull case: In this scenario, we assume US cloud service providers and consumer electronics brands begin sourcing CXMT DRAM amid severe global memory shortages, while CXMT achieves a breakthrough in 3D DRAM technology by 2028 (see Appendix 2: What is 3D DRAM Technology? Can China CXMT be the First Chip Maker to Adopt it?). Our checks with equipment vendors suggest CXMT could adopt VCT technology, a transitional architecture toward 3D DRAM, to advance to its Gen5 (~15nm-class) process node without relying on EUV lithography tools.

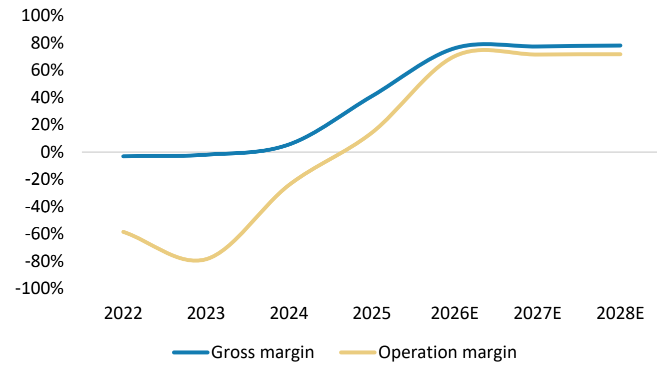
Bear case: In this scenario, we assume tighter export restrictions on semiconductor manufacturing equipment, particularly ASML's DUV lithography tools.

Exhibit 4: CXMT's revenue to rise at a 140% revenue CAGR, 2025-2028e



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

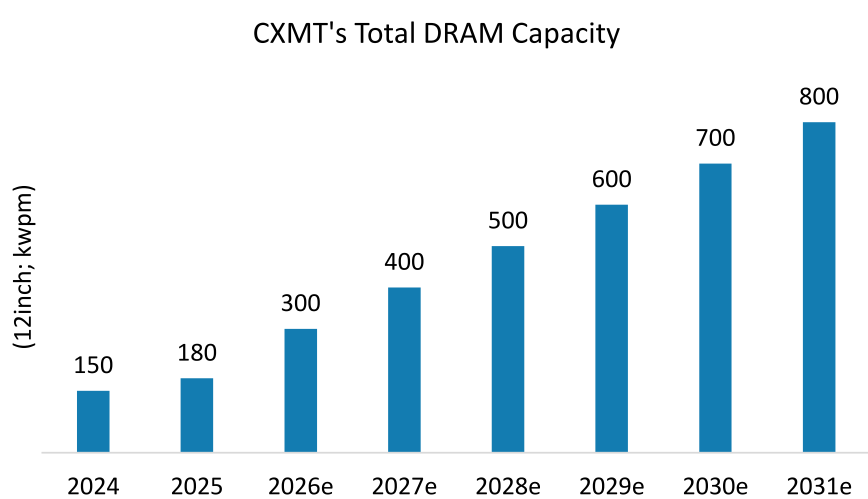
Exhibit 5: CXMT's OpM to be 70-72%, 2026-28e



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 6:

CXMT to expand capacity from 180kwpm in 2025 to 300kwpm in 2026, followed by annual capacity additions of approximately 100k wpm, reaching 800k wpm by 2031



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Implications for Global Memory and Semiconductor Capital Equipment Stocks

We remain constructive on global DRAM pricing and demand in 2027. While CXMT's supply is expanding rapidly, we do not expect incremental Chinese DRAM supply to materially affect industry pricing before 2028, given strong AI-driven demand and the time required for new capacity to reach meaningful scale. We also do not expect CXMT's capacity expansion to stop even if industry profitability levels or cash-flow generation start to moderate. If the DRAM industry enters a more cyclical phase after 2028, Chinese DRAM supply could become an increasingly important factor in global supply-demand dynamics and pricing.

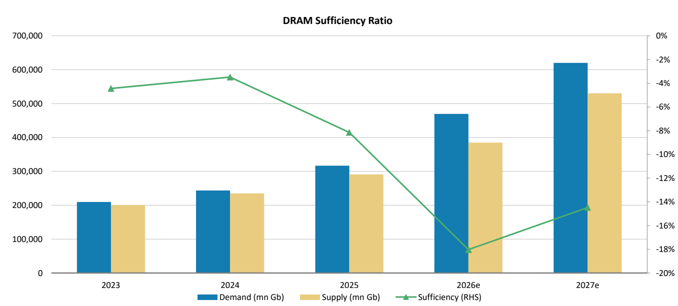
We think China's domestic DRAM supply remains insufficient to meet rapidly growing demand, providing CXMT with a substantial opportunity to expand market share (see Appendix 3: Assessing China DRAM Demand/Supply (Local Sufficiency) for a detailed supply-demand analysis).

So when should investors begin to focus on China DRAM as a competitive threat? As discussed in the previous section, the bull case for China DRAM rests on several developments:

- US CSPs and consumer electronics brands could begin sourcing DRAM from CXMT amid severe global memory shortages.
- Strong demand from China's domestic AI infrastructure buildout, including both GPU- and CPU-based systems, could drive sustained growth in local memory consumption.

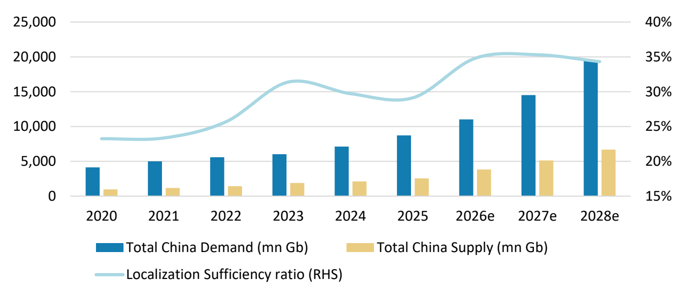
- Continued strength in global AI infrastructure spending could support favorable DRAM pricing and industry profitability for longer than currently expected. Finally, successful technology migration—potentially through VCT and ultimately 3D DRAM architectures—could help CXMT narrow the technology gap with global peers and accelerate its capacity expansion plans.

Exhibit 7: DRAM supply sufficiency ratio: Supply growth continues to lag demand



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 8: China's domestic supply is even more constrained, providing significant room for CXMT to grow



Source: TrendForce, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Global memory and equipment implications

Our Korea Technology analyst Shawn Kim maintains his OW ratings on both Samsung Electronics (Top Pick, 005930.KS, PT W381,000) and SK hynix (000660.KS, PT W2,600,000). He continues to view both companies as key beneficiaries to AI compute and agentic AI trends (see Agentic AI – The Surge Begins).

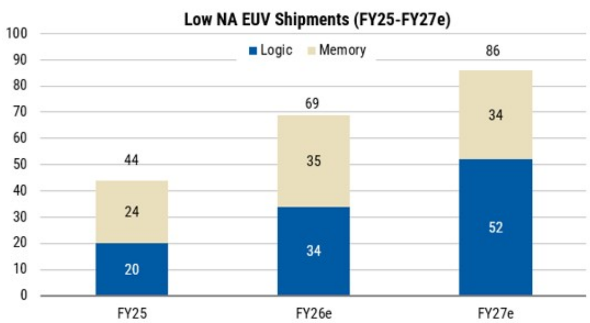
We estimate DRAM prices will increase by more than 20-30% in 3Q26, which should keep the YoY rate of change accelerating. Continued pricing strength is driving earnings upgrades and supporting valuation multiples, with both stocks trading at ~3.5-4.5x 2027e earnings. That said, we remain mindful that memory stocks are highly sensitive to changes in the rate of earnings and pricing momentum. With YoY DRAM pricing growth likely to plateau in 4Q26, and visibility on supply discipline beyond 2028 still evolving, we believe the sector may lack near-term cyclical catalysts until industry supply-demand dynamics become clearer. Nevertheless, valuation multiples could continue to re-rate higher, supported by favorable long-term agreements (LTAs) and sustained demand from AI infrastructure deployment.

Our North America Semiconductors analyst Joseph Moore reiterates his OW rating on Micron (MU.O, PT US$1,200). He believes a meaningful supply-demand imbalance remains unlikely as long as AI-related demand continues on its current trajectory. Currently, all DRAM suppliers, including CXMT, are expanding capacity as aggressively as possible. However, we estimate AI-related demand will continue to grow at a significantly faster rate than DRAM supply. At the same time, industry supply growth is constrained by the higher wafer intensity of high bandwidth memory (HBM) production and limited availability of EUV lithography tools. As a result, DRAM is increasingly becoming a key bottleneck in the broader AI infrastructure buildout. We expect CXMT to gain market share through faster capacity growth. However, we believe the company will prioritize domestic customers and demand within China, while geopolitical restrictions continue to limit access to more attractive international markets. That said, our investment case does not assume an indefinite period of memory shortages. We believe the current valuation already reflects a more normalized industry outlook. Micron currently trades at approximately 6x our estimate of peak run-rate EPS and 2.7x FY28E book value, while also offering significant capital return potential through the cycle. Our base-case model assumes approximately US$80bn of capital returns by mid-2028, equivalent to ~7% of the current market cap. However, with total FCF >US$300bn over the same period, capital returns could ultimately be 2-3x than our current assumption. In our view, the combination of substantial FCF generation, increasing earnings visibility from long-term agreements (LTAs), and a relatively low starting valuation should continue to support Micron's investment case, particularly if AI-related demand remains strong. For more details, see Semiconductors: Selloff of US memory stocks creates a compelling entry point (20 Jul 2026).

Our European Technology Semiconductors analyst Lee Simpson reiterates his OW rating on ASML (Top Pick, ASML.AX, PT €1,930). In his view, the industry's supply response ultimately depends on demand visibility and commercial commitment. Our Europe Tech Semis team expects ASML to continue expanding capacity in line with sustained EUV demand. The team model low NA EUV shipments increasing from 44 tools last year to more than 60 this year, with further capacity expansion possible as demand visibility improves. Importantly, the constraint is not simply ASML's lead times. Rather, the industry must navigate the entire capacity expansion chain: memory manufacturers need to commit capital, equipment suppliers need to deliver tools, fabs need to be prepared, processes need to be qualified, yields need to ramp, and HBM packaging and testing capacity needs to scale. See our Europe Semiconductors Team's latest ASML notes here and here.

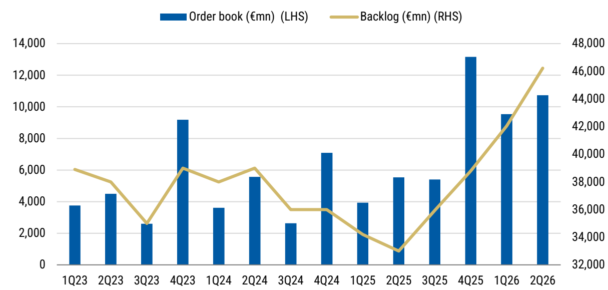
The post-Covid memory cycle offers an important lesson. Both Intel and Samsung deferred EUV-related orders after demand weakened and execution challenges delayed capacity ramps, resulting in order volatility across the semi equipment supply chain. In the current cycle, memory makers and hyperscalers may need to provide stronger LTAs, prepayments, and capacity commitments before suppliers undertake meaningful capacity expansion. This dynamic supports our view that supply relief is likely to materialize gradually rather than immediately. As a result, we believe memory market tightness could persist into 2027, even as EUV capacity continues to expand.

Exhibit 9: ASML Low NA EUV Shipments



Source: Company data, Morgan Stanley Research estimates (e)

Exhibit 10: ASML Order Book and Backlog



Source: Company data, Morgan Stanley Research

On the supply side, access to advanced lithography tools remains the key constraint. EUV systems have effectively been unavailable to China since 2023, and ASML has never shipped an EUV tool to the country. We believe this is a major factor behind the apparent plateau in CXMT's leading-edge DRAM capacity expansion.

The proposed MATCH Act could further tighten these constraints by designating CXMT, YMTC, SMIC, Hua Hong, and Huawei as "covered facilities." The proposed legislation would restrict not only sales of DUV immersion lithography tools, but also servicing and maintenance of existing installed equipment.This is particularly important because DUV tool fleets require ongoing maintenance and support to sustain production yields. While the legislation remains under consideration and has not been enacted, it highlights the direction of policy risk. In our view, the regulatory trajectory continues to point toward tighter, rather than looser, constraints on China's ability to scale advanced memory production.

According to our Japan Semiconductor Production Equipment analyst Suzune Tamura, demand for Japanese SPE companies remains quite strong. Customers continue to accelerate equipment purchases and, in some cases, are willing to pay premiums for earlier delivery, particularly when their orders carry lower priority than those of other customers. Equipment suppliers and component vendors have been expanding production capacity since late last year. Even so, lead times continue to lengthen, underscoring the fact that demand remains ahead of available supply.

Among Japanese semiconductor equipment companies in Tamura-san's coverage universe, Kokusai Electric (OW, 6525.T, PT JPY8,300) has the highest exposure to CXMT. Sales to CXMT accounted for 20% of total sales in F3/25 and an estimated 6% in F3/26. Tokyo Electron (OW, 8035.T, PT JPY65,000) also has meaningful exposure to CXMT, with its share of CXMT's semiconductor equipment spending broadly in line with its overall market share. SCREEN Holdings (OW, 7735.T, PT JPY17,000), on the other hand, has lost share at CXMT over the past several years, as Chinese semiconductor manufacturers increased adoption of domestically produced equipment in line with government localization initiatives. More recently, however, SCREEN has begun to regain share at CXMT. As CXMT advances its HBM development efforts, process complexity continues to increase and wafer cleaning steps have become more critical.

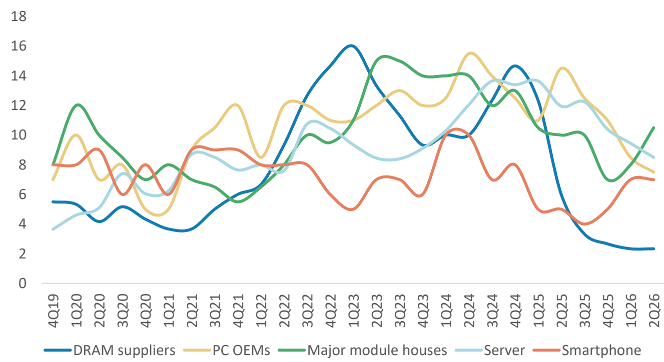
HBM capacity expansion in China should also benefit semiconductor back-end equipment companies. Disco (OW, 6146.T, PT JPY78,000) maintains a leading position in grinding equipment used in through-silicon via (TSV) manufacturing. The TSV process includes edge trimming and back grinding, both of which require high-cleanliness equipment. These tools typically command prices of several hundred million yen per unit, compared with tens of millions of yen for conventional back-end equipment.

In memory testing, Advantest (OW, Top Pick, 6857.T, PT JPY36,000) has a leading position in the HBM testers market. While Teradyne (EW, TER.O, PT US$397, covered by Shane Brett) has expanded into high-speed HBM testing in recent years, Avantest maintains a strong presence in memory testing and is working to regain market share through a new product platform scheduled for launch next year.

Our Japan Semiconductor analyst Kazuo Yoshikawa has an OW rating on Renesas Electronics (6723.T, PT JPY6,000). He believes the company is well positioned to benefit from AI-driven server demand through both CPU unit growth and rising memory interface content per server system. He forecasts a 29% CAGR in Renesas's memory interface revenue over CY25-28, driven by continued growth in server CPU shipments and increasing memory interface content as data-transfer speeds and bandwidth requirements between CPUs and memory continue to rise. He also estimates that memory interface products generate gross margins well above the company average, making this segment an important contributor to Renesas's overall gross margin expansion and earnings growth over the medium term.

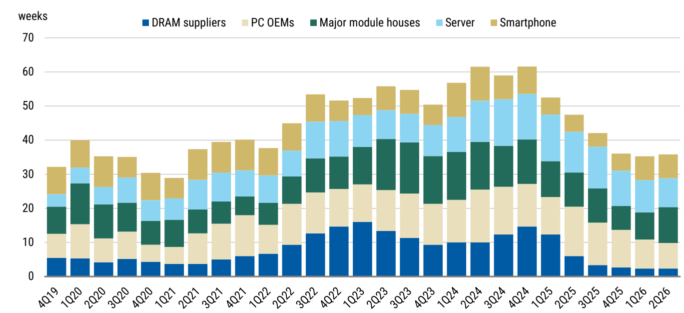
The company is also well positioned for multiplexed rank dual in-line memory module (MRDIMM) adoption, with each module incorporating one multiplexed rank registering clock driver (MRCD) and ten multiplexed rank data buffers (MDB), representing ~3x the content of a conventional DDR5 RDIMM. Renesas has already introduced a full MRDIMM chipset solution and commenced mass production in 2025.

Exhibit 11: DRAM inventory levels



Source: TrendForce, Morgan Stanley Research.

Exhibit 12: Consolidated DRAM inventory level across different end-markets



Source: TrendForce, Morgan Stanley Research.

Greater China semiconductors – Any winners or losers?

Taiwan DRAM vendors Nanya Tech and Winbond should remain among the beneficiaries: We maintain our OW ratings on Nanya Tech (2408.TW, NT$580) and Winbond (2344.TW, NT$288), given the industry's strong profitability and the continued re-rating of global memory stocks.

We reiterate our OW ratings on CXMT's equipment vendors NAURA, AMEC, and, ACMR: As CXMT expands capacity and migrates toward more advanced DRAM technologies, we expect its equipment intensity to rise. More complex, higher-aspect-ratio capacitor structures require greater etch and deposition intensity, while tighter process tolerances increase demand for cleaning, thermal processing, and inspection tools. On the back end, continued interconnect scaling should also support demand for deposition, wet-cleaning, and electrochemical-plating equipment. At the same time, export restrictions and growing manufacturing experience are likely to accelerate qualification of domestic equipment. In our view, the combination of capacity expansion, technology migration, and rising localization should provide continued support for domestic wafer fabrication equipment (WFE) demand over the coming years.

- **NAURA (OW, 002371.SZ, PT Rmb818):** We see NAURA as one of the broadest beneficiaries of CXMT's expansion. Its ICP etchers can address selected conductor and dielectric-related etch applications, while its PVD tools should benefit from increasing requirements for additional metal deposition and barrier-layer formation. NAURA also has exposure to back-end wet-cleaning processes, where rising interconnect complexity is increasing contamination-control requirements. Capacity expansion should support tool volume growth, while technology migration toward more advanced process nodes could expand the number of addressable process steps. In addition, NAURA's broad portfolio creates opportunities for cross-selling and may support further wallet-share gains following successful initial production qualification.
- **AMEC (OW, 688012.SS, PT Rmb488):** We see AMEC as a key beneficiary of rising capacitor-hole etch requirements in advanced DRAM manufacturing. CCP etch plays a critical role in DRAM production, as increasingly complex capacitor structures require high-aspect-ratio dielectric etching with stringent profile, selectivity, and uniformity requirements. As China's leading domestic etch

equipment supplier, we believe AMEC is well positioned to benefit from ongoing localization efforts as CXMT advances its process technology. Successful qualification in more demanding capacitor etch applications would represent an important strategic milestone, supporting repeat orders, higher chamber intensity, and expansion of the company's addressable etch opportunity per wafer.

- **ACMR (OW, ACMR.O, PT US$130):** ACMR should benefit from rising cleaning intensity across both front-end and back-end DRAM manufacturing processes. As process technologies advance, tighter control of particles, residues, and metallic contamination becomes increasingly important, supporting demand for single-wafer cleaning solutions. On the back end, growing interconnect complexity should create additional opportunities for bot wet-cleaning and electrochemical-plating (ECP) equipment. We believe ACMR's ECP platform is well positioned to address selected copper-plating applications as CXMT expands capacity and advances its technology roadmap. Successful qualification in more critical process steps could support repeat orders and increase ACMR's content per wafer.

We also expect Chinese Design houses in our coverage to benefit

- **GigaDevice (OW, 603986.SS, PT Rmb750, covered by Daniel Yen):** We view GigaDevice as a key beneficiary of CXMT's continued expansion, given its position as one of CXMT's major partners in the specialty memory market. GigaDevice disclosed a Rmb5.711bn budget for related-party transactions with CXMT in 2026, significantly above the Rmb1.161bn budget for 2025 and the previously disclosed Rmb1.547bn budget for 1H26 (link). We also believe CXMT could increase DDR4 capacity by an additional 10k wpm by 2027, on top of its current ~8k wpm, to address strong domestic demand in China (link). Beyond wafer-capacity growth, Gigadevice also benefits from CXMT's process migrations, which supports improvements in performance, density, and cost competitiveness. We believe CXMT's continued capacity expansion and process migration will provide ongoing support for GigaDevice's DRAM business. In addition, we understand that GigaDevice is collaborating with CXMT on wafer-on-wafer (WoW) technology development, where CXMT could provide meaningful technology support (link).
- **Montage (OW, 6809.HK, PT HK$432; 688008.SS, PT Rmb377, covered by Daniel Yen):** As a global leader in memory interface chips, including RCD/DB and MRCD/MDB solutions, Montage currently derives a significant portion of its memory-interface business from the three leading global memory suppliers. However, as CXMT continues to expand capacity and makes further inroads into the server DRAM market, we believe it could create a meaningful opportunity for Montage to increase its penetration within China's domestic memory supply chain. A growing domestic server DRAM ecosystem could enable Montage to further strengthen its leadership position in memory interfaces while reducing its reliance on the three global memory vendors, i.e., Samsung, SK hynix, and Micron.

What could the bull case be for China DRAM (or a bear case for global DRAM)?

(1) US cloud service providers (CSP) and consumer electronics brands start to source CXMT DRAM amid severe global memory shortages.

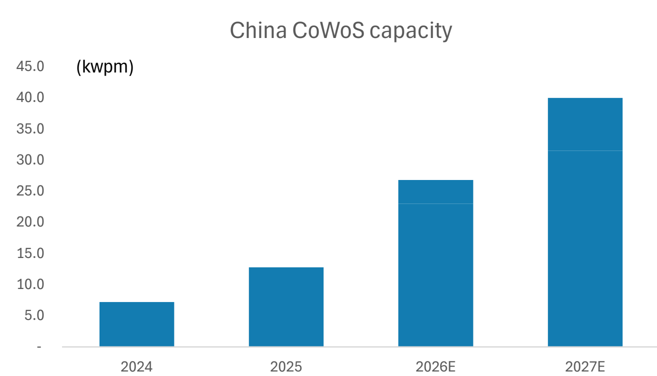
We expect CXMT to benefit from a broadening customer base as tight global memory supply continues to support demand for DRAM. In particular, select US CSPs and consumer electronics brands have begun evaluating or sourcing CXMT DRAM products, reflecting both ongoing supply constraints and a growing willingness among customers to diversify their supplier base. While qualification cycles and volume adoption may take time, incremental demand from international customers could improve CXMT's capacity utilization, strengthen its competitive position, and increase its strategic relevance within the global DRAM supply chain.

(2) Stronger demand from China's domestic AI server market, including both GPU and CPU

Demand from China's domestic AI server market also represents a key growth driver. Deployment of both GPU- and CPU-based AI servers is accelerating, supported by increasing investment in large language models (LLM), data centers, and sovereign AI infrastructure. This trend should drive sustained demand for high-capacity server DRAM, particularly as memory content per server increases. We believe CXMT is well positioned to capture a growing share of this demand given its status as a domestic supplier and improving product availability. Key variables remain product mix, performance requirements, and customer qualification timelines.

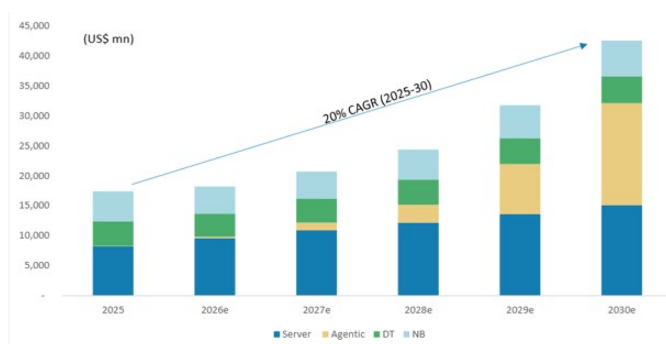
Besides DRAM-related development, we expect CXMT to explore additional growth opportunities, including 3D DRAM, NAND and advanced packaging technologies. While we have not seen any detailed capacity expansion plans for these initiatives, we believe they could provide additional revenue streams over the longer term and further diversify the company's business beyond conventional DRAM products.

Exhibit 13: China domestic CoWoS capacity to produce local AI GPU chips, which would need support from local HBM and LPDDR memory



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 14: We estimate China's CPU TAM to rise to US$42bn in 2030, which should drive substantial demand for CXMT DRAM



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

(3) Global AI infrastructure demand continues to support DRAM pricing

At the industry level, robust global AI infrastructure spending should continue to support DRAM pricing. AI servers require substantially more memory than traditional enterprise servers, while strong demand for HBM and high-density DDR5 is absorbing leading-edge wafer capacity. This shift in capacity allocation is tightening supply across the broader DRAM market and supporting a favorable pricing environment for both advanced and mainstream DRAM products. In our view, sustained AI-driven demand should extend the current DRAM upcycle, although the pace of price increases may moderate as suppliers gradually expand capacity.

(4) Technology migration: Progress toward next-generation DRAM architectures

While restrictions on extreme ultraviolet (EUV) lithography tools are likely to slow CXMT's migration to a 15nm-class node, comparable to the 1y node of leading global DRAM suppliers, the company continues to explore alternative technology pathways. Our industry checks, together with commentary from DRAMeXchange, suggest that CXMT may pursue 3D DRAM architectures to advance toward its Gen5 technology node without relying on EUV tools.

Technology migration remains the primary medium-term constraint for CXMT. Restrictions on access to EUV lithography equipment are likely to delay its transition toward a 15nm-class process node, broadly comparable with the 1y node of leading global DRAM suppliers. However, our industry checks, together with indications from DRAMeXchange, suggest that CXMT may explore a three-dimensional DRAM architecture as an alternative pathway to improve density and extend process scaling, although we note that CXMT has not made such public announcements. If confirmed and successful, we think this approach could support its migration toward a fifth-generation technology platform and partially offset the limitations of conventional planar scaling. That said, execution risk remains elevated, as 3D DRAM development will require advances across process integration, yield optimization, cost management, and high-volume manufacturing.

Core Debate: Can CXMT Produce HBM3E Soon?

Why this debate? HBM is the high-entry barrier DRAM product, and if China can enter this market, it may disrupt global DRAM value.

Our recent checks suggest HBM3E supply will be the major bottleneck for China'a AI GPU shipment in 2027. Global memory vendors' HBM4 conversion may further reduce their supply of HBM3E to China vendors. It would be critical for China to own a local HBM3E supply for its new-generation AI GPUs.

Market's view: It would take time. China CXMT needs to build HBM technology from scratch, such as HBM2, and 8-Hi stacking, before it can achieve the mainstream HBM3E 12-Hi stacking.

Our view: Our checks suggest CXMT can produce 12-Hi stacking with wafer-on-wafer technology. We believe CXMT can produce 3-4mn units of HBM3E stacks in 2027 despite low yields, sufficient to support ~0.8-1.0mn units of China AI GPU shipment. That said, we believe the global HBM supply will remain tight given strong global AI demand.

HBM: A high-end global DRAM segment that CXMT could potentially enter over time

CXMT has evolved from a portfolio focused primarily on legacy DRAM products toward a broader range of advanced memory solutions, including:

- DDR5 chips and PC memory modules.
- LPDDR5/LPDDR5X products for Chinese smartphones and mobile computing applications.
- Server memory offerings, including products supporting RDIMM and MRDIMM platforms.
- Longer-term ambitions in HBM4 and 3D/vertical-channel DRAM technologies.

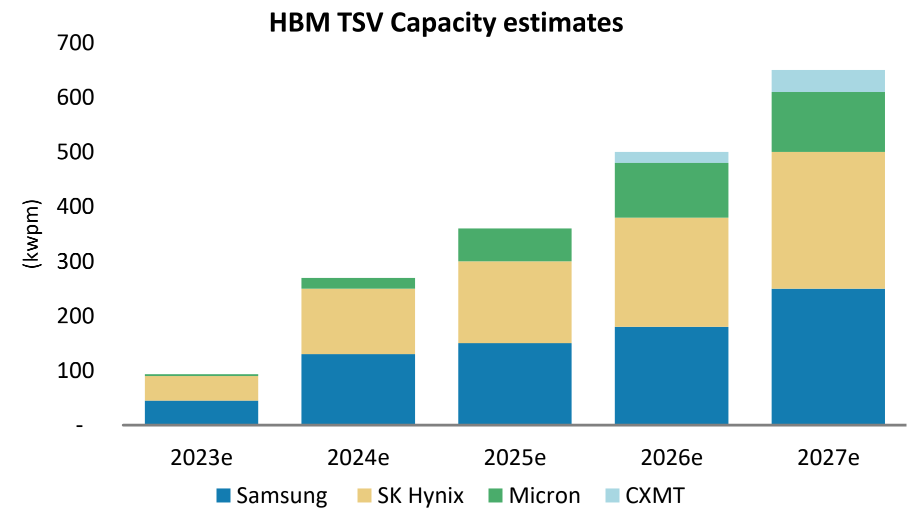
The core earnings driver remains conventional DDR and LPDDR products. While HBM is strategically important and represents a potential longer-term growth opportunity, we do not yet view it as a proven earnings pillar. We forecast HBM3E mass production in 2027 and HBM4 in 2028. However, successful HBM commercialization depends on far more than producing a functional memory stack. Key factors include known-good-die yield, TSV yield, thermal behavior, packaging, test capacity, and accelerator qualification.

Exhibit 15: HBM roadmap

Company	2020	2021	2022	2023	2024	2025	2026E	2027E	2028E
SK hynix	HBM2E		HBM3		HBM3E (1st)	HBM3E	HBM4	HBM4E	HBM5
Samsung	HBM2E		HBM3		HBM3E	HBM3E	HBM4	HBM4E	HBM5
Micron	HBM2E				HBM3E	HBM3E	HBM4	HBM4E	HBM5
CXMT					HBM2	HBM2E	HBM2E	HBM3E 12H	HBM4

Source: Company data, Techinsights, Morgan Stanley Research. E = Morgan Stanley Research estimates

Exhibit 16: CXMT's HBM TSV capacity to reach 20kwpm in 2026e, 40kwpm in 2027e and 70kwpm in 2028e



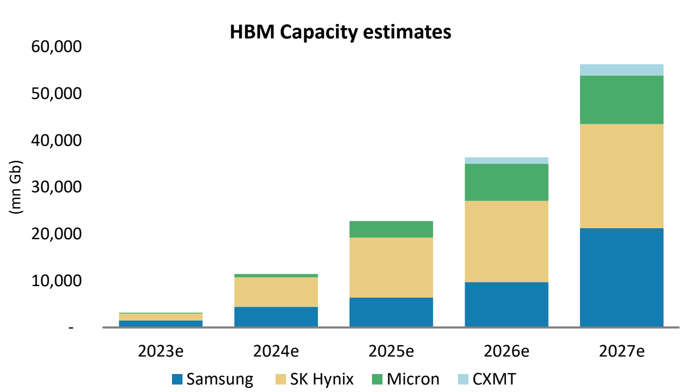
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

What's CXMT's edge in HBM?

HBM capacity is ramping rapidly, but the market remains highly concentrated among Samsung, SK hynix, and Micron. We forecast combined year-end HBM capacity among the three vendors will rise from ~500kwpm in 2026e to ~600kwpm in 2027e. Current export restrictions prohibit shipments of standalone raw HBM stacks into China. However, processors and accelerators that incorporate HBM can still be shipped provided they comply with applicable performance thresholds and export-control requirements.

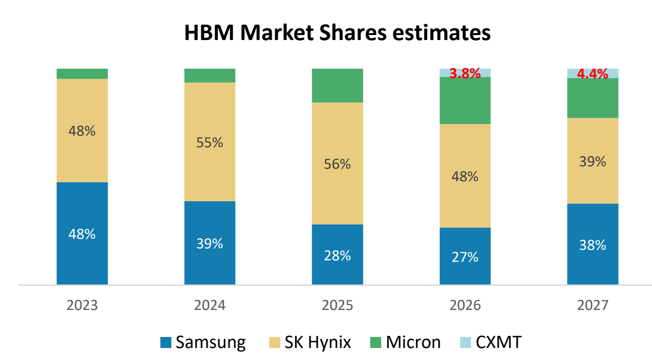
We believe CXMT is accelerating its HBM roadmap to support growing demand from domestic AI accelerator developers. Our forecasts assume HBM2E production in 2026, followed by HBM3E 12-Hi in 2027e and HBM4 in 2028e. Our checks suggest CXMT can produce 12-Hi stacking with wafer-on-wafer technology, so 24GB per HBM chip is possible.

Exhibit 17: Global HBM capacity build



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 18: CXMT's HBM to account for 3.8%/4.4% if the total HBM market in 2026e and 2027e in terms of bit shipment



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

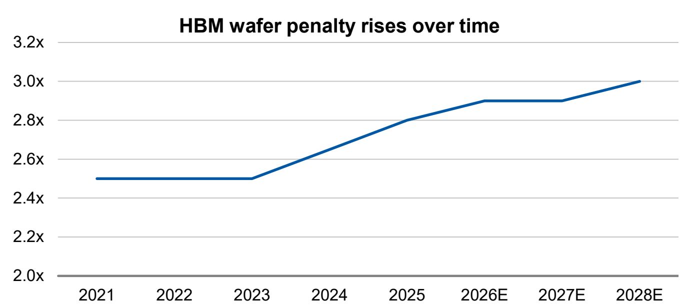
HBM consumes a disproportionate share of DRAM capacity

HBM is more than just a source of incremental DRAM demand. It also imposes a meaningful throughput penalty on the memory manufacturing ecosystem. Compared with conventional DRAM, HBM requires significantly larger die sizes and introduces additional complexity through 3D stacking, TSV integration, advanced packaging, extensive testing, and customer qualification. These additional manufacturing steps reduce effective bit output per wafer and increase demands on packaging and testing infrastructure.

Our model suggests the HBM bit-output penalty will increase from ~2.5x conventional DRAM wafers in 2021–23 to ~3.0x by 2028e. In other words, each unit of HBM bit output consumes materially more wafer capacity than conventional DRAM output.

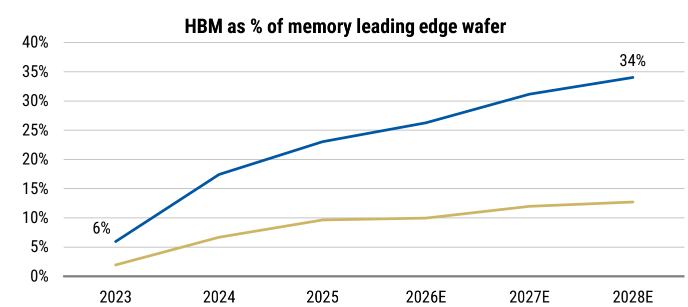
At the same time, HBM is absorbing a growing share of the industry's most constrained wafer capacity. We estimate HBM's share of leading-edge DRAM wafer consumption will increase from ~6% of leading-edge memory wafers in 2023 to ~34% by 2028e. As AI adoption accelerates, AI customers are not only creating incremental memory demand, but also consuming an increasing share of the leading-edge capacity that would otherwise support DDR, LPDDR, and conventional server DRAM products.

Exhibit 19: HBM die penalty – bit-output is reduced due to much larger DRAM dies



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 20: HBM cannibalization: Share of advanced DRAM wafer is rising sharply



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Even with this capacity expansion, supply is likely to remain constrained because HBM availability depends on far more than wafer starts. Effective HBM supply is also a function of yield, utilization, TSV capability, advanced packaging capacity, testing infrastructure, and customer qualification. Qualification remains particularly important, as not all HBM output can be deployed across every AI accelerator platform or customer environment.

After incorporating CXMT's projected HBM capacity expansion into our supply estimates and factoring in expected growth in China's AI GPU shipments, we still forecast a tight HBM market. Our model indicates an HBM supply sufficiency ratio of approximately 17% in 2026e and 15% in 2027e.

Exhibit 21: HBM TAM: Base case scenario

Base case	2023	2024	2025	2026e	2027e	23-27e CAGR
NVIDIA calculation						
GP GPU volume (k units)	1,474	3,609	6,359	9,565	11,478	*67%*
NVIDIA market share (%)	*90%*	*90%*	*88%*	*92%*	*90%*	
Total GPU volume (k units)	1,638	4,010	7,226	10,397	12,753	*67%*
ASIC calculation						
Total ASIC volume (k units)	2,556	5,298	5,053	5,951	9,819	*40%*
Y/Y %		107%	-5%	18%	65%	
HBM calculation						
HBM usage per GPGPU (GB)	80	143	208	288	317	*41%*
HBM usage per ASIC (GB)	40	70	141	150	238	*56%*
HBM ASP per GB (US$)	15	12	13	13	13	*-3%*
Inventory Buffer		20%	10%	10%	10%	
Total HBM Usage (mn GB)	233	1,132	2,440	4,276	7,011	*134%*
Y/Y %		*385%*	*115%*	*75%*	*64%*	
Total HBM market (US$bn)	**3**	**13**	**31**	**57**	**94**	*128%*
Total HBM market (mn Gb)	**1,866**	**9,059**	**19,516**	**34,205**	**56,085**	

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 22: Even with CXMT's capacity adding in, the overall HBM market remains undersupply

Sufficiency estimates	2023	2024	2025	2026e	2027e
HBM TSV Capacity (K wpm)					
Samsung	45	130	150	180	250
SK Hynix	45	120	150	200	250
Micron	3	20	60	100	110
CXMT				20	40
Total	**93**	**270**	**360**	**500**	**650**
Yield rate assumptions				**38.9%**	**30.0%**
Samsung		*50%*	*60%*	*55%*	*70%*
SK Hynix		*60%*	*75%*	*65%*	*70%*
Micron		*50%*	*70%*	*65%*	*70%*
CXMT				*40%*	*50%*
UTR assumptions					
Samsung		*80%*	*60%*	*70%*	*100%*
SK Hynix		*100%*	*100%*	*100%*	*100%*
Micron		*100%*	*100%*	*100%*	*100%*
CXMT				*100%*	*100%*
Implied HBM production (mn Gb)					
Samsung	1,500	4,435	6,387	9,696	21,239
SK Hynix	1,500	6,273	12,830	17,363	22,226
Micron	150	729	3,548	7,937	10,372
CXMT				1,382	2,462
Total	**3,150**	**11,436**	**22,765**	**36,379**	**56,300**
Sufficiency Ratio				**59.8%**	**54.8%**
HBM Demand (mn Gb)	1,866	9,059	19,516	34,205	56,085
Commodity DRAM market demand (mn Gb)	209,527	243,423	316,640	469,675	619,970
Total HBM+DRAM demand (mn Gb)	211,393	252,481	336,156	503,879	676,055
Commodity DRAM market supply (mn Gb)	200,214	228,086	301,257	382,107	519,666
Total HBM+DRAM supply (mn Gb)	203,364	239,523	324,022	418,486	575,965
Total DRAM sufficiency	***-4%***	***-5%***	***-4%***	***-17%***	***-15%***

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

CXMT (688825.SS): Key Beneficiary Of China's Memory Localization Trend; Initiate At OW

CXMT has evolved from China's first large-scale domestic DRAM producer into the world's fourth-largest DRAM manufacturer and a key beneficiary of the country's memory localization strategy. The company is expanding across DDR5, LPDDR5/5X, server DRAM, and HBM, supported by rapid capacity growth, improving yields and sustained R&D investment.

We forecast total DRAM capacity to rise from 180k wpm in 2025 to 500k wpm by 2028. At the same time, the transition toward G4B production and the planned ramp of HBM3e and HBM4 should enhance CXMT's product mix and strengthen its competitive position across higher-value memory segments. In addition, growing adoption by Chinese CSPs, smartphone manufacturers, and PC OEMs, together with structurally strong demand for AI-server memory, should support utilization rates and market-share gains. However, CXMT's earnings are subject to several risks, including DRAM pricing, elevated capex requirements, rising depreciation expenses, export-controls, and execution risk associated with next-generation process technology migration.

Earnings: Our strong 2026–28 earnings outlook is supported by strong capacity additions, elevated DRAM pricing, and substantial operating leverage. We project revenue to rise at a 140% CAGR during 2025-2028e, while maintaining gross margin above 75% and operating margin of ~70–72%. We expect the company's increasing exposure to higher-value products, including DDR5, server DRAM, and HBM, to further support earnings growth, while scale should help offset higher depreciation expenses and continued R&D investment associated with capacity expansion and technology migration.

Valuation: We derive our price target of Rmb88 using a residual income model, which we believe best captures CXMT's long-term value creation potential and ability to expand market share. Our price target implies approximately 18.5x 2027E P/E, representing a premium to global memory peers. We believe this premium is justified by CXMT's stronger growth profile, supported by faster capacity expansion, advantages from China's memory localization trend, and exposure to growing demand from the country's AI infrastructure ecosystem.

Key downside risks include (1) a sharper-than-expected decline in DRAM prices, (2) delays in G5, HBM or 3D DRAM technology development, (3) tighter restrictions on semiconductor equipment, materials, or servicing, (4) weaker-than-expected demand from major customers, and (5) higher-than-expected depreciation, inventory write-downs, or capital intensity.

CXMT: Financial Summary

Exhibit 23: CXMT: Financial summary

Income Statement

Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Net sales	61,799	388,869	643,077	850,096
COGS	(36,465)	(92,599)	(144,853)	(185,478)
Gross profit	25,334	296,271	498,224	664,618
Operating expenses	(16,561)	(22,789)	(37,942)	(54,491)
Operating income	**8,773**	**273,482**	**460,282**	**610,127**
Non-operating income	(1,071)	(1,078)	(1,600)	(1,600)
Pre-tax income	7,701	272,404	458,682	608,527
Income tax	(557)	(19,485)	(33,025)	(43,814)
Minority interests	5,269	63,226	106,414	141,178
Reported net Income	**1,875**	**189,693**	**319,243**	**423,535**
Adj.wtd.avg.shrs(m)	59,018	63,216	66,881	66,881
MW EPS (Rmb)	0.03	3.00	4.77	6.33
Reported EPS (Rmb)	0.03	3.00	4.77	6.33
EPS for consensus (Rmb)	**0.03**	**3.00**	**4.77**	**6.33**

Balance Sheet

Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cash	37,798	275,606	652,716	1,184,866
Mkt Securities	0	0	0	0
AR/NR	1,524	10,602	17,532	23,176
Inventory	28,568	59,824	93,583	119,828
Other	43,096	43,096	43,096	43,096
Current Assets	**110,986**	**389,128**	**806,928**	**1,370,966**
Long-term investments	4,384	4,384	4,384	4,384
Fixed assets	183,024	225,774	244,382	253,419
Deferred assets	1,006	1,006	1,006	1,006
Other assets	37,385	37,385	37,385	37,385
Total Assets	**336,785**	**657,677**	**1,094,085**	**1,667,160**
S/T borrowings	9,676	9,686	9,696	9,706
AP/NP	8,702	19,015	29,746	38,089
Other ST liabilities	34,465	34,465	34,465	34,465
LT debt	118,825	118,835	118,845	118,855
Other LT liabilities	11,017	11,017	11,017	11,017
Total Liabilities	**182,685**	**193,019**	**203,769**	**212,132**
Common shares	60,193	66,881	66,881	66,881
Additional capital	33,104	84,054	84,054	84,054
Retained earning	(36,650)	153,043	472,286	895,821
Other shareholders' equity	97,454	160,680	267,095	408,273
Total Equity	**154,100**	**464,658**	**890,315**	**1,455,028**
Total Liab. & Shrhldr's Equity	**336,785**	**657,677**	**1,094,085**	**1,667,160**

E = Morgan Stanley Research Estimates
Source: Morgan Stanley Research, Company Data

Cash Flow Statement

Rmb mn (Years End Dec)	2025	2026E	2027E	2028E
Cashflow from Operations	**36,520**	**261,720**	**444,090**	**599,129**
Net profits	7,144	252,919	425,657	564,713
Depreciation	24,680	38,820	48,392	57,963
Working Capital Change	(48,395)	(30,020)	(29,959)	(23,547)
Other adjustments	53,090	0	0	0
Cashflow from Investing	**(86,823)**	**(81,570)**	**(67,000)**	**(67,000)**
Capex	(49,739)	(81,570)	(67,000)	(67,000)
Change of LT Investment	2,434	0	0	0
Change of ST Investment	194	0	0	0
Other adjustments	(39,712)	0	0	0
Cashflow from financing	**46,121**	**57,658**	**20**	**20**
Increase in L/T debt	22,832	10	10	10
Increase in S/T debt	7,879	10	10	10
Cash Dividend Paid	0	0	0	0
Dir& Emp Bonus Paid	0	0	0	0
Issuance of stock	2,422	57,638	0	0
Other adjustments	12,988	0	0	0
Exchange rate adjustment	(528)	0	0	0
Net change in cash	**-4,710**	**237,808**	**377,110**	**532,149**

Financial Ratios

	2025	2026E	2027E	2028E
Growth(%)				
Turnover	155.6	529.2	65.4	32.2
Operating profits	-250.2	3017.4	68.3	32.6
EPS	-125.7	9133.0	59.1	32.7
Margins (%)				
Gross Margin	41.0	76.2	77.5	78.2
Operating Margin	14.2	70.3	71.6	71.8
Pretax Margin	12.5	70.1	71.3	71.6
Net Margin	3.0	48.8	49.6	49.8
Return (%)				
ROAE	1.2	61.3	47.1	36.1
ROAA	0.6	38.1	36.4	30.7
Gearing (%)				
Net Debt/Equity	58.9	(31.7)	(58.9)	(72.6)
Liabilities/Equity	118.5	41.5	22.9	14.6
Ratios (X)				
Current ratio	2.1	6.2	10.9	16.7
Quick ratio	0.7	4.5	9.1	14.7
Others				
AR/NR Turnover (days)	10	10	10	10
Inventory Turnover (days)	236	236	236	236
AP Turnover (days)	75	75	75	75
Cash Conversion (days)	171	171	171	171

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Risk Reward – CXMT Corp (688825.SS)

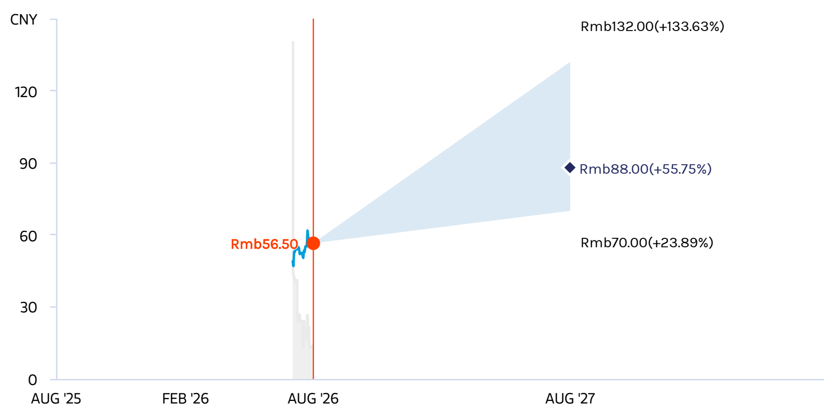
China DRAM Going Mainstream; OW

PRICE TARGET Rmb88.00

Base case, residual income model. Key assumptions:
- 9.9% CoE (1.2 beta, 2.0% risk-free rate, and 6.6% risk premium)
- 51% payout ratio
- 11.0% medium-term growth rate
- 3% terminal growth rate

We believe these assumptions are justified, given CXMT's fast growth, capacity expansion, localization tailwinds and AI-memory exposure. Base case PT implies an 18.5x 2027e P/E vs global peers' 4.4x 2027e P/E; the premium reflects CXMT's stronger growth profile.

RISK REWARD CHART



Key: — Historical Stock Performance ● Current Stock Price ◆ Price Target

Source: Refinitiv, Morgan Stanley Research

OVERWEIGHT THESIS

- CXMT is now the world's #4 DRAM maker and a key beneficiary of China's memory localization push.
- We forecast that DRAM capacity will rise from 180k wpm in 2025 to 500k by 2028, with G4b, DDR5, LPDDR5/5X, server DRAM and HBM3e/HBM4 improving mix.
- We expect 2025–28E revenue CAGR of 140%, 2026-28E GM >75% and OPM of ~70–72%, supported by pricing, scale and higher-value products.
- Our Rmb88 price target is based on a residual income model and implies 18.5x 2027E P/E. We see the premium to global memory peers as justified by CXMT's faster growth, capacity expansion, localization tailwinds and AI-memory exposure.
- **Risks:** DRAM pricing, capex/depreciation, export controls and migration execution.

Risk Reward Themes

Pricing Power:	*Positive*
Secular Growth:	*Positive*

View descriptions of Risk Rewards Themes here

BULL CASE Rmb132.00

27.7x 2027e P/E

We assume stronger sales/earnings growth, with faster global market share expansion:

(1) 200% revenue CAGR, 2025-28e, as we assume a stronger-than-expected DRAM commodity cycle, supported by AI inference demand moving into 2027 and beyond.

(2) faster-than-expected HBM ramp-up.

(3) Gross margin improves to over 90%, driven by economies of scale and CXMT's technology competitiveness.

(4) >800% earnings CAGR, 2025-28e.

BASE CASE Rmb88.00

18.5x 2027e P/E

We forecast strong revenue/earnings growth, with global market share expansion.

(1) 140% revenue CAGR, 2025-28e, as we assume DRAM pricing remains strong, supported by AI demand

(2) moderate pace of HBM ramp-up

(3) 2026-28e gross margin of over 75%, driven by economies of scale and CXMT's technology competitiveness

(4) 480% earnings CAGR, 2025-28e

BEAR CASE Rmb70.00

14.8x 2027e P/E

We assume revenue/earnings growth is slower, with slower global market share expansion.

(1) 80% revenue CAGR, 2025-28e, as we assume a period of AI computing digestion in 2028, leading to more aggressive margin erosion for the HBM segment owing to competitors' aggressive capacity expansion and pricing competition, as well as faster DRAM pricing drops with muted demand recovery.

(2) customer bargaining power pushes gross margin below 40%.

(3) 200% earnings CAGR, 2025-28e.

Risk Reward – CXMT Corp (688825.SS)

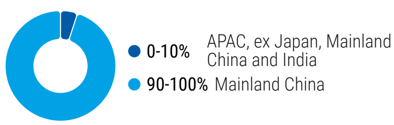
KEY EARNINGS INPUTS

Drivers	2025	2026e	2027e	2028e
Revenue (Rmb, mn)	180	300	400	500
Gross profit (Rmb, mn)	25,334	296,271	498,224	664,618
EPS (Rmb, mn)	0	3	5	6

INVESTMENT DRIVERS

- Commodity DRAM pricing
- HBM capacity expansion and tech migration
- Demand from major customers

GLOBAL REVENUE EXPOSURE



Source: Morgan Stanley Research Estimate
View explanation of regional hierarchies here

RISKS TO PT/RATING

RISKS TO UPSIDE

- resilient DRAM pricing, supported by AI demand
- faster-than-expected tech migration
- export control relief
- stronger demand from major customers

RISKS TO DOWNSIDE

- earlier-than-expected global DRAM price declines
- CXMT delays in new products, HBM or 3D DRAM development
- tighter restrictions on China's fabs for semi equipment, materials, or servicing
- weaker demand from major China customers

MS ESTIMATES VS. CONSENSUS

FY 2027e

EPS (Rmb)
◆ 4.8
Note: There are not sufficient brokers supplying consensus data for this metric

Sales / Revenue (Rmb, mn)
◆ 643,077.2
Note: There are not sufficient brokers supplying consensus data for this metric

Net income (Rmb, mn)
◆ 319,243
Note: There are not sufficient brokers supplying consensus data for this metric

◆ Mean ◆ Morgan Stanley Estimates

Source: Refinitiv, Morgan Stanley Research

CXMT: Investment Positives and Concerns

Investment positives

China memory localization and AI demand create a structural growth runway: CXMT has evolved into the world's fourth-largest DRAM supplier, supported by an expanding product portfolio that includes DDR5, LPDDR5/5X, server DRAM, and emerging HBM products. Adoption by leading Chinese cloud service providers, smartphone manufacturers, and PC OEMs, combined with improving product performance and production yields, suggests that CXMT is becoming increasingly competitive across mainstream DRAM applications rather than serving primarily lower-end markets.

Rapid capacity expansion should drive significant revenue growth and operating leverage: We forecast total DRAM capacity to increase from 180k wafers per month (wpm) in 2025 to 500k wpm by 2028, while bit shipments grow at a 46% CAGR over 2025-28e. This scale expansion, combined with favorable DRAM pricing and a richer product mix, should support substantial earnings growth. Following its return to profitability in 2025, we project gross margins above 75% and operating margins of approximately 70–72% during 2026-28e.

A structurally tight global DRAM market provides a favorable pricing backdrop for CXMT's capacity ramp: AI servers are becoming increasingly memory intensive, with server applications projected to account for 59% of DRAM bit demand by 2028. At the same time, HBM production is absorbing a growing share of leading-edge DRAM wafer capacity, tightening supply across the broader memory market. As a result, we expect the global DRAM market to remain undersupplied through 2027, supporting both pricing and utilization even as CXMT expands capacity. CXMT's position within China's domestic memory supply chain, its planned HBM3E and HBM4 roadmap, and its potential adoption of vertical-channel transistor (VCT) or 3D DRAM technologies could provide further upside through market-share gains and technology advancement. These factors should strengthen CXMT's competitive position as China's memory localization strategy and AI infrastructure investment continue to drive demand growth.

Investment concerns

Accumulated losses and a capital-intensive business model could constrain financial flexibility: As of the end of 2025, CXMT had accumulated losses of ~Rmb36.7bn, reflecting the substantial fixed-asset investment and sustained R&D spending required in the DRAM industry. Although the company returned to profitability in 4Q25, its asset-heavy business model continues to require significant investment in capacity expansion and could limit balance-sheet flexibility if industry conditions weaken.

Rising depreciation and inventory exposure could amplify earnings volatility through the cycle: The carrying value of fixed assets increased from Rmb84.5bn in 2023 to Rmb183.0bn in 2025, while annual depreciation expense rose from Rmb10.6bn to Rmb24.7bn over the same period. Continued capacity expansion is likely to further increase depreciation charges. At the same time, DRAM's commodity characteristics expose the company to inventory impairment risk during periods of pricing weakness. As demonstrated during the 2023 industry downturn, sharp declines in memory prices can lead to significant inventory write-downs and earnings volatility.

Technology, customer concentration, and external supply-chain constraints remain important execution risks: DRAM development involves high technical barriers, long qualification cycles, and rapid product iteration. As a result, delays or execution challenges in process migration could weaken CXMT's competitiveness. The company also derives a relatively high proportion of sales from its five largest customers. Any reduction in orders from its key customers could have a disproportionate impact on revenue growth and utilization. In addition, restrictions affecting semiconductor equipment, materials, technology transfer or servicing could disrupt production, slow capacity expansion, and complicate CXMT's longer-term technology roadmap.

CXMT: Financial Outlook

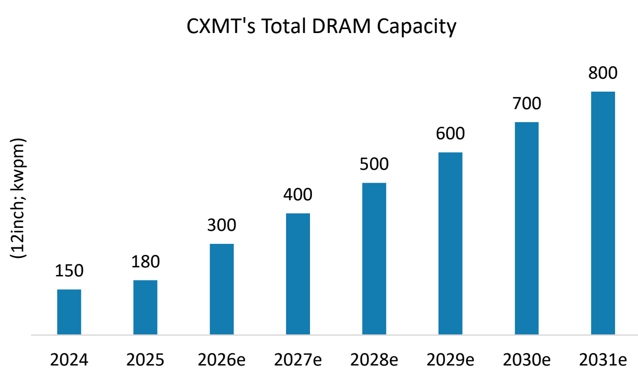
Capacity and production

CXMT is now expanding its DRAM manufacturing footprint, with near-term production concentrated in Hefei and Beijing, while Shanghai represents a potential longer-term growth location. We forecast CXMT's DRAM capacity to increase from 180k wpm in 2025 to 300k wpm in 2026, followed by annual capacity additions of ~100k wpm, reaching 800k wpm by 2031.

CXMT previously focused on Gen3 (~18nm-class) and Gen4 (~17nm-class) DRAM production, where we estimate yields have exceeded 90% for both nodes. Beginning in 2Q26, we believe the company completed its transition to 100% Gen4B (~16nm-class) production, with yields also reaching approximately 92% or higher. The transition to Gen5 (~15nm-class) is likely to be more challenging, as it requires help from DUV-based scaling techniques or successful adoption of alternative architectures such as VCT technology. Given both pathways face meaningful hurdles, including export-control constraints on advanced DUV tools and process-integration challenges associated with next-generation device architectures. As a result, we expect Gen5 mass production to begin in 2H28e.

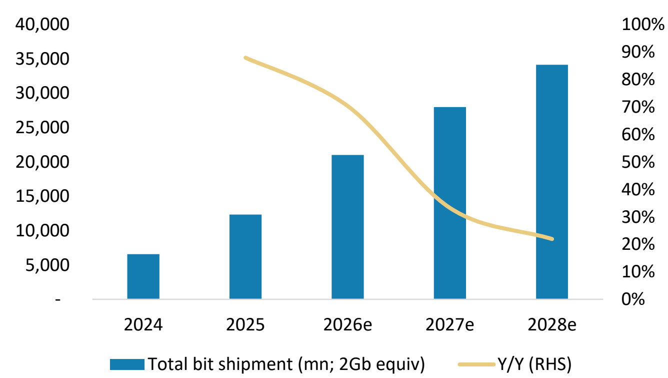
On the other hand, we also expect CXMT to ramp its HBM capacity rapidly, from 20k wpm in 2026e to 70k wpm in 2028e. Our projections assume the company's HBM roadmap progresses from HBM2E in 2026, to HBM3E 12-Hi in 2027, and ultimately HBM4 in 2028.

Exhibit 24: CXMT to ramp its capacity from 180k wpm in 2025, 300k wpm in 2026 and expanding at an annual pace of 100k wpm to 800k wpm in 2031.



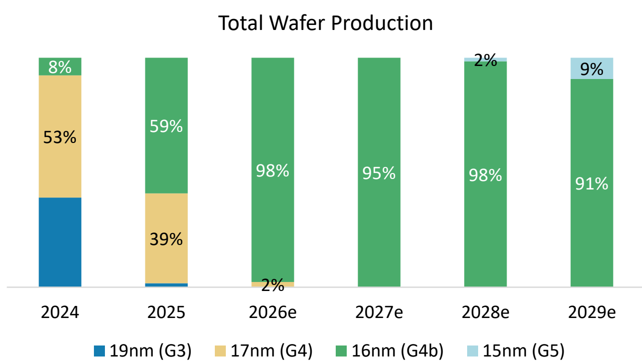
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 25: Total bit shipments to rise at a 40% CAGR during 2025-2028e



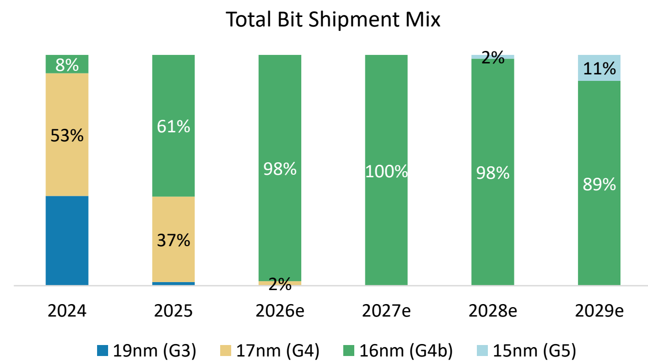
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 26: G4b capacity to be CXMT's main focus in the next few years ...



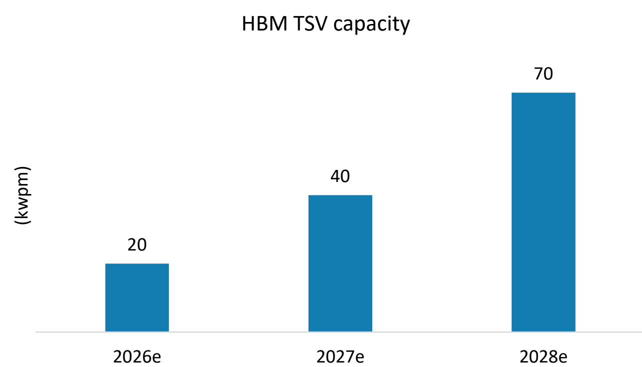
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 27: G5 capacity to start contributing from 2028/29



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 28: CXMT to ramp up HBM capacity from 20k wpm in 2026e to 70k wpm in 2028e



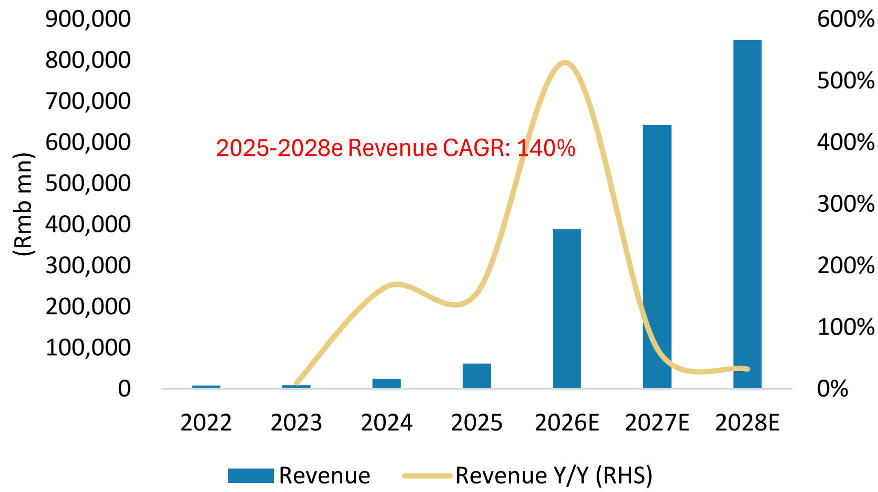
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Revenue

From 2023 to 2025, CXMT delivered rapid revenue growth, with revenue of Rmb9.087bn, Rmb24.178bn, and Rmb61.799bn, respectively. Net profit attributable to shareholders improved from a loss of Rmb16.340bn in 2023 to a loss of Rmb7.145bnin 2024, before turning positive at Rmb1.875bn, marking the company's first profitable year. On a non-recurring-items-adjusted basis, net profit attributable to shareholders reached Rmb5.3bn in 2025, suggesting a stronger underlying earnings profile than reported net income alone. Momentum appears to have continued into 2026. In 1Q26, CXMT reported Rmb50.8bn of revenue, up 719.13% YoY.

The company guides for 1H26 revenue of Rmb110-120bn. However, we project revenue of at least Rmb140bn, supported by continued capacity expansion and further increases in DRAM ASPs. According to the prospectus, the company's revenue outlook reflects the sharp rise in DRAM prices since 2H25, growing production and shipment volumes, and ongoing product-mix improvement. Looking beyond 2026, we expect CXMT to benefit from a combination of rapid capacity expansion, favorable DRAM pricing, and increasing contributions from DDR5, LPDDR5X, server DRAM, and HBM products. As a result, we forecast revenue growth at a 140% CAGR over 2025-28e.

Exhibit 29: CXMT's revenue to rise at a 140% CAGR, 2025-28e



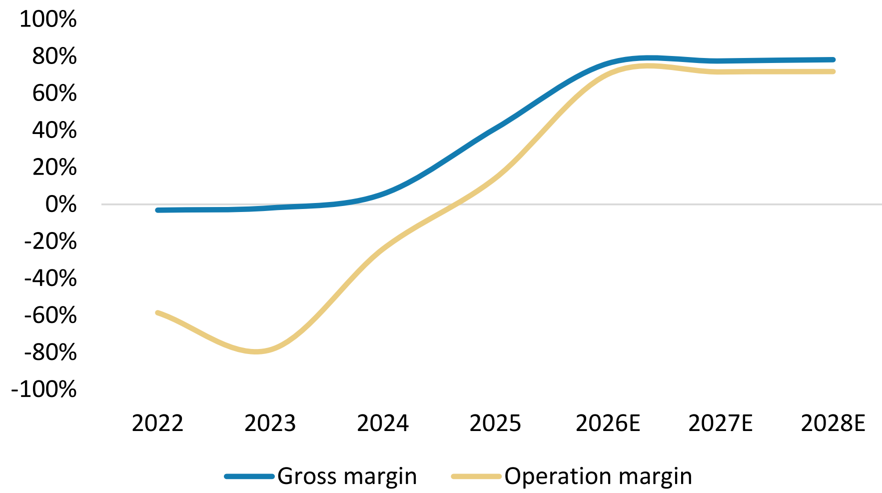
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Gross profit and operating margin

CXMT's gross margin is highly sensitive to the DRAM industry cycle. During the reporting period, gross margin from its core business was -2.2% in 2023, 5.0% in 2024, and 41.0% in 2025. The sharp improvement in 2025 was driven by rising DRAM product prices, economies of scale, and product mix improvements.

However, DRAM remains a cyclical industry with commodity-like characteristics. If macroeconomic conditions weaken, end-market demand softens, or industry supply eventually exceeds demand, DRAM pricing and operating performance could fluctuate significantly. That said, we currently see no indication of meaningful DRAM price declines in the near term. Despite the higher depreciation burden associated with CXMT's planned capacity additions of approximately 100k wpm annually, we forecast the company will maintain gross margins above 70% during 2026-28e. Assuming disciplined operating-expense management and continued operating leverage from higher utilization and production scale, we forecast operating margins to be 70-72% during 2026-28e.

Exhibit 30: CXMT's OpM to be 70-72% during 2026-28e



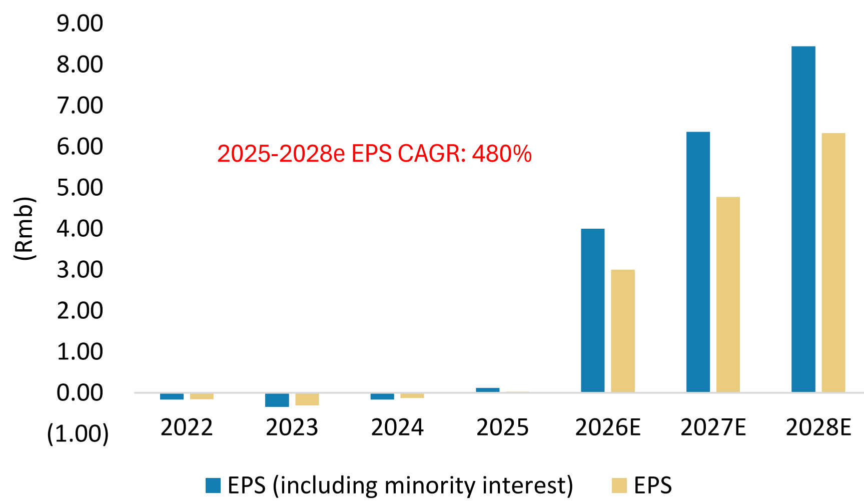
Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Net profit and EPS

From 2023 to 2025, CXMT's net profit attributable to shareholders improved from a loss of Rmb16.3bn to a profit of Rmb1.9bn, with the company achieving profitability in 2025 amid a favorable DRAM upcycle. Net profit attributable to shareholders was Rmb-16.3bn in 2023, Rmb-7.1bn in 2024, and Rmb1.9bn in 2025. We expect earnings to continue benefiting from operating leverage as revenue growth outpaces fixed-cost growth.

We expect CXMT to gradually consolidate ownership of externally financed capacity and facilities through potential buybacks or other ownership-integration arrangements. As a result, we use net profit and EPS including minority interests in our valuation framework.

Exhibit 31: CXMT's EPS to rise at a 480% CAGR, 2025-2028e



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

CXMT: Quarterly Financials

Exhibit 32: CXMT: Quarterly financials

Rmb mn	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	1Q28E	2Q28E	3Q28E	4Q28E	2024	2025	2026E	2027E	2028E
Total Revenues	**50,800**	**85,204**	**114,857**	**138,009**	**143,509**	**156,281**	**167,792**	**175,496**	**205,400**	**211,209**	**216,341**	**217,146**	**24,178**	**61,799**	**388,869**	**643,077**	**850,096**
Sequential Change	*71.0%*	*67.7%*	*34.8%*	*20.2%*	*4.0%*	*8.9%*	*7.4%*	*4.6%*	*17.0%*	*2.8%*	*2.4%*	*0.4%*					
Change vs Year Ago	*719.1%*	*822.5%*	*590.0%*	*364.4%*	*182.5%*	*83.4%*	*46.1%*	*27.2%*	*43.1%*	*35.1%*	*28.9%*	*23.7%*	*166.1%*	*155.6%*	*529.2%*	*65.4%*	*32.2%*
Cost of Sales	12,773	22,486	26,934	30,406	33,624	35,540	37,267	38,422	45,301	46,172	46,942	47,063	22,830	36,465	92,599	144,853	185,478
Percent of Revenues	*25%*	*26%*	*23%*	*22%*	*23%*	*23%*	*22%*	*22%*	*22%*	*22%*	*22%*	*22%*	*94%*	*59%*	*24%*	*23%*	*22%*
Gross Profit	**38,027**	**62,718**	**87,923**	**107,602**	**109,885**	**120,741**	**130,525**	**137,073**	**160,099**	**165,037**	**169,399**	**170,083**	**1,349**	**25,334**	**296,271**	**498,224**	**664,618**
Percent of Revenues	*74.9%*	*73.6%*	*76.6%*	*78.0%*	*76.6%*	*77.3%*	*77.8%*	*78.1%*	*77.9%*	*78.1%*	*78.3%*	*78.3%*	*5.6%*	*41.0%*	*76.2%*	*77.5%*	*78.2%*
Incremental Margin	*180%*	*72%*	*85%*	*85%*	*41%*	*85%*	*85%*	*85%*	*77%*	*85%*	*85%*	*85%*	*10%*	*64%*	*83%*	*79%*	*80%*
Total Opex	2,715	5,155	6,777	8,143	8,467	9,221	9,900	10,354	13,166	13,538	13,867	13,919	7,190	16,561	22,789	37,942	54,491
Percent of Revenues	*5.3%*	*6.1%*	*5.9%*	*5.9%*	*5.9%*	*5.9%*	*5.9%*	*5.9%*	*6.4%*	*6.4%*	*6.4%*	*6.4%*	*29.7%*	*26.8%*	*5.9%*	*5.9%*	*6.4%*
R&D	2,154	2,982	4,020	4,830	5,023	5,470	5,873	6,142	7,805	8,026	8,221	8,252	4,607	9,593	13,986	22,508	32,304
Percent of Revenues	*4.2%*	*3.5%*	*3.5%*	*3.5%*	*3.5%*	*3.5%*	*3.5%*	*3.5%*	*3.8%*	*3.8%*	*3.8%*	*3.8%*	*19.1%*	*15.5%*	*3.6%*	*3.5%*	*3.8%*
General & administrative	483	2,045	2,642	3,174	3,301	3,594	3,859	4,036	5,135	5,280	5,409	5,429	2,358	6,631	8,344	14,791	21,252
Percent of Revenues	*1.0%*	*2.4%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.3%*	*2.5%*	*2.5%*	*2.5%*	*2.5%*	*9.8%*	*10.7%*	*2.1%*	*2.3%*	*2.5%*
Selling & marketing	78	128	115	138	144	156	168	175	226	232	238	239	224	337	459	643	935
Percent of Revenues	*0.2%*	*0.2%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.1%*	*0.9%*	*0.5%*	*0.1%*	*0.1%*	*0.1%*
Operating Income	**35,312**	**57,563**	**81,147**	**99,460**	**101,418**	**111,521**	**120,625**	**126,719**	**146,933**	**151,498**	**155,532**	**156,164**	**(5,841)**	**8,773**	**273,482**	**460,282**	**610,127**
Percent of Revenues	*69.5%*	*67.6%*	*70.7%*	*72.1%*	*70.7%*	*71.4%*	*71.9%*	*72.2%*	*71.5%*	*71.7%*	*71.9%*	*71.9%*	*-24.2%*	*14.2%*	*70.3%*	*71.6%*	*71.8%*
Change vs Year Ago	*0.0%*	*0.0%*	*0.0%*	*626.5%*	*187.2%*	*93.7%*	*48.7%*	*27.4%*	*44.9%*	*35.8%*	*28.9%*	*23.2%*	*nm*	*nm*	*3017%*	*68%*	*33%*
Total Non-operating Income(Loss)	122	(400)	(400)	(400)	(400)	(400)	(400)	(400)	(400)	(400)	(400)	(400)	(3,208)	(1,071)	(1,078)	(1,600)	(1,600)
Profit Before Taxes	**35,434**	**57,163**	**80,747**	**99,060**	**101,018**	**111,121**	**120,225**	**126,319**	**146,533**	**151,098**	**155,132**	**155,764**	**(9,049)**	**7,701**	**272,404**	**458,682**	**608,527**
Percent of Revenues	*70%*	*67%*	*70%*	*72%*	*70%*	*71%*	*72%*	*72%*	*71%*	*72%*	*72%*	*72%*	*-37%*	*12%*	*70%*	*71%*	*72%*
Taxes	(2,423)	(4,116)	(5,814)	(7,132)	(7,273)	(8,001)	(8,656)	(9,095)	(10,550)	(10,879)	(11,170)	(11,215)	(2)	(557)	(19,485)	(33,025)	(43,814)
Tax Rate	*6.8%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*	*0.0%*	*7.2%*	*7.2%*	*7.2%*	*7.2%*
Net Income, Cont Ops	33,012	53,048	74,933	91,927	93,744	103,120	111,569	117,224	135,983	140,219	143,962	144,549	(9,051)	7,144	252,919	425,657	564,713
Percent of Revenues	*65%*	*62%*	*65%*	*67%*	*65%*	*66%*	*66%*	*67%*	*66%*	*66%*	*67%*	*67%*	*-37%*	*12%*	*65%*	*66%*	*66%*
Reported Income (Rmb)	**24,762**	**39,786**	**56,200**	**68,946**	**70,308**	**77,340**	**83,677**	**87,918**	**101,987**	**105,164**	**107,972**	**108,412**	**(7,145)**	**1,875**	**189,693**	**319,243**	**423,535**
Percent of Revenues	*49%*	*47%*	*49%*	*50%*	*49%*	*49%*	*50%*	*50%*	*50%*	*50%*	*50%*	*50%*	*-30%*	*3%*	*49%*	*50%*	*50%*
Change vs Year Ago	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*0%*	*NM*	*NM*	*10018%*	*68%*	*33%*
Reported EPS (Rmb, cent)	**0.41**	**0.66**	**0.86**	**1.03**	**1.05**	**1.16**	**1.25**	**1.31**	**1.52**	**1.57**	**1.61**	**1.62**	**(0.13)**	**0.03**	**3.00**	**4.77**	**6.33**
Change vs Year Ago	*0%*	*0%*	*0%*	*767%*	*156%*	*75%*	*46%*	*28%*	*45%*	*36%*	*29%*	*23%*	*NM*	*NM*	*9133%*	*59%*	*33%*

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

CXMT: Valuation Methodology and Peer Comparison

Our price target is Rmb88: We derive our price target from a residual income (RI) model, as we believe this methodology is best suited to capturing CXMT's long-term value. We assume a 9.9% cost of equity (beta 1.2, risk-free rate 2.0% and risk premium 6.6%), a payout ratio of 51%, a medium-term growth rate of 11.0%, and a terminal growth rate of 3.0%, all of which are in line with other Greater China foundry and Asia DRAM names under our coverage universe.

Our price target implies 18.5x 2027e P/E, representing a premium to global memory players. We believe this premium is justified by CXMT's stronger long-term growth profile, supported by sustained market-share gains in the global DRAM industry and growing participation in China's domestic memory market.

Exhibit 33: CXMT: RI model

Rmb mn	2026E	2027E	2028E	2029E	2030E	2031E	2032E	2033E	2034E	2035E	2036E	2037E
Total Equity	**304,085**	**623,328**	**1,046,863**	**1,277,224**	**1,532,924**	**1,816,751**	**2,131,800**	**2,481,504**	**2,869,675**	**3,300,545**	**3,778,810**	**4,309,685**
Net Profit	**189,693**	**319,243**	**423,535**	**470,124**	**521,837**	**579,240**	**642,956**	**713,681**	**792,186**	**879,326**	**976,052**	**1,083,418**
ROAE	**105.1%**	**68.8%**	**50.7%**	**40.5%**	**37.1%**	**34.6%**	**32.6%**	**30.9%**	**29.6%**	**28.5%**	**27.6%**	**26.8%**
Residual Income	54,041 184,063	179,185	254,299	319,677	347,654	378,093	411,434	448,107	488,556	533,251	582,698	637,452
Spread	**95.2%**	**58.9%**	**40.8%**	**30.5%**	**27.2%**	**24.7%**	**22.6%**	**21.0%**	**19.7%**	**18.6%**	**17.7%**	**16.9%**
Ending Equity Capital	304,085											
PV of Forecast Period	2,235,514											
PV of Continuing Value	3,352,236											
Equity Value	**5,891,835**											
No. of Shares	66,881											
Projected Price	**88**											

Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 34: Global DRAM supply chain valuation comparison

Ticker	Company	Rating	Price Target (LC)	Market Cap (US$mn)	EPS (Local currency)				BV (US$mn)				P/E (X)				P/B (X)			
					2025	2026e	2027e	2028e	2025	2026e	2027e	2028e	2025	2026e	2027e	2028e	2025e	2026e	2027e	2028e
688825.SS	CXMT	O	88	577,199	0.03	3.00	4.77	6.33	21,433	68,357	132,883	217,168								
Global/Greater China IDM Player																				
000660.KS	SK Hynix	O	2,600,000	909,603	59,300	361,277	451,280	506,170	83,662	276,963	508,279	767,759	29.2	4.8	3.8	3.4	10.9	3.3	1.8	1.2
005930.KS	Samsung Electronics	O	381,000	1,333,445	6,562	48,652	72,528	74,583	294,558	539,296	860,811	1,193,046	42.9	5.8	3.9	3.8	4.5	2.5	1.5	1.1
MU.O	Micron	O	1,200	1,091,872	8	73	169	149	54,165	136,156	292,011	410,905	116.6	13.2	5.7	6.5	20.2	8.0	3.7	2.7
2408.TW	Nanya Tech	O	580	57,168	2	62	112	74	5,438	11,828	23,908	31,903	223.6	8.5	4.7	7.1	10.5	4.8	2.4	1.8
2344.TW	Winbond	O	288	25,560	1	25	50	60	3,445	6,918	13,971	22,455	205.6	7.3	3.6	3.0	7.4	3.7	1.8	1.1
Average													**123.6**	**7.9**	**4.4**	**4.8**	**10.7**	**4.5**	**2.3**	**1.6**
Greater China DRAM Player																				
603986.SS	GigaDevice	O	750	42,677	2	35	45	52	2,749	6,409	10,025	14,044	165.2	11.8	9.1	7.9	15.5	6.7	4.3	3.0
6770.TW	PSMC	O	111	10,114	(2)	6	7	8	2,617	6,304	7,947	9,771	NM	12.1	10.5	8.4	3.9	1.6	1.3	1.0
6531.TW	AP Memory	O	1,666	4,512	8	18	35	65	391	449	569	781	114.1	49.9	25.0	13.5	11.5	10.1	7.9	5.8
Average													**139.7**	**24.6**	**14.8**	**9.9**	**10.3**	**6.1**	**4.5**	**3.3**

Source: Company data, FactSet, Morgan Stanley Research. E = Morgan Stanley Research estimates. Note: As of the market close on Aug 21, 2026.

Bull/bear case scenarios

Our bull case scenario value is Rmb132, implying 28x 2027e P/E: In this scenario, we assume stronger sales/earnings growth, with faster global market share expansion:

(1) 200% revenue CAGR, 2025-28e, as we assume a stronger-than-expected DRAM commodity cycle, supported by AI inference demand moving into 2027 and beyond.

(2) faster-than-expected HBM ramp-up.

(3) gross margin improves to over 90%, thanks to economies of scale and CXMT's technology competitiveness.

(4) >800% earnings CAGR, 2025-28e.

Our bear case scenario value is Rmb70, implying 15x 2027e P/E: In this scenario, we assume revenue/earnings growth is slower, with slower global market share expansion.

(1) 80% revenue CAGR, 2025-28e, as we assume a period of AI computing digestion in 2028, leading to more aggressive margin erosion for the HBM segment owing to competitors' aggressive capacity expansion and pricing competition, as well as faster DRAM pricing drops with muted demand recovery.

(2) customer bargaining power pushes gross margin below 40%.

(3) 200% earnings CAGR, 2025-28e.

CXMT: Company Overview

CXMT is China's largest and most advanced integrated DRAM company, with capabilities spanning DRAM R&D, design, manufacturing, and sales under an IDM model. Since its establishment in 2016, the company has focused exclusively on DRAM and has successfully advanced through four process generations, achieving mass production at each stage. Its product portfolio has evolved from DDR4 and LPDDR4X to DDR5 and LPDDR5/5X. According to the prospectus, the company’s core products and process technologies have reached internationally advanced levels..

CXMT operates under an IDM model and retains control over the most critical parts of the DRAM value chain, particularly design, process technology development, and wafer manufacturing. The company strongly emphasizes process control, manufacturing learning, and yield improvement. Packaging is primarily outsourced, while testing follows a hybrid model that combines internal capabilities with external partners.A portion of testing activities is conducted by the company's wholly owned subsidiary, CXMT Product Hefei. Module assembly and testing similarly combine in-house operations with outsourced providers, allowing the company to balance operational control with capacity flexibility.

CXMT adopts a “leapfrog R&D” strategy and has successfully advanced through multiple process generations and product iterations within a relatively short period. Its R&D process follows a structured framework comprising four stages: concept, planning, development, and verification. This process spans the full product-development cycle, from initial product specification through commercial deployment. Key activities include product requirement definition, technical feasibility assessment, layout and circuit design, simulation verification, tape-out, engineering verification, and customer verification.

CXMT places strong emphasis on independent R&D. Its R&D expenditure increased from Rmb4.7bn in 2023 to Rmb6.3bn in 2024 and Rmb9.6bn in 2025, totaling approximately Rmb20.6bn over the three-year period. This represented 21.7% of cumulative revenue during the same period. As of the end of 2025, CXMT employed 6,259 R&D personnel, accounting for 32.4% of total headcount.

In terms of intellectual property, as of the end of 2025, CXMT held 3,929 domestic patents, including 3,165 invention patents, as well as 3,043 overseas patents. The company has stated that its core technologies cover DRAM product design, manufacturing processes, packaging and testing, module design, and applications.

Key shareholders

Immediately after CXMT's July 27 IPO (before exercise of the over-allotment option), CXMT’s share capital is approximately 66.881 billion shares. Qinghui Jidian remains the largest shareholder with a 19.50% interest, followed by Changxin Integrated at 10.54%, National IC Industry Investment Fund II at 7.86%, Hefei Jixin at 7.53% and Anhui Investment Group at 7.12%. These five shareholders collectively hold 52.55% of the post-IPO share capital.

The five largest shareholders represent a range of institutional ownership types. Qinghui Jidian is an investment partnership established solely to invest in CXMT and is disclosed as having no actual controller. Its partnership interests are held by Xinrui Investment, Changxin Integrated and its general partner, with respective interests of 51.09%, 48.90% and 0.01%. Changxin Integrated is wholly owned by Hefei Industrial Investment Holding Group and is ultimately controlled by the Hefei municipal state-owned assets authority. National IC Industry Investment Fund II is a national integrated-circuit investment fund and is disclosed as having no actual controller. Hefei Jixin is CXMT's direct employee shareholding platform, while Anhui Investment Group is wholly owned and controlled by the Anhui provincial state-owned assets authority. The principal shareholder group therefore includes municipal and provincial state-owned capital, national semiconductor investment capital, employee ownership and dedicated investment vehicles.

Immediately before the IPO, all 60 of CXMT's direct shareholders were institutional investors, with no natural person holding shares directly. Beyond the five largest shareholders, the post-IPO register includes technology and semiconductor companies, local industrial investment vehicles, insurers, financial asset investment companies, private equity funds and foreign institutions. Among the larger named investors, Alibaba Cloud holds 3.47% after the initial offering, Hefei Chantou No. 1 holds 1.66% and GigaDevice holds 1.62%. The shareholder base also includes Alibaba Network, insurance companies, bank-affiliated investment entities, national and regional investment funds, and two foreign institutional shareholders.

The IPO further broadens the shareholder base through strategic allocations. The final strategic placement comprises approximately 1.667 billion shares, equivalent to 24.93% of the initial offering. The strategic investors include companies with existing or prospective business cooperation with CXMT, long-term institutional investors such as insurance companies and national investment funds, subsidiaries of the joint sponsors participating in the mandatory co-investment arrangements, and four asset management plans established for CXMT's senior management and core employees. The strategic placement therefore adds a further group of industry participants, long-duration financial investors and employee-linked investors to the post-IPO ownership structure.

CXMT is disclosed as having no controlling shareholder and no actual controller. Its board consists of 11 directors, including four independent directors. Of the seven non-independent directors, two are nominated by Qinghui Jidian, two by National IC Industry Investment Fund II, one by Hefei Jixin, one by Anhui Investment Group and one is an employee representative. The two directors attributed to Qinghui Jidian are respectively recommended by its general partner and by Changxin Integrated. CXMT also has no special-voting shares, contractual control arrangement or similar governance structure.

Chairman Yiming Zhu beneficially owns 1,590,303,801 CXMT shares, comprising 858,879 shares held indirectly through Qinghui Jidian, 1,535,841,835 shares through Hefei Jixin No. 41 Enterprise Management Partnership, an upper-tier vehicle within the Hefei Jixin employee shareholding structure, and 53,603,087 shares through GigaDevice. This represents approximately 2.38% of CXMT's 66.881 billion shares post-IPO, before exercise of the over-allotment option.

Exhibit 35: Shareholding structure immediately after the IPO (before exercise of the over-allotment option)

Shareholder	Post-IPO shares (bn)	Post-IPO stake	Shareholder profile
Qinghui Jidian	13.044	19.5%	CXMT's largest shareholder. It is a single-asset investment partnership whose sole investment is CXMT and is disclosed as having no actual controller. Its partnership interests are held by Xinrui Investment at 51.09%, Changxin Integrated at 48.90%, and the general partner at 0.01%.
Changxin Integrated	7.048	10.5%	Hefei municipal state-owned investment and engineering-services platform. It is wholly owned by Hefei Industrial Investment Holding Group and ultimately controlled by Hefei SASAC.
National IC Industry Investment Fund II, or Big Fund II	5.256	7.9%	National integrated-circuit industry investment fund, backed by the Ministry of Finance, state financial institutions and regional government investment platforms. The prospectus identifies the fund as having no actual controller.
Hefei Jixin	5.037	7.5%	CXMT's direct employee shareholding platform, providing equity alignment for management, technical personnel and other employees. It is disclosed as having no actual controller.
Anhui Investment Group	4.76	7.1%	Anhui provincial state-owned investment and enterprise-management platform. It is wholly owned and ultimately controlled by Anhui SASAC.
Alibaba Cloud	2.319	3.5%	Alibaba-affiliated cloud-computing company and strategic technology investor. Alibaba Cloud and the existing shareholder Alibaba Network are under common control of Alibaba Group.
Hefei Chantou No. 1	1.111	1.7%	Hefei-based equity investment partnership, representing local industrial and institutional capital.
GigaDevice	1.086	1.6%	Strategic partner, providing industry and technology alignment within CXMT's ownership base.
Other 52 existing institutional shareholders	20.531	30.7%	Diversified institutional pool comprising government-backed funds, bank-affiliated investment vehicles, insurers, private equity and venture capital funds, corporate investors and foreign institutions. CXMT had 60 direct institutional shareholders before the IPO, with no direct natural-person shareholders.
IPO strategic placement investors	1.667	2.5%	The final strategic placement represents 24.93% of the initial IPO shares. Participants include national social security and basic pension portfolios, national investment funds, large insurers, 18 strategic corporate partners, four management and core-employee asset management plans, and two sponsor-affiliate co-investors.
Other IPO investors	5.021	7.5%	Residual IPO allocation distributed through offline bookbuilding and online public subscription.
Total post-IPO shares	66.881	100.0%	Institutionally anchored and diversified shareholder base, with no single shareholder owning 20% or more.

Source: Company data, Morgan Stanley Research.

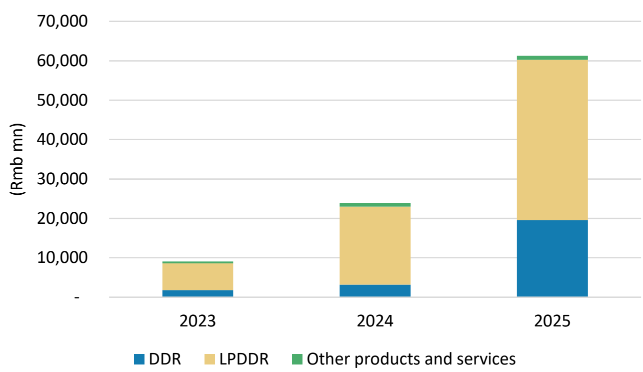
Key product portfolio

DDR series: This includes DDR4 and DDR5, mainly used in servers, desktops, laptops, and other consumer electronics. CXMT's DDR5 portfolio includes 16Gb, 24Gb, and 32Gb products, with data-transfer speeds of up to 8,000Mbps, supporting memory modules ranging from 8GB to 128GB. The company also notes that since the end of 2024, it discontinued production of its own DDR4 products and redirected resources toward newer-generation products such as DDR5.

LPDDR series: This includes LPDDR4X and LPDDR5/5X, mainly used in smartphones, tablets, laptops, wearable devices, and other consumer electronics. LPDDR5/5X features higher capacity, faster speed, lower power consumption, and enhanced data reliability. Based on our supply chain checks, relevant LPDDR5/5X products have already entered the supply chains of leading smartphone brands, including Xiaomi and Transsion.

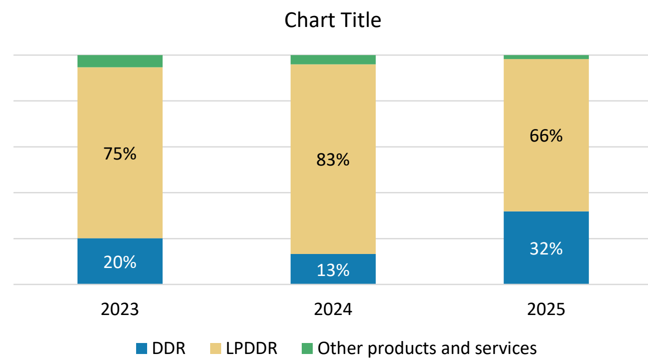
In addition to DRAM chips, CXMT offers a range of product solutions such as DRAM wafers and DRAM modules. Its module products include RDIMM, MRDIMM, UDIMM/CUDIMM, SODIMM/CSODIMM, and LPCAMM, serving applications such as servers, workstations, desktops, laptops, and high-performance mobile workstations.

Exhibit 36: Revenue breakdown, by product



Source: Company data, Morgan Stanley Research.

Exhibit 37: DRAM-related products remain CXMT's focus



Source: Company data, Morgan Stanley Research.

Key suppliers and customers

CXMT's products are used in servers, mobile devices, personal computers, and smart vehicles. According to the prospectus, core customers include Alibaba Cloud, ByteDance, Tencent, Lenovo, Xiaomi, Transsion, Honor, OPPO, and vivo.

CXMT uses a combination of direct sales and distribution channels, with the distribution model serving as one of its primary go-to-market strategies. According to the company, this approach helps accelerate channel development, expand market share, improve inventory and cash conversion, and enhance operational flexibility and responsiveness. During the reporting period, CXMT's largest customers consisted primarily of well-established distributors and leading downstream manufacturers. While no single customer accounted for more than 50% of revenue, customer concentration remained relatively high. Revenue contribution from the top five customers was approximately 74.1% in 2023, 67.3% in 2024, and 68.1% in 2025.

CXMT's main raw materials include chemicals, photoresists, silicon wafers, gases, targets, spare parts, and other semiconductor manufacturing materials. To manage supply-chain stability and quality control, the company has established a comprehensive supplier-management framework that includes supplier qualification, performance evaluation, regular audits, elimination procedures, and blacklist mechanisms. Suppliers are evaluated across multiple dimensions, including quality, delivery, cost, technology, service, and safety. We summarize CXMT's key suppliers and ecosystem partners across the semiconductor equipment, materials, packaging, testing, and memory-module segments in Exhibit 38 .

Exhibit 38: CXMT: Semi equipment supplier list (as of the date of this report)

Semi Equipment			
Process Segment	Company	Chinese Name	Ticker
Chinese suppliers			
Etching, deposition, furnace and cleaning equipment	NAURA Technology Group	北方華創	**002371.SZ**
Etching equipment	Advanced Micro-Fabrication Equipment	中微公司	**688012.SS**
CVD and ALD deposition	Piotech	拓荊科技	**688072.SS**
Cleaning, electroplating and furnace equipment	ACM Research Shanghai	盛美上海	**688082.SS/ACMR.O**
CMP equipment	Hwatsing Technology	華海清科	**688120.SS**
Coating and developing equipment	Kingsemi	芯源微	**688037.SS**
Metrology and inspection	Skyverse Technology	中科飛測	**688361.SS**
Semiconductor inspection equipment	Jingce Electronic Group	精測電子	**300567.SZ**
Precision semiconductor equipment components	Fortune Precision Equipment	富創精密	**688409.SS**
High-purity piping and components	Kinglai Hygienic Materials	新萊應材	**300260.SZ**
Ion implantation and semiconductor equipment investments	Kingstone Semiconductor / Wanye Enterprise	凱世通 / 萬業企業	**600641.SS**
International suppliers			
Coating, developing, deposition, etching and cleaning equipment	Tokyo Electron		**8035.T**
Metrology, inspection and semiconductor equipment	Hitachi High-Tech		**(Under Hitachi Ltd. 6201.T)**
HBM TCB	ASMPT		**0522.HK**

Source: Company data, Morgan Stanley Research.

Exhibit 39: CXMT: Semi material supplier list (as of the date of this report)

Semi Materials			
Material Category	Company	Chinese Name	Ticker
Chinese suppliers			
CMP slurry	Anji Microelectronics	安集科技	**688019.SS**
Precursors and photoresist-related materials	Yoke Technology	雅克科技	**002409.SZ**
Electronic specialty gases	PERIC Special Gases	中船特氣	**688146.SS**
Electronic specialty gases	Huate Gas	華特氣體	**688268.SS**
Electronic and bulk gases	Jinhong Gas	金宏氣體	**688106.SS**
Electronic and industrial gases	Guangzhou Guanggang Gases & Energy	廣鋼氣體	**688548.SS**
Metal-organic precursors and specialty gases	Nata Opto-electronic Material	南大光電	**300346.SZ**
High-purity sputtering targets	Jiangfeng Electronic Materials	江豐電子	**300666.SZ**
High-purity metals and target materials	Grinm Advanced Materials	有研新材	**600206.SS**
Photoresist and electronic materials	Red Avenue New Materials	彤程新材	**603650.SS**
Photoresist and electronic chemicals	Jingrui Electronic Materials	晶瑞電材	**300655.SZ**
Cleaning, electroplating and semiconductor chemicals	Shanghai Sinyang Semiconductor Materials	上海新陽	**300236.SZ**
300 mm silicon wafers	National Silicon Industry Group	滬硅產業	**688126.SS**
300 mm silicon wafers	Xi'an ESWIN Material Technology	西安奕材	**Private**
300 mm silicon wafers	Shanghai Xinsheng Semiconductor	上海新昇	**688126.SS**
300 mm silicon wafers	Shanghai Advanced Silicon Technology	上海超硅	**Private**
Silicon wafers and semiconductor materials	Zhejiang Liontech	立昂微	**605358.SS**
Silicon wafers	Zhonghuan Advanced Semiconductor Materials	中環領先	**002129.SZ**
Quartz materials and components	Hubei Feilihua Quartz Glass	菲利華	**300395.SZ**
Silicon components and silicon electrodes	Thinkon Semiconductor	神工股份	**688233.SS**
High-purity quartz materials	Jiangsu Pacific Quartz	石英股份	**603688.SS**
International suppliers			
Silicon wafers, photoresist-related materials and chemicals	Shin-Etsu Chemical		**4063.T**
Semiconductor silicon wafers	SUMCO Corporation		**3436.T**
Photoresists and electronic materials	JSR Corporation		**Private / delisted**
Photoresists and electronic chemicals	Tokyo Ohka Kogyo		**4186.T**
Electronic materials, specialty chemicals and gases	Merck KGaA		**MRK.DE**
Industrial and electronic gases	Air Liquide		**AI.PA**
Industrial and electronic gases	Linde plc		**LIN**

Source: Company data, Morgan Stanley Research.

Exhibit 40: CXMT: Packaging, testing and memory modules supplier list (as of the date of this report)

Packaging, Testing and Memory Modules			
Role	Company	Chinese Name	Ticker
Chinese suppliers			
Parent company of Payton Technology; memory packaging and testing	Shenzhen Kaifa Technology	深科技	**000021.SZ**
DRAM packaging and testing	Payton Technology	沛頓科技	**Private subsidiary**
Semiconductor packaging and testing	JCET Group	長電科技	**600584.SS**
Semiconductor packaging and testing	Huatian Technology	華天科技	**002185.SZ**
Semiconductor packaging and testing	Tongfu Microelectronics	通富微電	**002156.SZ**
Memory testing and inspection equipment	Shenzhen Seichitech Technology	精智达	**688627.SS**
Memory testing and inspection equipment	Hangzhou Changchun Technology	长川科技	**300604.SZ**
Display-driver and memory-related packaging and testing	Union Semiconductor Hefei	匯成股份	**688403.SS**
Memory modules and embedded storage products	Longsys Electronics	江波龍	**301308.SZ**
Memory modules, SSDs and embedded storage	Biwin Storage Technology	佰維存儲	**688525.SS**
Owner or operator of consumer-memory brands including Gloway	Jiahe Jinwei	嘉合勁威	**Private**
Consumer-memory brand	Gloway	光威	**Private brand**
Consumer-memory brand	KingBank	金百達	**Private brand**

Source: Company data, Morgan Stanley Research.

Key competitors

The DRAM industry is highly concentrated. Samsung Electronics, SK hynix, and Micron Technology have long held a combined global market share of over 90% in terms of revenue. According to Omdia data, in 2025, Samsung, SK hynix, and Micron had respective global DRAM revenue market shares of 33.96%, 34.48%, and 23.41%. Based on our estimates, CXMT's global DRAM revenue market share reached 7.67% in 4Q25, which we think indicates that it has gradually entered the ranks of major industry players.

We believe the company's competitive advantages mainly come from its position as China's domestic DRAM leader, capacity scale, comprehensive product portfolio, independent R&D capabilities, customer resources, and supply-chain ecosystem collaboration. However, we believe CXMT still lags behind the top three international manufacturers in terms of overall scale, technology accumulation, and product mix.

Management team

Yiming Zhu, the founder of CXMT, studied physics at Tsinghua University and later pursued graduate studies in electrical engineering at SUNY Stony Brook. After working in Silicon Valley, including at MoSys, he returned to China in 2005 and founded GigaDevice, which grew into a leading fabless supplier of NOR flash memory and microcontrollers. In 2016, he moved into the much larger but more demanding DRAM market through the Hefei-backed "506" project, which later became CXMT. Since DRAM manufacturing requires deep process expertise, advanced technology capabilities, and access to extensive intellectual property, CXMT initially benefited from technology and talent associated with the former European memory manufacturer Qimonda. By licensing a substantial portfolio of Qimonda patents and gaining access to extensive technical documentation, CXMT established an important foundation for its DRAM development efforts. This provided access to key DRAM technologies, including buried-wordline (BWL) and stacked-capacitor architectures. Building on this foundation, CXMT successfully scaled these technologies from approximately the 46nm-class to increasingly advanced process generations, ultimately progressing toward today's 16nm-class Gen4B production.

Still, CXMT's long-term progress depended not only on inherited intellectual property, but also on the engineering talent required to convert that knowledge into commercial production. The company recruited former Qimonda engineers, including veteran technologist Karl-Heinz Kuesters, whose manufacturing experience helped transform archived designs into technology concepts into commercially viable products with acceptable yields. CXMT also attracted professionals with backgrounds at Micron, SanDisk, Applied Materials, and semiconductor firms in South Korea and Taiwan. These hires strengthened the company's capabilities in process integration, materials, yield improvement, and next-generation memory. Over time, the combination of experienced international talent and returning Chinese engineers enabled CXMT to build an increasingly self-sufficient R&D and manufacturing organization. This internal capability supported the company's progression through multiple DRAM process generations, expansion from DDR4 and LPDDR4X into DDR5 and LPDDR5/5X, and more recently its development efforts in server DRAM and HBM. However, this transformation was neither quick nor inexpensive. It required nearly a decade of substantial capital investment, intensive R&D spending, process development, capacity expansion, and cumulative manufacturing learning before the company reached profitability.

Exhibit 41: Management team details (as of July 27, 2026)

Name	Year of Birth	Nationality / Residency	Education	Professional Experience	Current Role with the Issuer
Yiming Zhu	1972	Chinese national; Singapore permanent resident	Master's degree	Previously served as a senior engineer at iPolicy Networks Inc. and as a project manager at Monolithic System Technologies Inc. Founded GigaDevice in 2005. Served as General Manager of GigaDevice from Apr. 2005 to Jul. 2018 and has served as its Chairman since Apr. 2005. From Jul. 2018 to Apr. 2023, successively served as Director, Chairman and Chief Executive Officer of ChangXin Memory Technologies. From May 2020 to Apr. 2023, served as Chief Executive Officer of the issuer.	Chairman since Feb. 2021
Qiyu Cao	1971	Chinese national; no permanent residency outside China	Ph.D.	From May 2002 to Jun. 2005, worked at Philips Semiconductor in the United States and Hyperband Communications in the United States. From Jul. 2005 to Aug. 2013, successively founded Beijing Kuntian Kewei Electronics Co., Ltd. and Beijing Fanda Xun Technology Co., Ltd. From Nov. 2013 to Oct. 2017, served as General Manager of the NAND Flash Business Division of NoVoMem Inc. After joining the issuer in Nov. 2017, successively served as Executive Vice President of Ruili Integration, Executive Vice President of Product R&D at ChangXin Memory, Executive Vice President of Technology R&D, and General Manager of ChangXin Jidian.	Director since Jun. 2021; has successively served as Chief Executive Officer and President since Apr. 2023
Rui Zheng	1982	Chinese national; no permanent residency outside China	Ph.D.	From Jul. 2010 to Jul. 2017, successively served as Senior Supervisor of the Comprehensive Development Department, Project Manager of the Financial Investment Department, Manager of the Risk Control Department of Anhui Investment Group, and Assistant General Manager of Anhui Zhongan Capital Management Co., Ltd. From Jul. 2017 to Dec. 2021, served as Deputy General Manager of Anhui Wantou Asset Management Co., Ltd. From Dec. 2021 to Oct. 2024, served as Party Branch Secretary and Deputy General Manager of that company. Since Oct. 2024, has served as Party Branch Secretary, Executive Director and General Manager of the company.	Director since May 2025
Wei Fang	1969	Chinese national; no permanent residency outside China	College diploma; Senior Accountant	From Aug. 1991 to Jan. 2022, successively worked at the former Hefei Fertilizer Plant, Hefei Sifang Chemical Group Co., Ltd. and Hefei Industrial Investment Holding Co., Ltd. From Mar. 2015 to Jan. 2022, served as General Manager of the Planning and Finance Department of Hefei Industry Investment Holding (Group) Co., Ltd. Since Feb. 2022, has served as Chief Economist of Hefei Industry Investment Holding (Group) Co., Ltd.	Director since Aug. 2022
Jun Wei	1967	Chinese national; no permanent residency outside China	Master's degree	From Jul. 1990 to Mar. 2008, successively worked at the Electronic Industry Development Department of the Ministry of Machinery and Electronics Industry, the Comprehensive Planning Department of the Ministry of Electronics Industry, and the Comprehensive Planning Department of the Ministry of Information Industry. From Mar. 2008 to Jul. 2015, served as Deputy Director-General of the Planning Department of the Ministry of Industry and Information Technology. Since Jul. 2015, has served as Vice President of China Integrated Circuit Industry Investment Fund Co., Ltd. Since Oct. 2019, has served as Vice President of the second-phase fund. Since May 2025, has served as Vice President of China Integrated Circuit Industry Investment Fund Phase II Co., Ltd.	Director since Jun. 2023
Huawei Hou	1980	Chinese national; no permanent residency outside China	Master's degree	From Jul. 2001 to May 2006, worked at the Shipbuilding Industry Economic Research Center of China State Shipbuilding Corporation as a ship market analyst. From May 2006 to Oct. 2011, worked at China Development Bank, successively serving as a deputy-section-level officer and Deputy Division Director. From Oct. 2011 to Aug. 2013, served as Deputy Director of the Financial Street Subdistrict Office of Xicheng District, Beijing. From Aug. 2013 to Apr. 2023, worked at China Development Bank, successively serving as Deputy Division Director of the Second Appraisal Bureau.	Director since Aug. 2025

Source: Company data, Morgan Stanley Research.

Appendix 1: Global DRAM Cycle: A Multi-year Bottleneck

The explosive demand for AI infrastructure has caused supply constraints in all types of memory (DRAM, HBM, NAND, HDD). As AI increasingly moves beyond data centers into edge devices and the physical world, it could trigger a further surge in demand for memory chips. In that scenario, these components would no longer be seen as commodity chips but as critical pillars of the global digital infrastructure.

AI and data centers are becoming the dominant buyers of memory – we expect the server share of DRAM demand to rise from 37% in 2023 to 59% in 2028e. The result is a structurally tighter market where non-AI buyers may face weaker access to supply for an extended period.

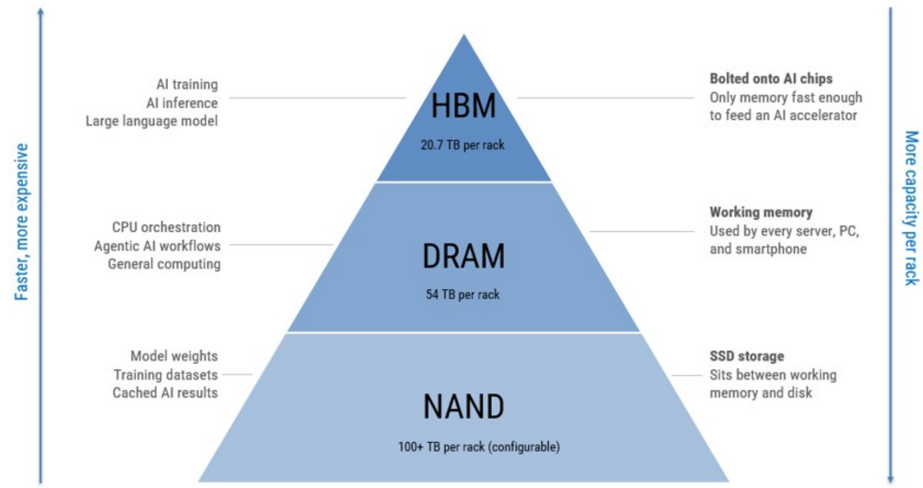
Memory is the working space a computer uses to hold data that it is actively processing – the digital equivalent of the desk you are working on right now, as opposed to the filing cabinet behind you. Three types of memory matter for the AI economy:

- **DRAM (Dynamic Random Access Memory)** is the fast but expensive working memory used by every server, PC and smartphone. It is what the CPU uses to hold and fetch the data it is computing.
- **HBM (High Bandwidth Memory)** is a premium type of DRAM focused on providing maximum data bandwidth. HBM is DRAM stacked vertically and bolted directly onto AI chips – it is the memory layer that makes large AI models possible, feeding data fast enough to limit AI accelerator downtime. Think of DRAM as a single lane road and HBM as a 12-lane highway for data transmission.
- **NAND (Not-And)** is the storage memory used in solid-state drives (SSDs), which sit between working memory and slower HDD disk storage. NAND holds the trained model weights, the training datasets, and the cached results that AI systems retrieve from. It is also used for picture, video and data storage.

All three types of memory matter as Al workloads use all three at once – and in massive quantities. Next-generation rack-scale AI systems, such as NVIDIA's Vera Rubin NVL72, contain 20.7TB of HBM attached to GPUs and 54TB of LPDDR5X serving the CPUs, before considering local or external SSD storage.

AI demand is scaling across three layers at once — more memory per chip, more chips per system, and more systems per cluster. Agentic AI adds another layer of pressure through longer context windows, larger conversation memory and repeated inference across multiple agents. The result is a structural increase in memory intensity across all aspects of the AI stack.

Exhibit 42: AI servers are becoming memory systems



Source: NVIDIA, Morgan Stanley Research. Note: Rubin NVL72 memory capacity based on platform disclosures. NAND/eSSD capacity varies by OEM and storage architecture.

Supplier map: Strategic memory remains led by the Big 3; China scales in commodity pools

The supplier landscape is highly segmented by product. HBM and advanced/server DRAM remain concentrated among Samsung, SK hynix and Micron, with SK hynix leading HBM today, Samsung positioned as the broadest DRAM/NAND supplier and potential HBM recovery candidate, and Micron emerging as the key US-headquartered strategic supplier. NAND supply is broader, with Samsung, Kioxia/SanDisk, SK hynix/Solidigm, Micron and YMTC all relevant. China is becoming more important through CXMT in commodity DRAM/LPDDR and YMTC in NAND, but neither is a near-term global HBM relief valve. The key implication is that supply relief is uneven: China can help in selected commodity memory markets, while frontier AI memory remains concentrated in the Big 3.

Exhibit 43: Supplier position, by memory type

Supplier position by memory type

Market position by product pool: HBM remains Big 3-led; China emerges in commodity DRAM/NAND
Market leader · Strong · Present / growing · Limited / emerging · Not present

	HBM AI accelerator	Server DRAM DDR5	PC & mobile LPDDR	Enterprise NAND | SSD	Consumer NAND
SK hynix South Korea	**Leader** Strong Nvidia exposure.	**Strong** DDR5 ahead of peers. Wafer shift to HBM tightens conventional supply.	**Strong** LPDDR5X for premium smartphones. Losing low-end to CXMT.	**Present** Enterprise SSD via Solidigm.	**Present** Lower priority -- HBM and server absorb most capacity.
Samsung South Korea	**Recovery candidate** HBM3E yield issues cost share. HBM4 is a critical recovery moment.	**Leader** Largest conventional DRAM capacity. Broadest customer base.	**Leader** Largest LPDDR supplier. Inside most premium Android phones.	**Leader** Largest NAND capacity globally. V-NAND technology advantage.	**Leader** eMMC, UFS, client SSDs across all tiers. Dominant share.
Micron USA	**Growing fast** HBM3E qualified for NVIDIA H200; only US-headquartered HBM supplier.	**Strong** DDR5 competitive. US manufacture differentiates for government buyers.	**Strong** LPDDR5X competitive. Apple historically significant customer.	**Strong** QLC NAND competitive for hyperscale SSDs. US manufacture a differentiator.	**Present** Consumer flash. Lower priority versus HBM/server mix shift.
Kioxia / SanDisk Japan / USA	NAND only -- no DRAM	--	--	**Strong** 2nd-largest NAND capacity globally. BiCS technology.	**Strong** USB, client SSDs, mobile UFS. Financially more fragile in downturns.
CXMT China, state-backed	**Attempting** HBM3 development / sampling, but not a near-term global HBM relief valve.	**Limited** Emerging in DDR4/DDR5; not yet qualified at scale for mainstream global hyperscale server DRAM	**Growing** Mid/low LPDDR5 for Chinese Android market. Supporting Huawei.	DRAM only -- no NAND	--
YMTC China, state-backed	NAND only -- no DRAM	--	--	**Emerging** Growing in Chinese market. Export restrictions limit global access.	**Growing fast** Domestic NAND ramp; China self-sufficiency ambition; export restrictions limit global access

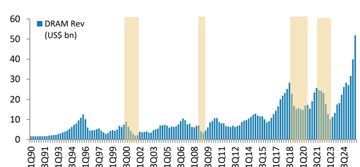
Source: Company commentary, Morgan Stanley Research estimates. Note: Market positions are illustrative and intended to frame supplier participation by memory type.

Memory cycle monitor

Where are we in the DRAM cycle?

Duration is getting shorter now. Historically, the memory segment tended to be 4-8 quarters in a downturn and 4-9 quarters in an upturn. The slowdown tends to be shorter than the upturn in the memory cycle because it is easy to cancel orders and rationalize inventory but harder to bring back capacity. The 2018 share-price correction lasted only four quarters, from 1Q18 to 1Q19. The most recent downcycle, in 2024, lasted for only 8 months before stocks found a bottom, and with tariffs and AI demand supporting an increase in pricing again, share prices also rebounded sharply. We are now 11 months into the current upcycle, which began in September 2025.

Exhibit 44: DRAM cycles in perspective over the past 30 years



Downturn	Duration (Q)	Rev	ASP	Reason
1996-98	10	-77%	-95%	Post 93-95 PC Boom
2000-03	11	-61%	-87%	Internet Bubble
2008-09	2	-51%	-46%	Global Financial Crisis
2010-12	8	-41%	-67%	Weak Global Economy
2014-16	5	-34%	-41%	Smartphone Growth Slowdown
2018-19	6	-48%	-59%	Excess Inventory; Capacity Expansion; Soft Demand
2021-23	6	-61%	-55%	Covid pull-in demand, geopolitical disruptions, U.S recession risks

Source: FactSet, Morgan Stanley Research.

How does the current supply-demand balance compare with previous cycles, and where are we in the cycle?

The current memory cycle is materially tighter than the 2023–24 cycle and is being driven less by a broad-based demand rebound than by AI-related capacity displacement.

In 2023, most conventional memory markets were in deep oversupply following the post-COVID inventory correction. Conditions began to normalize in 2024 as inventories cleared, pricing stabilized, and suppliers restored discipline. Today, however, HBM is effectively sold out, server DRAM is tight, and enterprise NAND is tightening as wafer capacity and investment are redirected toward higher-value AI products.

The key difference versus previous cycles is that the shortage is increasingly structural rather than purely cyclical. HBM demand is growing rapidly under multi-year commitments, hyperscalers are consuming more DRAM per system, and added HBM capacity can constrain supply for conventional DRAM and NAND. As a result, pricing pressure is spreading across the stack even though end-market demand remains uneven.

That said, the cycle is bifurcated. AI- and data-center-oriented products appear likely to remain tight, while PC, mobile, and consumer NAND are more exposed to demand destruction if higher prices cause customers to reduce units or spending. Overall, the market has moved from the broad oversupply of 2023, through recovery in 2024, into a tight and increasingly supply-constrained environment, with the strongest conditions concentrated in HBM and server-related memory.

The memory cycle is no longer uniform. AI-facing memory remains structurally tight, while much of the tightness in consumer-facing DRAM reflects capacity crowding-out rather than exceptionally strong end-demand. HBM is the most constrained segment because rapid AI accelerator growth, high memory content per system, complex packaging requirements, and multi-year supply commitments absorb both wafer capacity and capital expenditure. Server DRAM is also benefiting from AI and cloud infrastructure spending: AI servers require substantially more memory per system, while hyperscalers are increasing memory intensity across both accelerated and general-purpose compute. Together, HBM and high-capacity server DRAM are therefore setting the direction for industry allocation and pricing.

Enterprise demand provides an additional source of support as cloud providers and large data-center customers expand infrastructure, although the pace remains more dependent on deployment schedules and customer budgets than HBM demand. By contrast, PC and smartphone demand is recovering only gradually. Replacement cycles, AI-PC and premium handset features may raise DRAM content per device, but unit growth remains modest and is sensitive to higher component prices. Automotive DRAM is comparatively resilient because memory content continues to increase with ADAS, infotainment, connectivity, and centralized computing architectures. However, automotive is a smaller market with long qualification cycles, so it is more likely to provide stable incremental demand than to drive the overall DRAM cycle.

The implication for commodity DRAM is a tightening supply outlook despite uneven end-market demand. As suppliers prioritize HBM, high-capacity server DRAM, and other higher-value products, fewer wafers and less incremental investment are available for conventional DDR and LPDDR. This can push commodity DRAM pricing higher even without a broad-based boom in PC or smartphone shipments. AI and hyperscale customers are likely to receive priority allocation, while PC, smartphone, automotive, and industrial customers may respond through lower memory configurations, delayed product refreshes, inventory discipline, or margin compression. The outlook is therefore increasingly bifurcated: AI-related products remain structurally undersupplied, while commodity DRAM tightens indirectly through constrained supply, higher product mix, and limited capacity additions rather than through uniformly strong downstream demand.

Exhibit 45: Memory cycle conditions, by type

Memory cycle conditions by type

Supply/demand balance from oversupply to critically tight

Oversupply — Critically tight

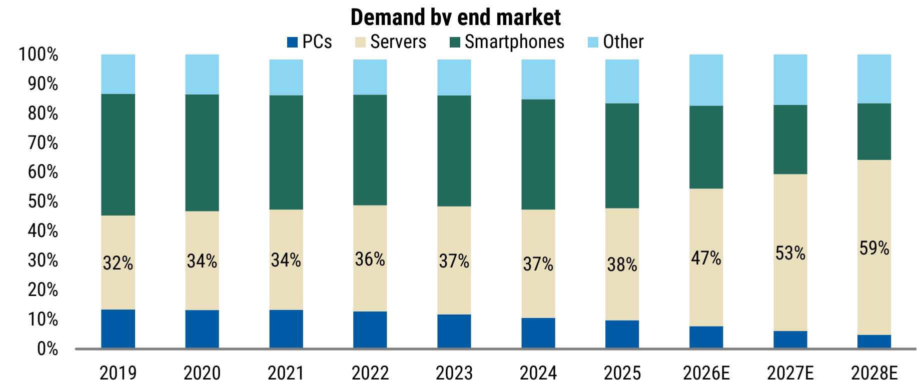
	2023	2024	Today (2025-26)	Forward
HBM AI accelerator	**Niche / nascent** HBM2E. Small TAM. Early AI clusters only. SK hynix dominant.	**Critically tight** HBM3 ramp for H100. Explosive demand. Samsung HBM3E yield issues. Wafer reallocation begins.	**Sold out** HBM3E mainstream. Entire market on multi-year LTAs. Hyperscalers at front of queue. Pricing opaque but structurally elevated.	**Structural shortage** HBM4 ramping. TAM growing at double digit annually.
Server DRAM DDR5 / RDIMM	**Oversupply** Post-COVID glut. Hyperscaler digestion. DDR5 transition early. Prices down 40–60%.	**Recovering** AI buildout driving demand. Hyperscaler share rising toward 50%. Prices stabilising.	**Tight** Agentic AI driving 1.5–3TB DRAM per CPU. Wafer pull to HBM cannibalises supply. Hyperscalers crowd out others.	**Continued pressure** Rising every quarter through CY26 per TrendForce. CXL emerging as next tier. No supply relief until HBM investment moderates.
PC & mobile DRAM LPDDR / DDR	**Deep oversupply** Prices fell 50%+. Worst PC/smartphone correction in years. Inventory correction through H1 2023. Manufacturers reporting heavy losses.	**Normalising** Win11 refresh beginning. AI PC content upgrades. Prices recovering. Demand stable.	**Crowded out** Supply redirected to HBM/server. Lenovo: DRAM doubled Q/Q. Gartner projects 10.4% PC shipment reduction. Elasticity 1.2–1.7x.	**Demand risk** VARs expect 60% of customers to cut PC spending if prices rise. ODM builds down 11% Y/Y in C1Q26. Unit declines likely.
Enterprise NAND Data centre SSDs	**Severe oversupply** Worst correction in NAND's history. Kioxia/WD in financial distress. Enterprise SSD prices fell 60%+. Industry-wide capex cuts.	**Recovery underway** AI data centre demand leads recovery. "Great Refresh of 2026" beginning. Supply discipline restoring balance.	**Tightening** NTAP: memory ~50% of product costs. VAR checks: 10–25% end-user price increases. NAND fab diverting wafers.	**Continued pressure** Prices rising through CY26. YMTC meaningful in NAND by 2030. Enterprise TAM expanding – but CIOs cutting storage budgets.
Consumer NAND Mobile / PC flash	**Deep oversupply** Synchronised with enterprise NAND – shared fab infrastructure. Smartphone OEMs destocking. Consumer electronics demand collapsed.	**Recovering** AI phone features driving NAND content upgrades. Recovery led by premium tier. YMTC gaining Chinese domestic share.	**Rising – most exposed** Prices rising with enterprise NAND. Highest elasticity – demand destruction most acute. Nintendo, CRSR flagging profitability risk.	**Most at risk** High elasticity will reduce unit volumes. YMTC partially offsets for Chinese OEMs. Low-end consumer exposure most vulnerable.

Source: Company commentary, TrendForce, Gartner, VAR checks, Morgan Stanley Research estimates. Note: Qualitative supply/demand conditions shown by product type; price increases refer to cited market checks where available.

Demand is shifting from consumer electronics to AI/server infrastructure

DRAM server share of bit demand rises from 37% in 2023 to 59% in 2028e, while the share of shipments for PCs and smartphones declines sharply. This matters because the dominant buyer increasingly has stronger purchasing power, longer-duration demand visibility and greater willingness to sign LTAs or prepay for supply. As a result, pricing and shipment allocation continue to gravitate toward AI/server infrastructure requirements at the expense of traditional consumer end-markets.

Exhibit 46: DRAM – AI server demand becomes dominant…



Source: TrendForce, Morgan Stanley Research. E = TrendForce estimates.

Supply slow to respond

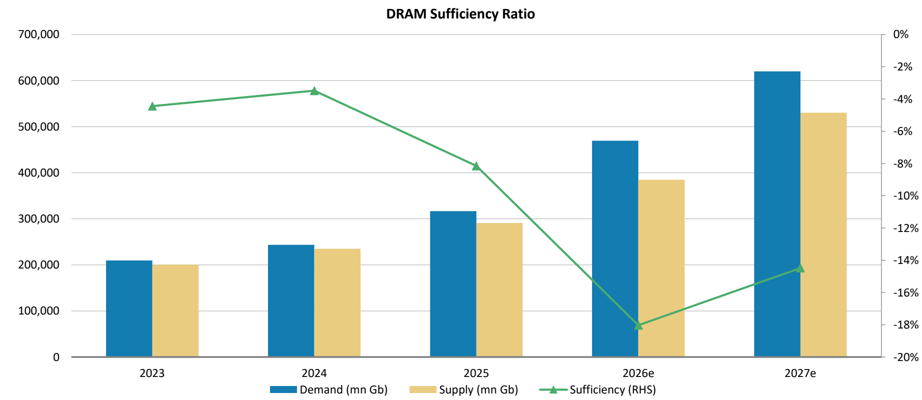
New memory capacity will come online from late 2027 but is still unable to meet demand for all markets. Strategically, most of the valuable capacity is likely to be dedicated to where pricing, margins and customer commitments are strongest – namely to HBM customers, server DRAM and enterprise SSDs. While AI semiconductors are forecast to grow at a ~50% CAGR by 2030 (as per TSMC forecasts at its annual technology symposium on May 14, 2026), we expect DRAM production to grow at a pace that will meet only about 60% of demand, given annual bit shipments are unlikely to rise much above the 30% level due to EUV constraints.

We model DRAM supply remaining tight even as capacity additions accelerate. DRAM undersupply widens from ~3–4% in 2023–24 to ~18% in 2026, before easing but still remaining tight, at ~14%, in 2027e. The key issue is timing.

- Annual DRAM wafer capacity additions rise from ~65kwpm in 2025e to ~498kwpm in 2027e and ~563kwpm in 2028e, led by Samsung, SK hynix, Micron and CXMT. This represents meaningful supply growth by historical standards, but arrives after the period of maximum tightness.
- Of these, CXMT's acceleration represents a meaningful shift in the supply-side landscape, and we expect CXMT's market shares to reach 15% in 2028e vs. 9% in 2025. From 2023 to 2028e, we expect Mainland China to account for ~24% of net global DRAM capacity additions, behind South Korea but ahead of the US. But even with the rapid increase in CXMT's capacity, we still see demand hard to be fulfilled by all the new supply added.

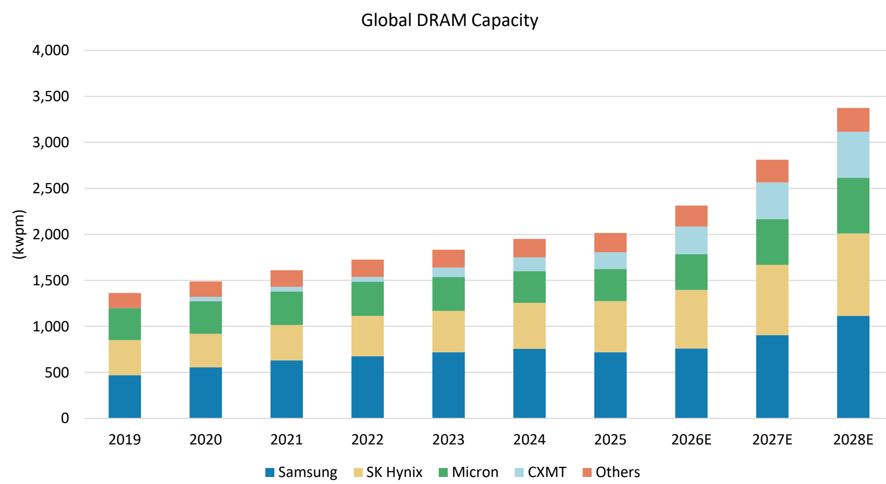
The timing issue in bringing on new capacity is compounded by execution risk. New memory capacity requires fab construction, tool delivery, process qualification, customer validation and yield ramp. Even when capacity is announced, usable output can lag by several quarters or years.

Exhibit 47: DRAM supply sufficiency ratio



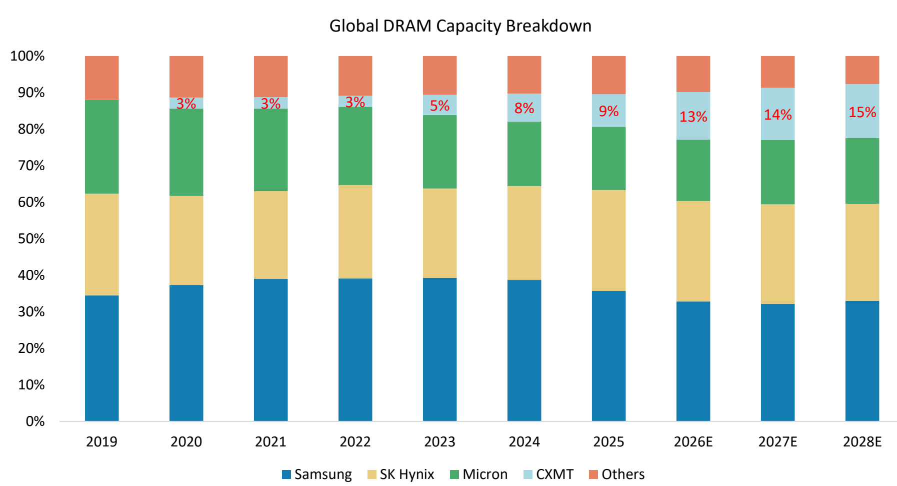
Source: TrendForce, Morgan Stanley Research. E = Morgan Stanley estimates.

Exhibit 48: Global DRAM capacity is expected to reach 3,374kwpm in 2028e from 2,313kwpm in 2026e



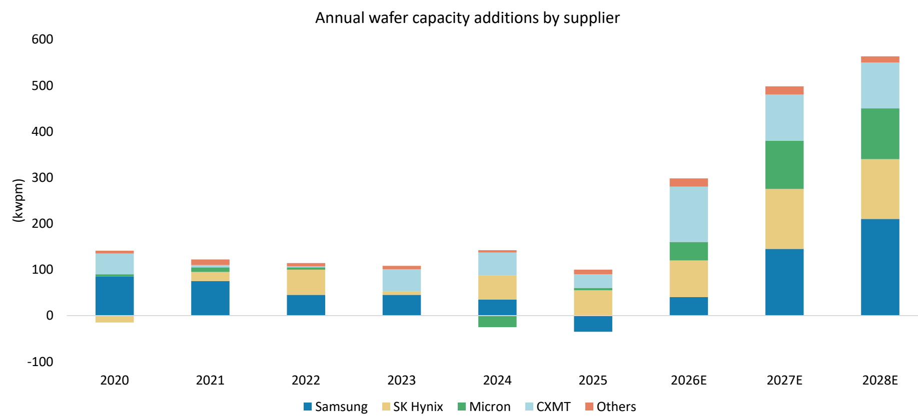
Source: TrendForce, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 49: CXMT's global capacity market share to extend from 9% in 2025 to 15% in 2028e



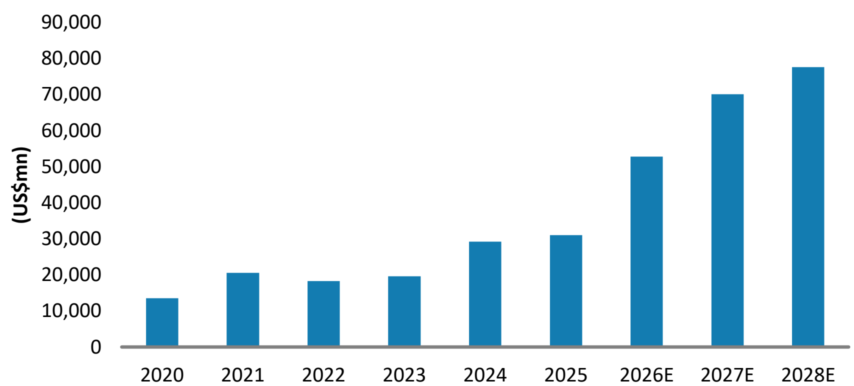
Source: TrendForce, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 50: DRAM annual wafer capacity additions are moving to all-time high



Source: TrendForce estimates, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 51: Global DRAM spending



Source: Company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

LTAs could create a two-tier memory market

As top-tier CSPs secure multi-year supply through LTAs, the residual memory market could become structurally tighter. Customers without LTAs may need to procure from a smaller pool of uncommitted supply, potentially at higher and more volatile prices. We view this as a potential structural cost borne by every buyer who is not at the front of the LTA queue.

Historically, weak PC or smartphone demand would help loosen DRAM markets and lower prices for everyone. In this cycle, AI demand keeps allocation tight even if consumer electronics demand softens, because HBM and server applications now set the priority order, and supplier incentives reinforce the allocation. After the last memory downcycle, suppliers have limited incentive to flood the market with commodity supply when AI customers offer stronger pricing, better margins and more durable contracts.

We run DRAM sufficiency analysis on the two largest non-server end-markets: PCs and smartphones

Starting from total 2027 DRAM + HBM supply of 573,503mn Gb based on our projection, we allocate 70% to servers and 30% to non-server applications. Within non-server, we assign 17% to PCs and 58% to smartphones, consistent with historical patterns. On the PC side, bottom-up demand of 34,386mn Gb implies a shortfall of 5,138mn Gb (-15%). The picture is comparably challenging for smartphones: total smartphone DRAM demand of 113,125mn Gb against an addressable supply of 99,790mn Gb produces a 13,335mn Gb gap (-12%).

The shortfall proves highly sensitive to content assumptions. On the PC side, our base case of 89Gb per device implies a -15% shortfall; holding content flat year-on-year at 83Gb narrows this to -9%, and reducing spec upgrade assumptions brings it to 77Gb, reducing the shortfall to just -2%. Smartphones follow the same pattern: against a 99Gb base case (-12%), flat content at 95Gb narrows the gap to -7%, and limiting spec refresh brings it to 90Gb, a -2% shortfall.

The read-through is twofold. First, the unit-volume risk is a function of memory content assumptions, not an absolute supply wall — OEMs have a clear lever to protect shipment volumes. Second, that lever is not free: every gigabyte of limiting rising specs that protects unit volumes is a gigabyte of forgone content growth, so the adjustment ultimately resolves the unit shortfall by suppressing memory bit demand. The market clears either through fewer devices or lower memory intensity per device — and the more likely outcome is a blend of the two.

This reinforces supplier pricing power and changes the cycle. The more supply is committed to strategic AI customers, the less flexible supply remains for opportunistic or non-LTA buyers. This could support ASP resilience even if parts of consumer demand weaken.

Exhibit 52: Our memory sufficiency framework suggests PC and smartphone could see 12-15% memory shortfall

PC memory sufficiency in 2027	
Total DRAM + HBM supply (mn Gb)	573,503
Server gets 70% of allocation	401,452
Non-Server gets 30% of allocation	172,051
PC gets 17% of non-server allocation	29,249
Total PC memory demand (mn Gb)	34,386
Shortfall (mn Gb)	**(5,138)**
	-15%
Memory content per device (Gb)	89
Shortfall (mn units)	**(58)**
vs. MS PC shipment forecast	388
Downside vs. MSe	**-15%**

Sensitivity analysis

	Memory content per device (Gb)	Shortfall
Current assumption	89	-15%
Flat content YoY	83	-9%
De-specing	77	-2%

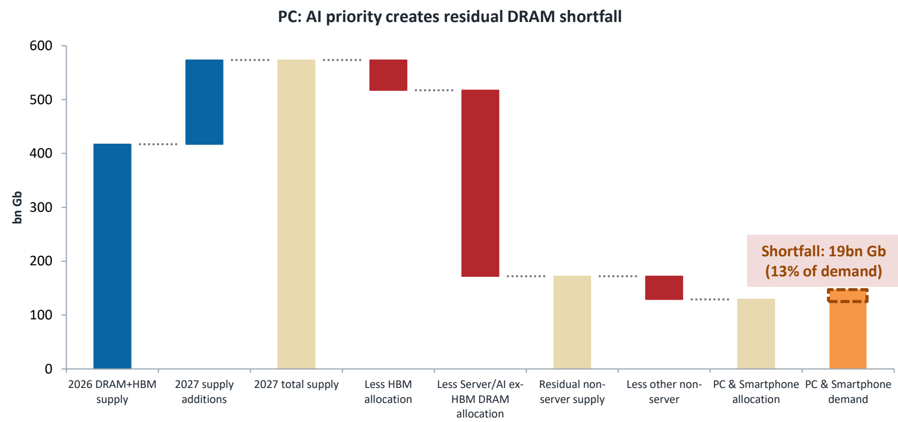
Smartphone memory sufficiency in 2027	
Total DRAM + HBM supply (2027)	573,503
Server gets 70% of allocation	401,452
Non-Server gets 30% of allocation	172,051
Smartphone gets 58% of non-server allocation	99,790
Total smartphone demand (2027)	113,125
Shortfall (mn Gb)	**(13,335)**
	-12%
Memory content per device (Gb)	99
Shortfall (mn units)	**(134)**
vs. MS smartphone forecast	1,137
Downside vs. Mse	**-12%**

Sensitivity analysis

	Memory content per device (Gb)	Shortfall
Current assumption	99	-12%
Flat content YoY	95	-7%
De-specing	90	-2%

Source: Morgan Stanley Research estimates. Note: PC includes laptops, desktops and tablets.

Exhibit 53: AI prioritization turns 2027 supply growth into a consumer memory shortfall

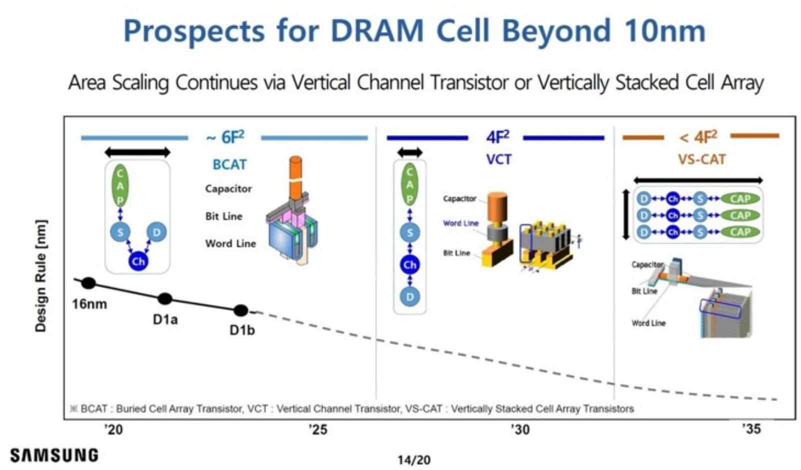


Source: Morgan Stanley Research estimates.

Appendix 2: What is 3D DRAM Technology? Can China CXMT be the First Chip Maker to Adopt it?

Our checks with equipment vendors suggest that CXMT may try VCT (vertical channel transistor), a transitional technology to 3D DRAM, to push its node to Gen5 (~15nm) without EUV tools. Our model assumes some production of this new DRAM technology in 2H28.

Exhibit 54: DRAM scaling through the transistor architecture change



Source: Samsung.

Planar DRAM: The 1c (1γ) inflection and the planar wall

DRAM scaling has historically followed a predictable cadence: reducing the physical dimensions of the transistor and capacitor to increase bit density and lower the cost per bit. The 10nm-class progression has been annotated sequentially as 1x, 1y, 1z, 1a, 1b, and now the latest 1c, or 1γ, node. Samsung, SK hynix, and Micron are now actively mass-producing or ramping up 1c DRAM.

The transition from the 14nm 1a node to the 12nm node delivered a significant geometric shrink. In contrast, the transition from 1b to 1c represents a shrink of less than 10%. This deceleration suggests that planar DRAM scaling is becoming less efficient as physical constraints around the transistor, capacitor, leakage, and process complexity intensify.

Planar DRAM: The EUV divergence - parametric yield vs theoretical density

EUV tools are essential for the 1c node. However, as the number of EUV-applied layers increases, the difficulty of subsequent processes, particularly etching, rises sharply. Major memory players have therefore made different strategic choices in balancing density, yield, cost, and process complexity.

- **Micron: capital-efficient EUV usage**. Micron has maintained a strategy of strict capital efficiency. It applies EUV to fewer critical layers while relying on advanced multi-patterning for others. This may allow Micron to achieve a roughly 30% density improvement over its 1b node without taking on the depreciation burden associated with a larger EUV tool base.

- **Samsung: reduced EUV layer count**. Samsung originally planned to use 8–9 EUV layers for its 1c node, but recently reduced its EUV layer-count target to 6–7 layers, a roughly 30% reduction. Samsung appears to believe that parametric yield losses associated with complex EUV etching can outweigh the theoretical density benefit. By substituting DUV lithography for certain layers, Samsung also avoids some of the capital intensity associated with additional EUV tools.

Exhibit 55: Comparison of 1B process specs across major vendors

	Samsung	SK hynix	Micron
Public node name	Calls it 12nm-class DRAM; reverse-engineering sources generally map Samsung's current 12nm-class parts to D1b / 1b-class. Samsung announced mass production of 16Gb DDR5 on this process in May 2023. (Samsung Semiconductor Global)	Calls it 1bnm, the fifth-generation 10nm-class process. SK hynix said it completed development, and completed 1bnm development and entered Intel's data center certified validation process in 2023. (SK hynix Newsroom)	Calls it 1β / 1-beta. Micron announced 1β LPDDR5X qualification samples and mass-production readiness in November 2022. (Micron Technology)
Process / lithography angle	Uses EUV and high-k materials; TechInsights says Samsung expanded EUV use to five or more masks in D1a/D1b and is among the most aggressive on cell scaling. (Samsung Semiconductor Global)	Uses EUV for D1a/D1b and HKMG in 1bnm DDR5, according to SK hynix and TechInsights. (SK hynix Newsroom)	No EUV at 1β: Micron says it used proprietary multi-patterning and immersion lithography; TechInsights similarly notes Micron stayed with ArF/ArFi through 1β and planned EUV for 1γ. (Micron Technology)
DDR5 speed disclosed	16Gb DDR5 up to 7.2Gbps on 12nm-class process. (Samsung Semiconductor Global)	Initial 1bnm server DDR5 disclosed at 6.4Gbps. That was the speed of the samples submitted to Intel for validation, not necessarily the process ceiling. (SK hynix Newsroom)	16Gb DDR5 up to 7200 MT/s, with 1β DDR5 shipping to data-center and PC customers. (Micron Technology)
DDR5 density / capacity path	16Gb DDR5 first, then 32Gb DDR5 on 12nm-class process. Samsung says 32Gb dies enable 128GB modules without TSV and future modules up to 1TB. (Samsung Semiconductor Global)	16Gb-class 1bnm DDR5 first, then 32Gb 1b die-based 256GB DDR5 RDIMM, validated for Intel Xeon 6 platforms. (SK hynix Newsroom)	1β DDR5 portfolio includes 16Gb, 24Gb, and 32Gb DRAM dies for RDIMM/MCRDIMM products. (Micron Technology)
Power / productivity claims	Samsung claims up to 23% lower power and up to 20% better wafer productivity versus its previous DRAM generation for 12nm-class DDR5. For 32Gb DDR5 modules, it claims about 10% lower power versus 16Gb-based TSV designs. (Samsung Semiconductor Global)	SK hynix claims over 20% lower power versus its 1anm generation for 1bnm DDR5. Its 32Gb 1b 256GB RDIMM is claimed to use about 18% less power than prior 256GB products based on 16Gb 1a DRAM. (SK hynix Newsroom)	Micron claims 1β gives about 15% better power efficiency and over 35% better bit density versus 1α. For 1β DDR5, Micron claims 50% higher performance and 33% better performance per watt versus its prior generation. (Micron Technology)
Mobile / LPDDR specs	Samsung's 12nm-class LPDDR5X reaches up to 10.7Gbps, with nearly 25% lower power versus the previous LPDDR5X generation and up to 32GB package capacity. (Samsung Semiconductor Global)	No discrete 1bnm mobile product announced. (SK hynix Newsroom)	Micron's first 1β LPDDR5X was 8.5Gbps; later 1β LPDDR5X production samples reached 9.6Gbps, with up to 16GB package capacity. (Micron Technology)

Source: Company data, TechInsights, Morgan Stanley Research.

Next steps: 3D overcoming the planar wall

Following the 1c node, memory makers project 1d or 1e nodes. However, the physical limitations of the capacitor may become an insurmountable barrier in the traditional $6F^2$ planar arrangement. As the lateral footprint shrinks, the capacitor requires a higher depth-to-width ratio to maintain sufficient capacitance. This creates severe charge leakage, higher refresh rates, lower power efficiency, and worsening thermal bottlenecks.

The next structural breakthrough is the transition to a $4F^2$ cell structure, also classified as the 0a node. This transition is expected to reach mass-production viability around 2027–28 and represents the next major step in extending DRAM density beyond the limits of conventional planar scaling.

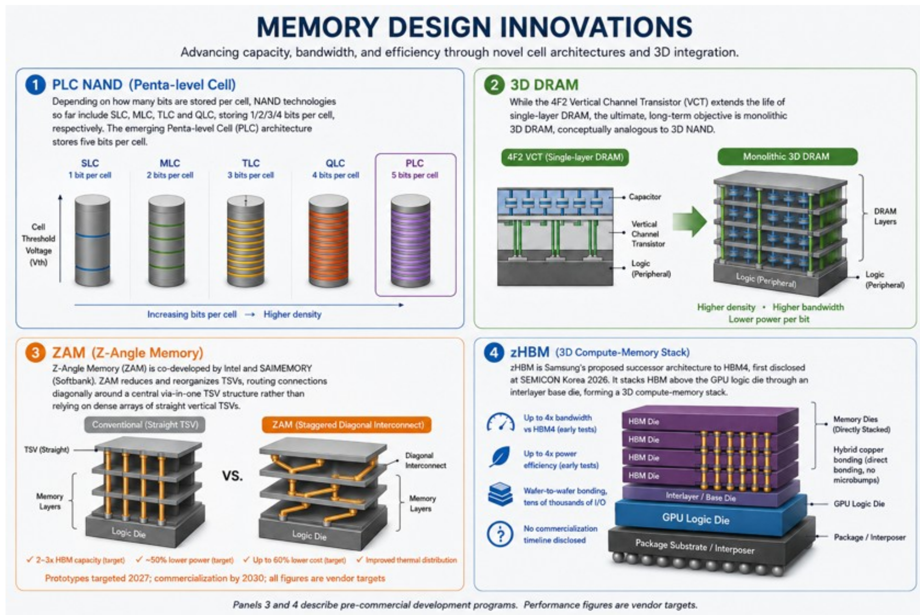
While $4F^2$ vertical-channel transistor structures can extend the life of single-layer DRAM, the longer-term architectural objective is monolithic 3D DRAM, conceptually analogous to the transition from planar NAND to 3D NAND. The design goal is to stack DRAM cells vertically within a single integrated manufacturing flow, thereby multiplying the number of bits per unit of die area and extending DRAM scaling beyond the limits of planar cell layouts.

1. Vertical-channel approaches. One candidate route is the vertical-channel approach, which extends the role of vertical transistor structures beyond single-layer DRAM. This route is intended to improve density while preserving a recognizable DRAM operating model.

2. Capacitor-inclusive stacked structures. A more difficult route is fully stacked, capacitor-inclusive 3D DRAM. The bottleneck is the integration of a sufficiently large capacitor into each stacked DRAM cell without making the stack too thick, costly, or yield-challenged.

Fully stacked, capacitor-inclusive 3D DRAM remains a longer-term technology. Early vertical-channel models and proof-of-concept silicon are appearing in R&D, but broad mass production is more likely a next-decade event. Manufacturability will depend on whether the industry can combine stacked channel formation, capacitor integration, high-aspect-ratio etching, and atomic-layer deposition at acceptable cost and yield.

Exhibit 56: Memory design innovation



Source: Company data.

Appendix 3: Assessing China DRAM Demand/Supply (Local Sufficiency)

Based on our supply chain checks, Alibaba has begun purchasing CXMT server DRAM. ByteDance and Tencent also appear to be procuring from CXMT.

CXMT's LPDDR products are particularly well aligned with domestic mobile substitution because Chinese smartphone brands can qualify the product within local platform and module ecosystems. Smartphone and mobile OEMs include Xiaomi, Transsion, Honor, Oppo, Vivo.

PC and system customers include Lenovo, and other domestic PC and module manufacturers; Dell and HP use CXMT in selecte China and ASEAN models.

CXMT also sells through distributors and module partners. Its top-five customer metric shows concentration of approximately 68% in 2025, which creates account risk even though no individual customer represents more than half of revenue.

What is the realistic addressable market?

The TAM should be separated into four pools.

Pool 1: Chinese domestic consumer and enterprise demand

We view this as the highest-probability TAM, i.e., CXMT's most accessible market:

- Smartphones
- PCs and notebooks
- Other consumer electronics
- Networking and industrial equipment
- Servers
- Government and state-linked infrastructure
- AI accelerators and CPUs

CXMT can address most of this market over time, subject to technical qualification and product availability.

Pool 2: Chinese operations of multinational OEMs

We view China-specific PC models or other consumer devices as the next group of penetrable markets. Customers can compartmentalize regulatory exposure by limiting CXMT content to particular geographies and models.

Pool 3: Non-US or geopolitically neutral export markets

ASEAN, emerging-market and selected regional brands may be accessible, particularly where price and supply availability outweigh US policy risk.

Pool 4: Global premium OEM, hyperscaler and AI markets

This is the least certain pool; barriers include:

- Potential US procurement restrictions
- OEM concern about future supply interruption
- IP and reputational risk
- Qualification timelines
- Lack of leading HBM products

Global shortages could temporarily widen this TAM, but we believe that investors should not capitalize the full global DRAM market as freely addressable.

CXMT's risk-adjusted commercial TAM will remain concentrated in China and selected non-US markets unless geopolitical conditions become more favorable for CXMT.

China domestic demand versus local supply

China is one of the largest consumers and manufacturing centers for memory because it hosts a substantial portion of global smartphone, PC, server, networking and electronics production. Historically, almost all advanced DRAM was imported.

Even with CXMT's rapid expansion, domestic supply remains below China's total effective demand. We forecast CXMT reaching approximately 15% of global bit shipments by 2030. China's share of global electronics manufacturing and semiconductor consumption is larger than that, implying room for domestic substitution even at that scale. According to our calculation, we think the sufficiency ratio reaches only ~35% in 2027e-2028e, a level at which we see no risk of oversupply.

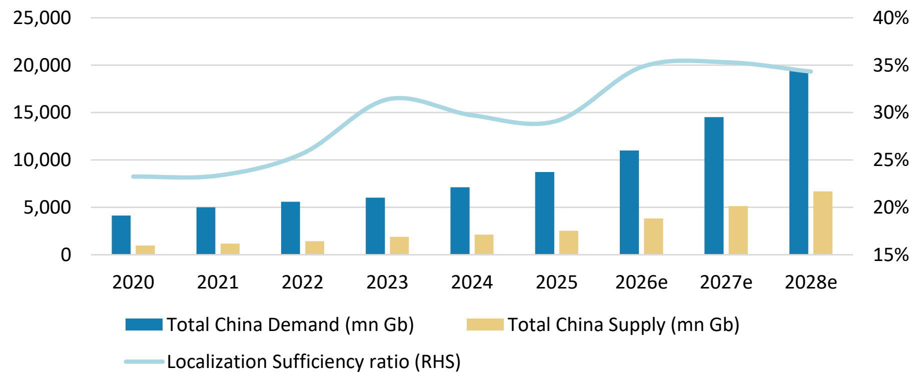
However, we also note that "China demand" is not fully captive. Multinational factories in China may be required by global product specifications to use Samsung, SK hynix or Micron.

Exhibit 57: Mapping China's supply and demand

	2020	2021	2022	2023	2024	2025	2026e	2027e	2028e
Total China Demand (mn Gb)	4,130	4,990	5,581	6,024	7,108	8,720	11,008	14,506	19,494
PCs	2,178	2,660	2,858	2,826	3,001	3,393	3,393	3,563	3,741
Servers	5,543	6,792	8,027	8,847	10,451	13,271	20,570	30,856	46,283
Smartphones	6,565	7,744	8,397	9,079	10,657	12,428	12,428	13,671	15,038
Other	2,235	2,762	3,043	3,344	4,325	5,789	7,642	9,934	12,914
Total China Supply (mn Gb)	960	1,165	1,433	1,891	2,112	2,539	3,831	5,121	6,690
Localization Sufficiency ratio (RHS)	**23%**	**23%**	**26%**	**31%**	**30%**	**29%**	**35%**	**35%**	**34%**

Source: TrendForce, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

Exhibit 58: We believe China's domestic supply is insufficient to support the strong demand



Source: TrendForce, company data, Morgan Stanley Research. E = Morgan Stanley Research estimates.

How much China demand could ultimately be served domestically?

A useful segmentation is:

High substitution potential: **50–80%** over time –

- Mainstream domestic PCs
- Domestic handset models
- Consumer electronics
- Modules and white-box systems
- Government and state-linked procurement
- Domestic cloud servers
- Industrial products with manageable qualification requirements

Moderate substitution potential: **25–50%** –

- Premium smartphones
- High-density enterprise servers
- Automotive and telecom infrastructure
- China-based models of multinational OEMs
- Selected AI accelerators using domestic HBM or high-capacity DDR

Low near-term substitution potential: below **20–30%** –

- Frontier HBM
- Globally standardized premium server platforms
- High-end automotive applications requiring long histories
- Products requiring leading global CPU/GPU vendor co-qualification
- Multinational products exposed to US government procurement or compliance restrictions

Aggregating these categories, CXMT and other domestic suppliers could plausibly serve a significant share of China's conventional DRAM demand. Further domestic market share upside, including leading HBM and premium international products, would require further progress in technology, qualification and packaging.

China's DRAM industry evolution

China's DRAM industry has progressed from effectively no commercially significant domestic production a decade ago to a meaningful position in mainstream memory, led by ChangXin Memory Technologies (CXMT). Founded in 2016, CXMT moved into volume production of domestically manufactured 8Gb DDR4 in 2019 and subsequently expanded into LPDDR4X, DDR5 and LPDDR5/5X. Its current portfolio addresses PCs, smartphones, tablets and servers, with published specifications increasingly comparable to mainstream international products.

The most important development has been scale. We estimate CXMT's monthly wafer capacity rose from approximately 40k wafers in 2020 to 70k in 2022, 120k in 2023, 150k by the end of 2024, and to 180k by the end of 2025. With CXMT's capacity expansion, China has moved from negligible DRAM supply to a position capable of materially affecting global legacy-memory pricing. We believe China's strongest competitive position is currently in mainstream DDR5 and LPDDR5 products sold into the domestic PC, handset, consumer-electronics and industrial ecosystems.

HBM development

In the previous section, we mentioned CXMT's progress in the HBM3E 12-Hi product. Other new entrants are also pursuing HBM development. For example, XMC (Wuhan Xinxin) is reportedly building HBM-related advanced-packaging capacity using 3D multiwafer stacking, with planned capacity of at least 3,000 12-inch wafers per month. Current activity appears to remain at the R&D and pilot-production stage, but capacity could increase over the coming years. Our supply-chain checks suggest that JHICC is also targeting longer-term participation in the HBM market. The company could expand its HBM capacity over the long term, initially starting with JBM2e

Exhibit 59: China's evolution of capacity, technology and market positioning

Period	Estimated CXMT capacity	Technology and products	Primary market position
2016–2018	Pre-volume production	Initial process development and fab construction	Technology acquisition and localization
2019–2020	~ 20-40kwpm	19nm-class 8Gb DDR4; early LPDDR4	Domestic consumer PCs and entry-level devices
2021–2022	~ 40-70kwpm	DDR4 and LPDDR4X; process and yield refinement	Import substitution in price-sensitive applications
2023	~ 120kwpm	17/18nm-class mainstream production; DDR5 development	Broader PC, handset and industrial exposure
2024	~ 160–200kwpm	High-volume DDR4/LPDDR4X; initial advanced products	Material influence on legacy DRAM pricing
2025 onward	~ 250–300kwpm	DDR5, LPDDR5/5X and early HBM development	Mainstream domestic supplier, with selective international qualification

Source: Company data, Morgan Stanley Research.

How has China's DRAM industry progressed over the past decade?

2015–2018: Policy-driven industry formation

China entered the previous decade as the world's largest consumer of semiconductors, but it remained heavily dependent on imported DRAM. Multiple domestic memory initiatives were launched as part of China's broader semiconductor self-sufficiency strategy, including CXMT, Fujian Jinhua and other smaller projects.

CXMT was established in Hefei in 2016. Unlike several Chinese semiconductor ventures that attempted to develop a technology platform from scratch, CXMT built its initial DRAM architecture partly around intellectual property originating from Germany's Qimonda. This provided a more defensible starting point and reduced some early intellectual-property risk relative to alternative technology-transfer routes.

The company also benefited from local-government financing, inexpensive land, infrastructure support and access to engineers recruited from established Asian memory manufacturers. Industry assessments have identified talent recruitment from Samsung and SK hynix operations in China as an important contributor to the rapid development of China's memory capabilities.

2019–2021: Commercial validation at 19nm-class DDR4

CXMT announced volume production of an 8Gb DDR4 device in 2019, generally associated with its 10G1, or approximately 19nm-class, manufacturing process. This represented China's first meaningful domestically manufactured DRAM product at commercial scale. By 2020, estimated capacity had reached approximately 40,000 300mm wafers per month, with the company producing DDR4 and subsequently adding LPDDR4 products.

At this stage, CXMT's positioning was primarily consumer and entry-level PC DRAM, lower-end to mid-range smartphone memory, domestic module manufacturers, and industrial and embedded applications with less demanding qualification requirements.

CXMT's technology remained several generations behind the leading global producers, but the products were sufficiently competitive for portions of the Chinese domestic market. The key achievement was therefore not technology leadership but the establishment of repeatable manufacturing, yield improvement and a domestic customer base.

2022–2024: Rapid capacity expansion and pressure on legacy DRAM pricing

From 2022, CXMT accelerated its capacity build-out. We estimate its monthly wafer capacity expanded from roughly 70k wafers in 2022 to around 120k in 2023.

During the earlier phase of this expansion, incremental output was concentrated largely in DDR4 and LPDDR4X. This capacity growth came amid weak global consumer-electronics demand and industry-wide inventory corrections, adding supply to legacy DRAM segments that were already experiencing significant pricing pressure.

Industry research, including TrendForce, linked CXMT's expansion in older-generation DRAM to weaker pricing in DDR4 and LPDDR4X. The effect remained visible into early 2025, when continued softness in DDR4 spot prices was partly attributed to abundant supply from CXMT. By late 2024, however, CXMT had begun shifting its product mix toward DDR5, marking the start of a transition away from legacy DDR4 production.

2024-2025: Transition into mainstream DDR5 and LPDDR5X

CXMT began transitioning its mainstream DRAM portfolio toward newer-generation products in late 2024, when DDR5 entered mass production, followed by LPDDR5X in 2025. CXMT's 8,533Mbps and 9,600Mbps LPDDR5X products entered mass production in May 2025, while its 10,667Mbps variant was in customer sampling by October 2025.

In November 2025, CXMT publicly showcased a broader DDR5 portfolio with speeds of up to 8,000Mbps and die densities of up to 24Gb, together with multiple module formats including server-oriented RDIMM and MRDIMM products. Its LPDDR5X portfolio included 12Gb and 16Gb dies and extended to a published maximum data rate of 10,667Mbps.

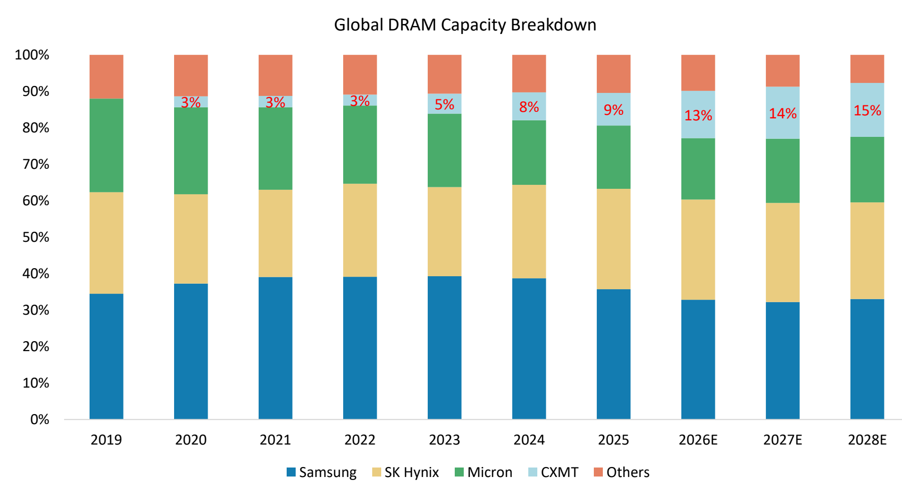
Where CXMT fits within the broader China DRAM story

CXMT is China's most advanced DRAM producer and the clearest vehicle for China's push toward memory self-sufficiency. There are other Chinese players developing DRAM and HBM, but CXMT combines proven high-volume manufacturing, established relationships with domestic cloud, PC, and smartphone customers, and a comparatively mature process and R&D organization.

China's DRAM industry has moved through three stages: establishing volume production with CXMT's ~19nm-class 8Gb DDR4 in 2019, scaling DDR4 and LPDDR4X, and entering the mainstream market with DDR5 and LPDDR5X for PCs, mobile devices, and select server applications.

The product gap with global leaders has narrowed, but CXMT still trails in cost per bit, process maturity, qualification breadth, manufacturing scale, and advanced AI-memory integration.

Exhibit 60: CXMT's global capacity market share to extend from 9% in 2025 to 15% in 2028e



Source: TrendForce, Morgan Stanley Research. E = Morgan Stanley Research estimates.

This report references U.S. Executive Order 14032 and/or entities or securities that are designated thereunder. U.S. persons may be prohibited from buying certain securities of entities named in this report. Readers are solely responsible for ensuring that their investment activities are carried out in compliance with applicable laws.

This report references export controls and/or entities that may be subject to export control restrictions. Readers are solely responsible for ensuring that their investment or trade activities are carried out in compliance with applicable laws.

This report references U.S. Executive Order 14105 and/or entities that may be in scope of such order. U.S. persons may be prohibited from engaging in certain transactions or otherwise require certain other transactions be notified to the U.S. Department of Treasury. Readers are solely responsible for ensuring that their investment or trade activities are carried out in compliance with applicable laws.

Risk Reward Reference links

1. View explanation of Options Probabilities methodology - Options_Probabilities_Exhibit_Link.pdf

2. View descriptions of Risk Rewards Themes - RR_Themes_Exhibit_Link.pdf

3. View explanation of regional hierarchies - GEG_Exhibit_Link.pdf

4. View explanation of Theme/Exposure methodology - ESG_Sustainable_Solutions_External_Link.pdf

5. View explanation of HERS methodology - ESG_HERS_External_Link.pdf

Disclosure Section

The information and opinions in Morgan Stanley Research were prepared or are disseminated by Morgan Stanley Asia Limited (which accepts the responsibility for its contents) and/or Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research), and/or Morgan Stanley Taiwan Limited and/or Morgan Stanley & Co International plc, Seoul Branch, and/or Morgan Stanley Australia Limited (A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents), and/or Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents), and/or Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com which accepts the responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research, and their affiliates (collectively, "Morgan Stanley"). Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts.

For important disclosures, stock price charts and equity rating histories regarding companies that are the subject of this report, please see the Morgan Stanley Research Disclosure Website at www.morganstanley.com/eqr/disclosures/webapp/generalresearch, or contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY, 10036 USA.

For valuation methodology and risks associated with any recommendation, rating or price target referenced in this research report, please contact the Client Support Team as follows: US/Canada +1 800 303-2495; Hong Kong +852 2848-5999; Latin America +1 718 754-5444 (U.S.); London +44 (0)20-7425-8169; Singapore +65 6834-6860; Sydney +61 (0)2-9770-1505; Tokyo +81 (0)3-6836-9000. Alternatively you may contact your investment representative or Morgan Stanley Research at 1585 Broadway, (Attention: Research Management), New York, NY 10036 USA.

Analyst Certification

The following analysts hereby certify that their views about the companies and their securities discussed in this report are accurately expressed and that they have not received and will not receive direct or indirect compensation in exchange for expressing specific recommendations or views in this report: Shane Brett; Charlie Chan; Daisy Dai, CFA; Cindy Huang; Ethan Jia; Shawn Kim; Duan Liu; Joseph Moore; Lee Simpson; Suzune Tamura, CFA; Tiffany Yeh; Daniel Yen, CFA; Kazuo Yoshikawa, CFA.

.

Global Research Conflict Management Policy

Morgan Stanley Research has been published in accordance with our conflict management policy, which is available at www.morganstanley.com/institutional/research/conflictpolicies. A Portuguese version of the policy can be found at www.morganstanley.com.br

Important Regulatory Disclosures on Subject Companies

As of July 31, 2026, Morgan Stanley beneficially owned 1% or more of a class of common equity securities of the following companies covered in Morgan Stanley Research: ACM Research Inc, Advanced Micro-Fabrication Equipment Inc, Advanced Wireless Semiconductor Co, Advantest, Alchip Technologies Ltd, AllRing Tech Co., AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMPT Ltd, Aspeed Technology, Cambricon Technology Corporation, DISCO, eMemory Technology Inc, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Hangzhou Silan Microelectronics Co. Ltd., Hua Hong Semiconductor Ltd, King Yuan Electronics Co Ltd, KOKUSAI ELECTRIC, Macronix International Co Ltd, MediaTek, Micron Technology Inc., Montage Technology Co Ltd, NAURA Technology Group Co Ltd, Nuvoton Technology Corporation, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Renesas Electronics, SG Micro Corp., Shanghai Fudan Microelectronics, Silergy Corp., Silicon Motion, SK hynix, StarPower Semiconductor Ltd, Teradyne Inc, Tokyo Electron, TSMC, UMC, Vanguard International Semiconductor, WIN Semiconductors Corp, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley managed or co-managed a public offering (or 144A offering) of securities of Alchip Technologies Ltd, Iluvatar CoreX Semiconductor Co., Ltd., KOKUSAI ELECTRIC, Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co, Samsung Electronics, SK hynix.

Within the last 12 months, Morgan Stanley has received compensation for investment banking services from ASMPT Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Micron Technology Inc., Montage Technology Co Ltd, Powerchip Semiconductor Manufacturing Co, Silicon Motion, SK hynix, Tokyo Electron.

In the next 3 months, Morgan Stanley expects to receive or intends to seek compensation for investment banking services from Advanced Micro-Fabrication Equipment Inc, Advantest, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, CXMT Corp, DISCO, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, KOKUSAI ELECTRIC, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Micron Technology Inc., Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, Renesas Electronics, Samsung Electronics, SCREEN Holdings, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, SK hynix, Teradyne Inc, Tokyo Electron, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has received compensation for products and services other than investment banking services from ASE Technology Holding Co. Ltd., King Yuan Electronics Co Ltd, MediaTek, Micron Technology Inc., Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Samsung Electronics, Silicon Motion, SMIC, Tokyo Electron, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has provided or is providing investment banking services to, or has an investment banking client relationship with, the following company: Advanced Micro-Fabrication Equipment Inc, Advantest, Alchip Technologies Ltd, AP Memory Technology Corp, ASE Technology Holding Co. Ltd., ASMedia Technology Inc, ASMPT Ltd, Aspeed Technology, CXMT Corp, Espressif Systems, GigaDevice Semiconductor Beijing Inc, GlobalWafers Co Ltd, Gudeng Precision, Himax Technologies Inc, Hua Hong Semiconductor Ltd, Iluvatar CoreX Semiconductor Co., Ltd., Innoscience, King Yuan Electronics Co Ltd, KOKUSAI ELECTRIC, Macronix International Co Ltd, MediaTek, MetaX Integrated Circuits, Micron Technology Inc., Montage Technology Co Ltd, Novatek, Phison Electronics Corp, Powerchip Semiconductor Manufacturing Co, Realtek Semiconductor, Renesas Electronics, Samsung Electronics, SCREEN Holdings, SG Micro Corp., Shenzhen Longsys Electronics Co Ltd, SICC Co Ltd, Silergy Corp., Silicon Motion, SK hynix, Teradyne Inc, Tokyo Electron, TSMC, UMC, Universal Scientific Ind. (Shanghai), Vanguard International Semiconductor, Winbond Electronics Corp, WinWay Technology Co Ltd, WPG Holdings, WT Microelectronics Co. Ltd..

Within the last 12 months, Morgan Stanley has either provided or is providing non-investment banking, securities-related services to and/or in the past has entered into an agreement to provide services or has a client relationship with the following company: ASE Technology Holding Co. Ltd., Iluvatar CoreX Semiconductor Co., Ltd., King Yuan Electronics Co Ltd, MediaTek, Micron

Technology Inc., Montage Technology Co Ltd, Nanya Technology Corp., Novatek, Nuvoton Technology Corporation, Realtek Semiconductor, Renesas Electronics, Samsung Electronics, Silicon Motion, SK hynix, SMIC, Tokyo Electron, TSMC, UMC, Universal Scientific Ind. (Shanghai), Winbond Electronics Corp, WT Microelectronics Co. Ltd..

The equity research analysts or strategists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality of research, investor client feedback, stock picking, competitive factors, firm revenues and overall investment banking revenues. Equity Research analysts' or strategists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

Morgan Stanley and its affiliates do business that relates to companies/instruments covered in Morgan Stanley Research, including market making, providing liquidity, fund management, commercial banking, extension of credit, investment services and investment banking. Morgan Stanley sells to and buys from customers the securities/instruments of companies covered in Morgan Stanley Research on a principal basis. Morgan Stanley may have a position in the debt of the Company or instruments discussed in this report. Morgan Stanley trades or may trade as principal in the debt securities (or in related derivatives) that are the subject of the debt research report.

Certain disclosures listed above are also for compliance with applicable regulations in non-US jurisdictions.

STOCK RATINGS

Morgan Stanley uses a relative rating system using terms such as Overweight, Equal-weight, Not-Rated or Underweight (see definitions below). Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold and sell. Investors should carefully read the definitions of all ratings used in Morgan Stanley Research. In addition, since Morgan Stanley Research contains more complete information concerning the analyst's views, investors should carefully read Morgan Stanley Research, in its entirety, and not infer the contents from the rating alone. In any case, ratings (or research) should not be used or relied upon as investment advice. An investor's decision to buy or sell a stock should depend on individual circumstances (such as the investor's existing holdings) and other considerations.

Global Stock Ratings Distribution

(as of July 31, 2026)

The Stock Ratings described below apply to Morgan Stanley's Fundamental Equity Research and do not apply to Debt Research produced by the Firm.

For disclosure purposes only (in accordance with FINRA requirements), we include the category headings of Buy, Hold, and Sell alongside our ratings of Overweight, Equal-weight, Not-Rated and Underweight. Morgan Stanley does not assign ratings of Buy, Hold or Sell to the stocks we cover. Overweight, Equal-weight, Not-Rated and Underweight are not the equivalent of buy, hold, and sell but represent recommended relative weightings (see definitions below). To satisfy regulatory requirements, we correspond Overweight, our most positive stock rating, with a buy recommendation; we correspond Equal-weight and Not-Rated to hold and Underweight to sell recommendations, respectively.

	Coverage Universe		Investment Banking Clients (IBC)			Other Material Investment Services Clients (MISC)	
Stock Rating Category	Count	% of Total	Count	% of Total IBC	% of Rating Category	Count	% of Total Other MISC
Overweight/Buy	1529	42%	442	47%	29%	750	43%
Equal-weight/Hold	1582	43%	396	42%	25%	773	44%
Not-Rated/Hold	3	0%	1	0%	33%	1	0%
Underweight/Sell	560	15%	97	10%	17%	217	12%
Total	3,674		936			1741	

Data include common stock and ADRs currently assigned ratings. Investment Banking Clients are companies from whom Morgan Stanley received investment banking compensation in the last 12 months. Due to rounding off of decimals, the percentages provided in the "% of total" column may not add up to exactly 100 percent.

Analyst Stock Ratings

Overweight (O). The stock's total return is expected to exceed the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Equal-weight (E). The stock's total return is expected to be in line with the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Not-Rated (NR). Currently the analyst does not have adequate conviction about the stock's total return relative to the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Underweight (U). The stock's total return is expected to be below the average total return of the analyst's industry (or industry team's) coverage universe, on a risk-adjusted basis, over the next 12-18 months.

Unless otherwise specified, the time frame for price targets included in Morgan Stanley Research is 12 to 18 months.

Analyst Industry Views

Attractive (A): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be attractive vs. the relevant broad market benchmark, as indicated below.

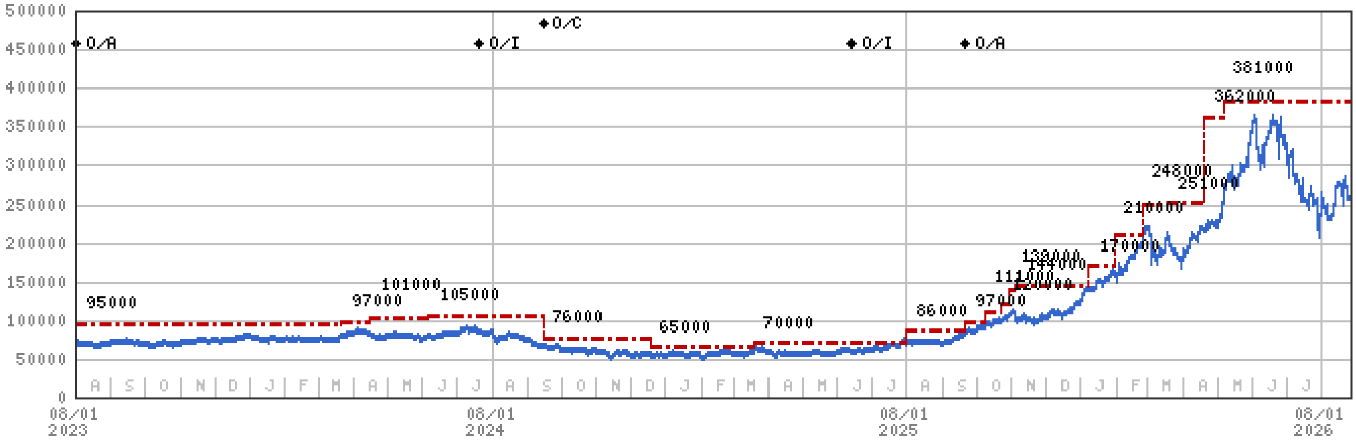
In-Line (I): The analyst expects the performance of his or her industry coverage universe over the next 12-18 months to be in line with the relevant broad market benchmark, as indicated below.

Cautious (C): The analyst views the performance of his or her industry coverage universe over the next 12-18 months with caution vs. the relevant broad market benchmark, as indicated below.

Benchmarks for each region are as follows: North America - S&P 500; Latin America - relevant MSCI country index or MSCI Latin America Index; Europe - MSCI Europe; Japan - TOPIX; Asia - relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Stock Price, Price Target and Rating History (See Rating Definitions)

Samsung Electronics (005930.KS) - As of 08/26/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : O/I; 8/12/21 : O/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : O/C; 6/13/25 : O/I; 9/21/25 : O/A

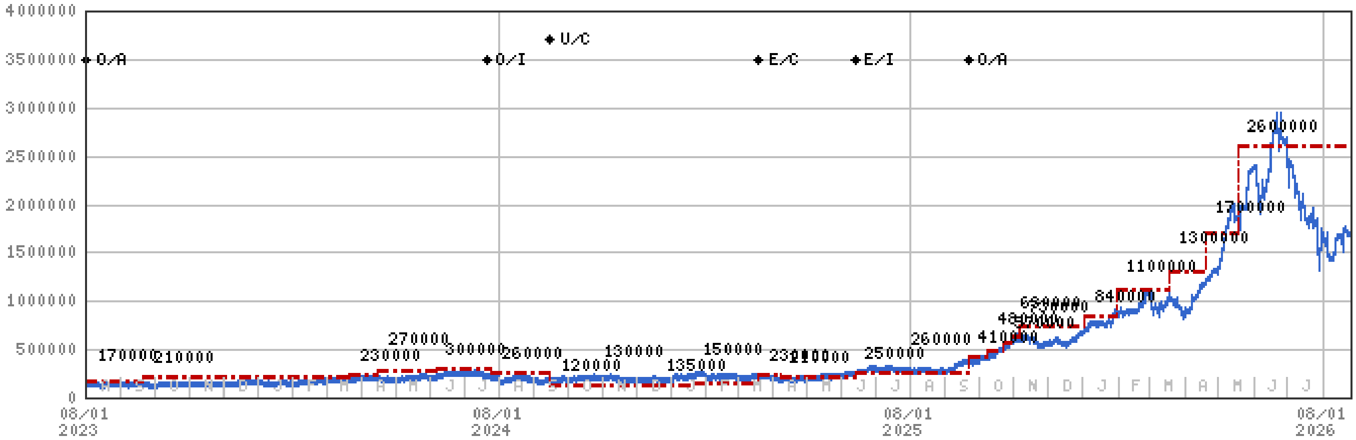
Price Target History: 6/8/21 : 98000; 8/12/21 : 89000; 9/15/21 : 95000; 12/3/21 : 97000; 3/18/22 : 95000; 4/28/22 : 85000; 6/10/22 : 80000; 7/5/22 : 75000; 7/22/22 : 70000; 9/17/22 : 68000; 3/21/23 : 70000; 5/30/23 : 90000; 7/7/23 : 95000; 3/22/24 : 97000; 4/16/24 : 101000; 6/6/24 : 105000; 9/15/24 : 76000; 12/18/24 : 65000; 3/19/25 : 70000; 8/1/25 : 86000; 9/21/25 : 97000; 10/8/25 : 111000; 10/23/25 : 120000; 10/30/25 : 139000; 11/5/25 : 144000; 1/8/26 : 170000; 1/30/26 : 210000; 2/24/26 : 248000; 3/18/26 : 251000; 4/19/26 : 362000; 5/6/26 : 381000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target -- No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

SK hynix (000660.KS) - As of 08/26/26 GMT in KRW
Industry : S. Korea Technology



Stock Rating History: 8/1/21 : O/I; 8/12/21 : U/C; 12/3/21 : E/C; 2/11/22 : O/C; 7/22/22 : E/C; 10/4/22 : O/A; 7/21/24 : O/I; 9/15/24 : U/C; 3/19/25 : E/C; 6/13/25 : E/I; 9/21/25 : O/A

Price Target History: 6/8/21 : 156000; 8/12/21 : 80000; 9/15/21 : 88000; 12/3/21 : 110000; 12/23/21 : 125000; 1/24/22 : 130000; 1/28/22 : 136000; 2/11/22 : 155000; 3/18/22 : 150000; 4/27/22 : 130000; 6/10/22 : 120000; 7/5/22 : 110000; 7/22/22 : 105000; 10/4/22 : 130000; 12/7/22 : 120000; 3/21/23 : 110000; 5/30/23 : 140000; 7/7/23 : 170000; 9/21/23 : 210000; 3/22/24 : 230000; 4/16/24 : 270000; 6/6/24 : 300000; 7/25/24 : 260000; 9/15/24 : 120000; 10/24/24 : 130000; 12/18/24 : 135000; 1/20/25 : 150000; 3/19/25 : 230000; 4/7/25 : 210000; 6/13/25 : 250000; 7/24/25 : 260000; 9/21/25 : 410000; 10/8/25 : 480000; 10/23/25 : 570000; 10/29/25 : 630000; 11/5/25 : 730000; 1/2/26 : 840000; 1/30/26 : 1100000; 3/18/26 : 1300000; 4/19/26 : 1700000; 5/18/26 : 2600000

Source: Morgan Stanley Research Date Format : MM/DD/YY Price Target No Price Target Assigned (NA)
Stock Price (Not Covered by Current Analyst) Stock Price (Covered by Current Analyst)
Stock and Industry Ratings (abbreviations below) appear as ◆ Stock Rating/Industry View
Stock Ratings: Overweight (O) Equal-weight (E) Underweight (U) Not-Rated (NR) No Rating Available (NA)
Industry View: Attractive (A) In-line (I) Cautious (C) No Rating (NR)

Effective January 13, 2014, the stocks covered by Morgan Stanley Asia Pacific will be rated relative to the analyst's industry (or industry team's) coverage.

Effective January 13, 2014, the industry view benchmarks for Morgan Stanley Asia Pacific are as follows: relevant MSCI country index or MSCI sub-regional index or MSCI AC Asia Pacific ex Japan Index.

Important Disclosures for Morgan Stanley Smith Barney LLC Customers

Important disclosures regarding any material conflict of interest that can reasonably be expected to have influenced Morgan Stanley Smith Barney LLC's choice of a third-party research provider or the subject company of a third-party research report, are available on the Morgan Stanley Wealth Management disclosure website at www.morganstanley.com/online/researchdisclosures. For Morgan Stanley specific disclosures, you may refer to https://www.morganstanley.com/eqr/disclosures/webapp/generalresearch.

Each Morgan Stanley research report is reviewed and approved on behalf of Morgan Stanley Smith Barney LLC. This review and approval is conducted by the same person who reviews the research report on behalf of Morgan Stanley. This could create a conflict of interest.

Other Important Disclosures

Morgan Stanley Research policy is to update research reports as and when the Research Analyst and Research Management deem appropriate, based on developments with the issuer, the sector, or the market that may have a material impact on the research views or opinions stated therein. In addition, certain Research publications are intended to be updated on a regular periodic basis (weekly/monthly/quarterly/annual) and will ordinarily be updated with that frequency, unless the Research Analyst and Research Management determine that a different publication schedule is appropriate based on current conditions.

Morgan Stanley is not acting as a municipal advisor and the opinions or views contained herein are not intended to be, and do not constitute, advice within the meaning of Section 975 of the Dodd-Frank Wall Street Reform and Consumer Protection Act.

Morgan Stanley produces an equity research product called a "Tactical Idea." Views contained in a "Tactical Idea" on a particular stock may be contrary to the recommendations or views expressed in research on the same stock. This may be the result of differing time horizons, methodologies, market events, or other factors. For all research available on a particular stock, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Morgan Stanley Research is provided to our clients through our proprietary research portal on Matrix and also distributed electronically by Morgan Stanley to clients. Certain, but not all, Morgan Stanley Research products are also made available to clients through third-party vendors or redistributed to clients through alternate electronic means as a convenience. For access to all available Morgan Stanley Research, please contact your sales representative or go to Matrix at http://www.morganstanley.com/matrix.

Any access and/or use of Morgan Stanley Research is subject to Morgan Stanley's Terms of Use (http://www.morganstanley.com/terms.html). By accessing and/or using Morgan Stanley Research, you are indicating that you have read and agree to be bound by our Terms of Use (http://www.morganstanley.com/terms.html). In addition you consent to Morgan Stanley processing your personal data and using cookies in accordance with our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html), including for the purposes of setting your preferences and to collect readership data so that we can deliver better and more personalized service and products to you. To find out more information about how Morgan Stanley processes personal data, how we use cookies and how to reject cookies see our Privacy Policy and our Global Cookies Policy (http://www.morganstanley.com/privacy_pledge.html). Please use the provided link to review the Terms and Conditions and Most Important Terms and Conditions for Morgan Stanley India Company Private Limited (https://www.morganstanley.com/assets/pdfs/about-us-global-offices/india/Terms_and_conditions.pdf) and the following link to review the audit report (https://ny.matrix.ms.com/eqr/research/webapp/researchdocs/

MSICPL_Morgan_Stanley_Research_Audit_Report.pdf).

If you do not agree to our Terms of Use and/or if you do not wish to provide your consent to Morgan Stanley processing your personal data or using cookies please do not access our research. Morgan Stanley Research does not provide individually tailored investment advice. Morgan Stanley Research has been prepared without regard to the circumstances and objectives of those who receive it. Morgan Stanley recommends that investors independently evaluate particular investments and strategies, and encourages investors to seek the advice of a financial adviser. The appropriateness of an investment or strategy will depend on an investor's circumstances and objectives. The securities, instruments, or strategies discussed in Morgan Stanley Research may not be suitable for all investors, and certain investors may not be eligible to purchase or participate in some or all of them. Morgan Stanley Research is not an offer to buy or sell or the solicitation of an offer to buy or sell any security/instrument or to participate in any particular trading strategy. The value of and income from your investments may vary because of changes in interest rates, foreign exchange rates, default rates, prepayment rates, securities/instruments prices, market indexes, operational or financial conditions of companies or other factors. There may be time limitations on the exercise of options or other rights in securities/instruments transactions. Past performance is not necessarily a guide to future performance. Estimates of future performance are based on assumptions that may not be realized. If provided, and unless otherwise stated, the closing price on the cover page is that of the primary exchange for the subject company's securities/instruments.

The fixed income research analysts, strategists or economists principally responsible for the preparation of Morgan Stanley Research have received compensation based upon various factors, including quality, accuracy and value of research, firm profitability or revenues (which include fixed income trading and capital markets profitability or revenues), client feedback and competitive factors. Fixed Income Research analysts', strategists' or economists' compensation is not linked to investment banking or capital markets transactions performed by Morgan Stanley or the profitability or revenues of particular trading desks.

The "Important Regulatory Disclosures on Subject Companies" section in Morgan Stanley Research lists all companies mentioned where Morgan Stanley owns 1% or more of a class of common equity securities of the companies. For all other companies mentioned in Morgan Stanley Research, Morgan Stanley may have an investment of less than 1% in securities/instruments or derivatives of securities/instruments of companies and may trade them in ways different from those discussed in Morgan Stanley Research. Employees of Morgan Stanley not involved in the preparation of Morgan Stanley Research may have investments in securities/instruments or derivatives of securities/instruments of companies mentioned and may trade them in ways different from those discussed in Morgan Stanley Research. Derivatives may be issued by Morgan Stanley or associated persons.

With the exception of information regarding Morgan Stanley, Morgan Stanley Research is based on public information. Morgan Stanley makes every effort to use reliable, comprehensive information, but we make no representation that it is accurate or complete. We have no obligation to tell you when opinions or information in Morgan Stanley Research change apart from when we intend to discontinue equity research coverage of a subject company. Facts and views presented in Morgan Stanley Research have not been reviewed by, and may not reflect information known to, professionals in other Morgan Stanley business areas, including investment banking personnel.

Morgan Stanley Research personnel may participate in company events such as site visits and are generally prohibited from accepting payment by the company of associated expenses unless pre-approved by authorized members of Research management.

Morgan Stanley may make investment decisions that are inconsistent with the recommendations or views in this report.

To our readers based in Taiwan or trading in Taiwan securities/instruments: Information on securities/instruments that trade in Taiwan is distributed by Morgan Stanley Taiwan Limited ("MSTL"). Such information is for your reference only. The reader should independently evaluate the investment risks and is solely responsible for their investment decisions. Morgan Stanley Research may not be distributed to the public media or quoted or used by the public media without the express written consent of Morgan Stanley. Any non-customer reader within the scope of Article 7-1 of the Taiwan Stock Exchange Recommendation Regulations accessing and/or receiving Morgan Stanley Research is not permitted to provide Morgan Stanley Research to any third party (including but not limited to related parties, affiliated companies and any other third parties) or engage in any activities regarding Morgan Stanley Research which may create or give the appearance of creating a conflict of interest. Information on securities/instruments that do not trade in Taiwan is for informational purposes only and is not to be construed as a recommendation or a solicitation to trade in such securities/instruments. MSTL may not execute transactions for clients in these securities/instruments.

Certain information in Morgan Stanley Research was sourced by employees of the Shanghai Representative Office of Morgan Stanley Asia Limited for the use of Morgan Stanley Asia Limited. Morgan Stanley is not incorporated under PRC law and the research in relation to this report is conducted outside the PRC. Morgan Stanley Research does not constitute an offer to sell or the solicitation of an offer to buy any securities in the PRC. PRC investors shall have the relevant qualifications to invest in such securities and shall be responsible for obtaining all relevant approvals, licenses, verifications and/or registrations from the relevant governmental authorities themselves. Neither this report nor any part of it is intended as, or shall constitute, provision of any consultancy or advisory service of securities investment as defined under PRC law. Such information is provided for your reference only.

Morgan Stanley Research is disseminated in Brazil by Morgan Stanley C.T.V.M. S.A. located at Av. Brigadeiro Faria Lima, 3600, 6th floor, São Paulo - SP, Brazil; and is regulated by the Comissão de Valores Mobiliários; in Mexico by Morgan Stanley México, Casa de Bolsa, S.A. de C.V which is regulated by Comision Nacional Bancaria y de Valores. Paseo de los Tamarindos 90, Torre 1, Col. Bosques de las Lomas Floor 29, 05120 Mexico City; in Japan by Morgan Stanley MUFG Securities Co., Ltd. and, for Commodities related research reports only, Morgan Stanley Capital Group Japan Co., Ltd; in Hong Kong by Morgan Stanley Asia Limited (which accepts responsibility for its contents) and by Morgan Stanley Bank Asia Limited; in Singapore by Morgan Stanley Asia (Singapore) Pte. (Registration number 199206298Z) and/or Morgan Stanley Asia (Singapore) Securities Pte Ltd (Registration number 200008434H), regulated by the Monetary Authority of Singapore (which accepts legal responsibility for its contents and should be contacted with respect to any matters arising from, or in connection with, Morgan Stanley Research) and by Morgan Stanley Bank Asia Limited, Singapore Branch (Registration number T14FC0118J); in Australia to "wholesale clients" within the meaning of the Australian Corporations Act by Morgan Stanley Australia Limited A.B.N. 67 003 734 576, holder of Australian financial services license No. 233742, which accepts responsibility for its contents; in Australia to "wholesale clients" and "retail clients" within the meaning of the Australian Corporations Act by Morgan Stanley Wealth Management Australia Pty Ltd (A.B.N. 19 009 145 555, holder of Australian financial services license No. 240813, which accepts responsibility for its contents; in Korea by Morgan Stanley & Co International plc, Seoul Branch; in India by Morgan Stanley India Company Private Limited having Corporate Identification No (CIN) U22990MH1998PTC115305, regulated by the Securities and Exchange Board of India ("SEBI") and holder of licenses as a Research Analyst (SEBI Registration No. INH000001105); Stock Broker (SEBI Stock Broker Registration No. INZ000244438), Merchant Banker (SEBI Registration No. INM000011203), and depository participant with National Securities Depository Limited (SEBI Registration No. IN-DP-NSDL-567-2021) having registered office at Altimus, Level 39 & 40, Pandurang Budhkar Marg, Worli, Mumbai 400018, India; Telephone no. +91-22-61181000; Compliance Officer Details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: tejarshi.hardas@morganstanley.com; Grievance officer details: Mr. Tejarshi Hardas, Tel. No.: +91-22-61181000 or Email: msic-compliance@morganstanley.com. Morgan Stanley India Company Private Limited (MSICPL) may use AI tools in providing research services. All recommendations contained herein are made by the duly qualified research analysts; in Canada by Morgan Stanley Canada Limited; in Germany and the European Economic Area where required by Morgan Stanley Europe S.E., authorised and regulated by Bundesanstalt fuer Finanzdienstleistungsaufsicht (BaFin) under the reference number 149169; in the US by Morgan Stanley & Co. LLC, which accepts responsibility for its contents. Morgan Stanley & Co. International plc, authorized by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority, disseminates in the UK research that it has prepared, and research which has been prepared by any of its affiliates, only to persons who (i) are investment professionals falling within Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Order"); (ii) are persons who are high net worth entities falling within Article 49(2)(a) to (d) of the Order; or (iii) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of section 21 of the Financial Services and Markets Act 2000, as amended) may otherwise lawfully be communicated or caused to be communicated. RMB Morgan Stanley Proprietary Limited is a member of the JSE Limited and A2X (Pty) Ltd. RMB Morgan Stanley Proprietary Limited is a joint venture owned equally by Morgan Stanley International Holdings Inc. and RMB Investment Advisory (Proprietary) Limited, which is wholly owned by FirstRand Limited. The information in Morgan Stanley Research is being disseminated by Morgan Stanley Saudi Arabia, regulated by the Capital Market Authority in the Kingdom of Saudi Arabia, and is directed at Sophisticated investors only.

Morgan Stanley Hong Kong Securities Limited is the liquidity provider/market maker for securities of ASMPT Ltd, GigaDevice Semiconductor Beijing Inc, Hua Hong Semiconductor Ltd, Montage Technology Co Ltd listed on the Stock Exchange of Hong Kong Limited. An updated list can be found on HKEx website: http://www.hkex.com.hk.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (DIFC Branch), regulated by the Dubai Financial Services Authority (the DFSA) or by Morgan Stanley & Co. International plc (ADGM Branch), regulated by the Financial Services Regulatory Authority Abu Dhabi (the FSRA), and is directed at Professional Clients only, as defined by the DFSA or the FSRA, respectively. The financial products or financial services to which this research relates will only be made available to a customer who we are satisfied meets the regulatory criteria of a Professional Client. A distribution of the different MS Research ratings or recommendations, in percentage terms for Investments in each sector covered, is available upon request from your sales representative.

The information in Morgan Stanley Research is being communicated by Morgan Stanley & Co. International plc (QFC Branch), regulated by the Qatar Financial Centre Regulatory Authority (the QFCRA), and is directed at business customers and market counterparties only and is not intended for Retail Customers as defined by the QFCRA.

As required by the Capital Markets Board of Turkey, investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Investment advisory service is provided exclusively to persons based on their risk and income preferences by the authorized firms. Comments and recommendations stated here are general in nature. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations.

The following companies do business in countries which are generally subject to comprehensive sanctions programs administered or enforced by the U.S. Department of the Treasury's Office of Foreign Assets Control ("OFAC") and by other countries and multi-national bodies: Samsung Electronics.

The trademarks and service marks contained in Morgan Stanley Research are the property of their respective owners. Third-party data providers make no warranties or representations relating to the accuracy, completeness, or timeliness of the data they provide and shall not have liability for any damages relating to such data. The Global Industry Classification Standard (GICS) was developed by and is the exclusive property of MSCI and S&P.

Morgan Stanley Research, or any portion thereof may not be reprinted, sold or redistributed without the written consent of Morgan Stanley.

Indicators and trackers referenced in Morgan Stanley Research may not be used as, or treated as, a benchmark under Regulation EU 2016/1011, or any other similar framework.

The issuers and/or fixed income products recommended or discussed in certain fixed income research reports may not be continuously followed. Accordingly, investors should regard those fixed income research reports as providing stand-alone analysis and should not expect continuing analysis or additional reports relating to such issuers and/or individual fixed income products.

Morgan Stanley may hold, from time to time, material financial and commercial interests regarding the company subject to the Research report.

Registration granted by SEBI and certification from the National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Investment in securities market are subject to market risks. Read all the related documents carefully before investing.

INDUSTRY COVERAGE: Greater China Technology Semiconductors

COMPANY (TICKER)	RATING (AS OF)	PRICE* (08/25/2026)
Charlie Chan		
ACM Research Inc (ACMR.O)	O (03/07/2023)	US$79.77
Advanced Micro-Fabrication Equipment Inc (688012.SS)	O (11/06/2023)	Rmb354.00
Advanced Wireless Semiconductor Co (8086.TWO)	U (07/14/2025)	NT$115.50
Alchip Technologies Ltd (3661.TW)	O (05/14/2021)	NT$3,960.00
ASE Technology Holding Co. Ltd. (3711.TW)	O (09/15/2024)	NT$592.00
Cambricon Technology Corporation (688256.SS)	O (04/27/2026)	Rmb1,010.48
CXMT Corp (688825.SS)	O (08/26/2026)	Rmb56.50
Global Unichip Corp (3443.TW)	O (06/24/2026)	NT$6,125.00
GlobalWafers Co Ltd (6488.TWO)	E (05/19/2026)	NT$980.00
Gudeng Precision (3680.TWO)	O (11/25/2025)	NT$461.50
Hua Hong Semiconductor Ltd (1347.HK)	E (03/12/2026)	HK$110.40
Iluvatar CoreX Semiconductor Co., Ltd. (9903.HK)	O (04/27/2026)	HK$370.00
King Yuan Electronics Co Ltd (2449.TW)	O (03/03/2023)	NT$244.00
Maxscend Microelectronics Co Ltd (300782.SZ)	U (01/11/2021)	Rmb75.85
MediaTek (2454.TW)	O (11/28/2025)	NT$3,945.00
MetaX Integrated Circuits (688802.SS)	E (04/27/2026)	Rmb655.25
Nanya Technology Corp. (2408.TW)	O (05/28/2026)	NT$517.00
NAURA Technology Group Co Ltd (002371.SZ)	O (11/06/2023)	Rmb700.40
OmniVision Integrated Circuits Group Inc (603501.SS)	E (11/17/2025)	Rmb85.55
Phison Electronics Corp (8299.TWO)	E (02/25/2026)	NT$2,085.00
SG Micro Corp. (300661.SZ)	E (11/03/2025)	Rmb114.86
Silergy Corp. (6415.TW)	U (05/19/2026)	NT$469.00
SMIC (0981.HK)	O (10/21/2025)	HK$67.85

TSMC (2330.TW)	O (02/07/2022)	NT$2,415.00
UMC (2303.TW)	O (05/19/2026)	NT$123.50
Vanguard International Semiconductor (5347.TWO)	E (01/14/2026)	NT$150.50
WIN Semiconductors Corp (3105.TWO)	U (07/14/2025)	NT$379.00
Daisy Dai, CFA		
ASMPT Ltd (0522.HK)	O (07/24/2025)	HK$162.50
China Resources Microelectronics Limited (688396.SS)	U (03/02/2026)	Rmb58.64
Empyrean Technology Co Ltd (301269.SZ)	E (01/17/2025)	Rmb92.40
Hangzhou Silan Microelectronics Co. Ltd. (600460.SS)	U (08/25/2025)	Rmb34.32
Hygon Information Technology Co., Ltd. (688041.SS)	O (07/03/2026)	Rmb232.06
Innoscience (2577.HK)	E (10/13/2025)	HK$44.20
JCET Group Co Ltd (600584.SS)	E (01/16/2026)	Rmb74.05
Shanghai Fudan Microelectronics (1385.HK)	O (03/07/2025)	HK$25.18
SICC Co Ltd (688234.SS)	O (03/20/2026)	Rmb106.41
StarPower Semiconductor Ltd (603290.SS)	E (05/14/2026)	Rmb92.99
Unigroup Guoxin Microelectronics Co Ltd (002049.SZ)	U (01/10/2023)	Rmb61.46
Universal Scientific Ind. (Shanghai) (601231.SS)	O (11/05/2025)	Rmb26.45
Yangjie Technology (300373.SZ)	O (06/10/2022)	Rmb93.25
Daniel Yen, CFA		
AP Memory Technology Corp (6531.TW)	O (07/11/2025)	NT$916.00
ASMedia Technology Inc (5269.TW)	U (10/03/2025)	NT$1,455.00
Aspeed Technology (5274.TWO)	O (06/09/2025)	NT$15,400.00
Egis Technology Inc (6462.TWO)	E (01/28/2026)	NT$99.40
Elan Microelectronics Corp (2458.TW)	O (10/03/2025)	NT$139.50
eMemory Technology Inc (3529.TWO)	O (08/18/2026)	NT$2,195.00
Espressif Systems (688018.SS)	O (05/15/2023)	Rmb109.00
GigaDevice Semiconductor Beijing Inc (603986.SS)	O (05/15/2025)	Rmb395.51
Macronix International Co Ltd (2337.TW)	O (09/18/2025)	NT$125.50
Montage Technology Co Ltd (6809.HK)	O (03/18/2026)	HK$262.40
Montage Technology Co Ltd (688008.SS)	O (03/18/2026)	Rmb194.42
Novatek (3034.TW)	U (02/04/2026)	NT$557.00
Nuvoton Technology Corporation (4919.TW)	U (11/10/2025)	NT$121.00
Parade Technologies Ltd (4966.TWO)	O (05/27/2026)	NT$556.00
Powerchip Semiconductor Manufacturing Co (6770.TW)	O (10/27/2025)	NT$70.20
Realtek Semiconductor (2379.TW)	E (01/30/2026)	NT$734.00
Shenzhen Goodix Technology Co Ltd (603160.SS)	U (07/14/2025)	Rmb48.02
Winbond Electronics Corp (2344.TW)	O (05/28/2026)	NT$181.50
WPG Holdings (3702.TW)	O (03/16/2026)	NT$108.00
WT Microelectronics Co. Ltd. (3036.TW)	O (01/27/2026)	NT$196.50
Duan Liu		
Dosilicon Co Ltd (688110.SS)	U (09/06/2024)	Rmb113.97
Shenzhen Longsys Electronics Co Ltd (301308.SZ)	E (02/25/2026)	Rmb369.66
Tiffany Yeh		
AllRing Tech Co. (6187.TWO)	O (09/23/2025)	NT$1,270.00
FOCI Fiber Optic Communications Inc (3363.TWO)	O (01/15/2025)	NT$623.00
Himax Technologies Inc (HIMX.O)	E (02/04/2026)	US$13.67
Hon Precision (7769.TW)	O (04/17/2026)	NT$6,335.00
MPI Corporation (6223.TWO)	O (04/17/2026)	NT$5,600.00
Silicon Motion (SIMO.O)	O (05/06/2024)	US$241.83
WinWay Technology Co Ltd (6515.TW)	O (04/17/2026)	NT$6,025.00

Stock Ratings are subject to change. Please see latest research for each company.
* Historical prices are not split adjusted.